Model Parallelism in LLM Training

Autumn 2025

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Disclaimer

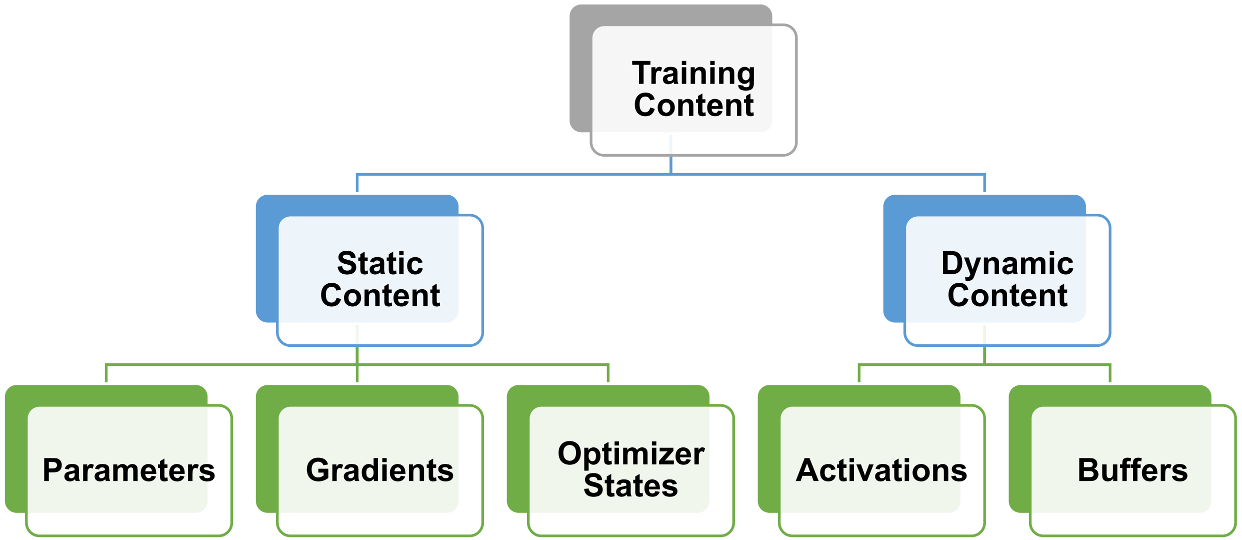
Machine learning systems is a broad and rapidly evolving field. The course material has been developed using a broad spectrum of resources, including research papers, lecture slides, blogposts, research talks, tutorial videos, and other materials shared by the research community.

Distributed LLM Training: Outline

- Data Parallelism
  - Parameter-Server
  - All-Reduce
  - Memory Optimization
- Model Parallelism
  - **Pipeline Parallelism**
  - Tensor Parallelism
  - Sequence Parallelism
- Mixture of Experts

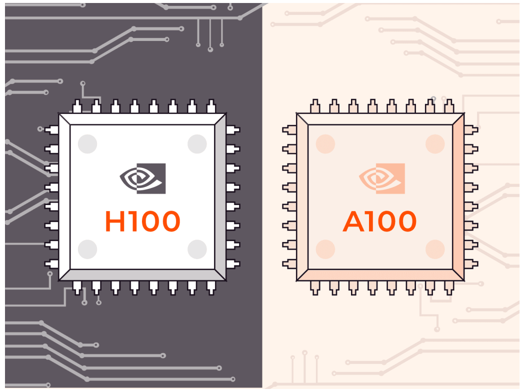
Pipeline Training

- Retrospect: GPU HBM Content During Training

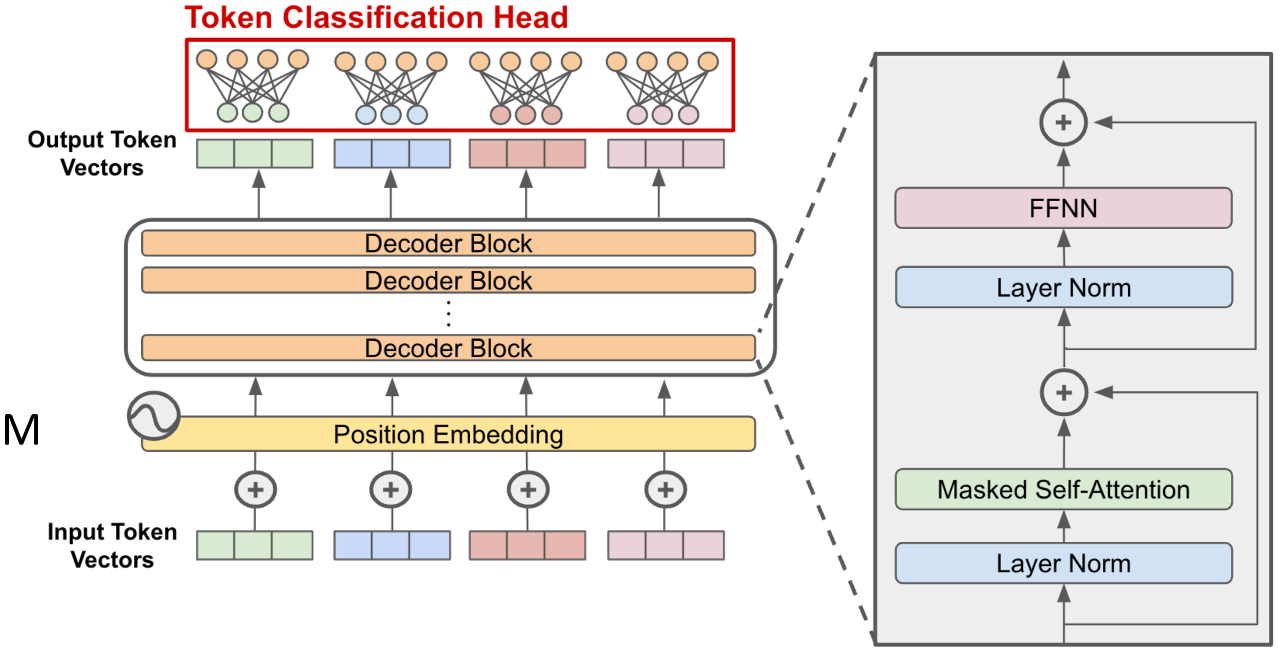


- An example of GPT-3 175B
  - **Optimizer States**
    - 32-bit Parameter (700 GB)
    - Adam Moment (700 GB)
    - Adam Variance (700 GB)
  - **16-bit Parameter (350 GB)**
  - **16-bit Gradient (350 GB)**
  - **Activations (depending on batch size)**
  - Buffer and Fragmentation

Pipeline Training

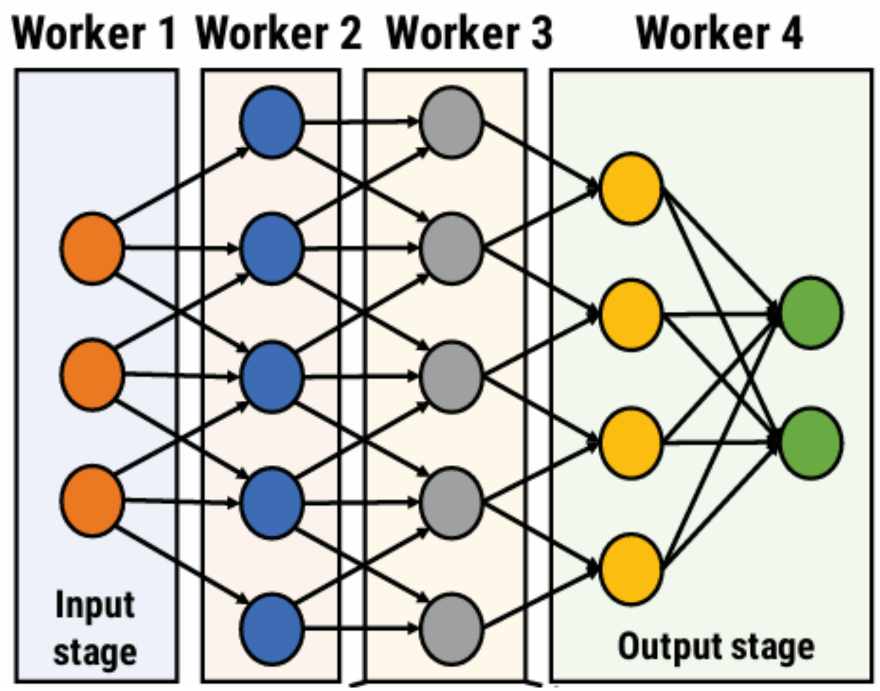


Insufficient HBM



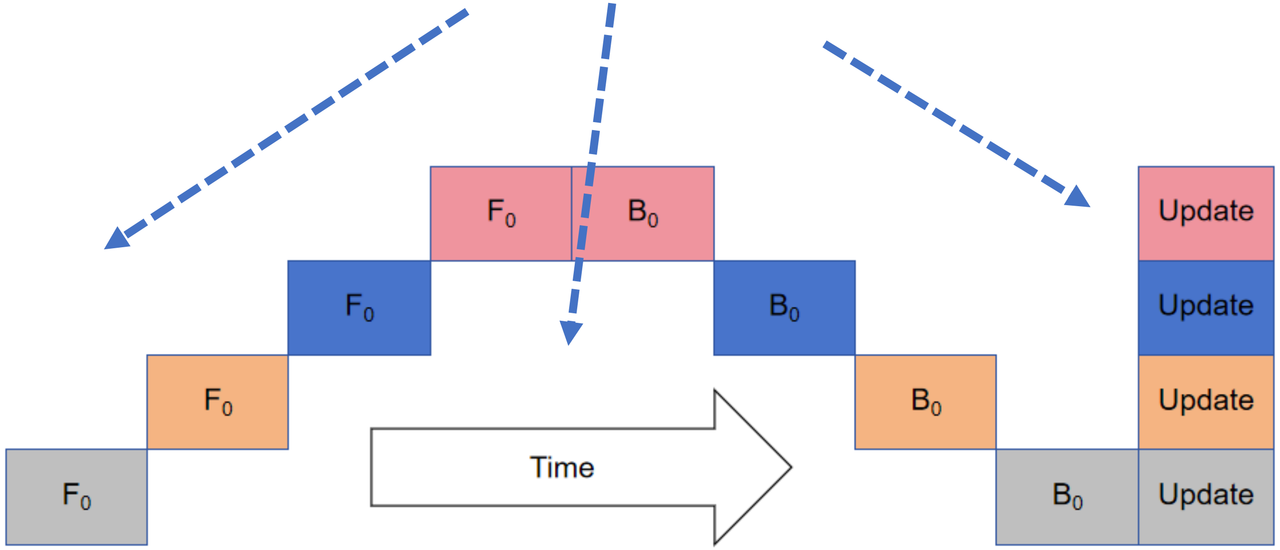
When LLM model is too LARGE!

Pipeline Training



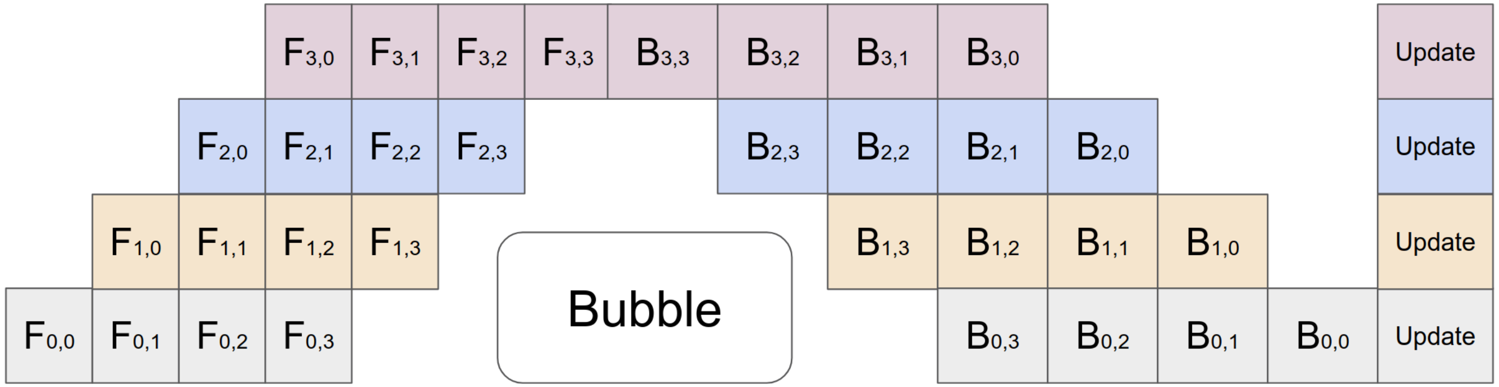
Layerwise partitioning on multiple GPUs

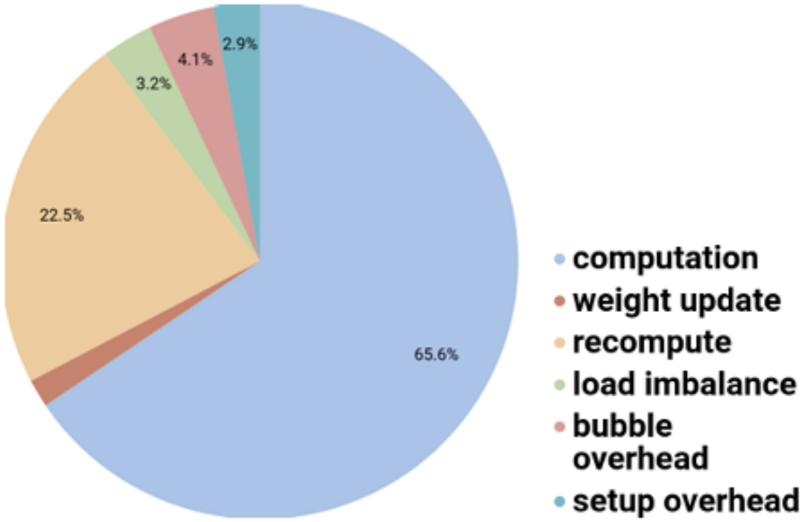
Large bubble $O(\frac{K-1}{K})$ → underutilized GPU



Naïve model parallelism
(F for forward； B for backward)

Pipeline Training



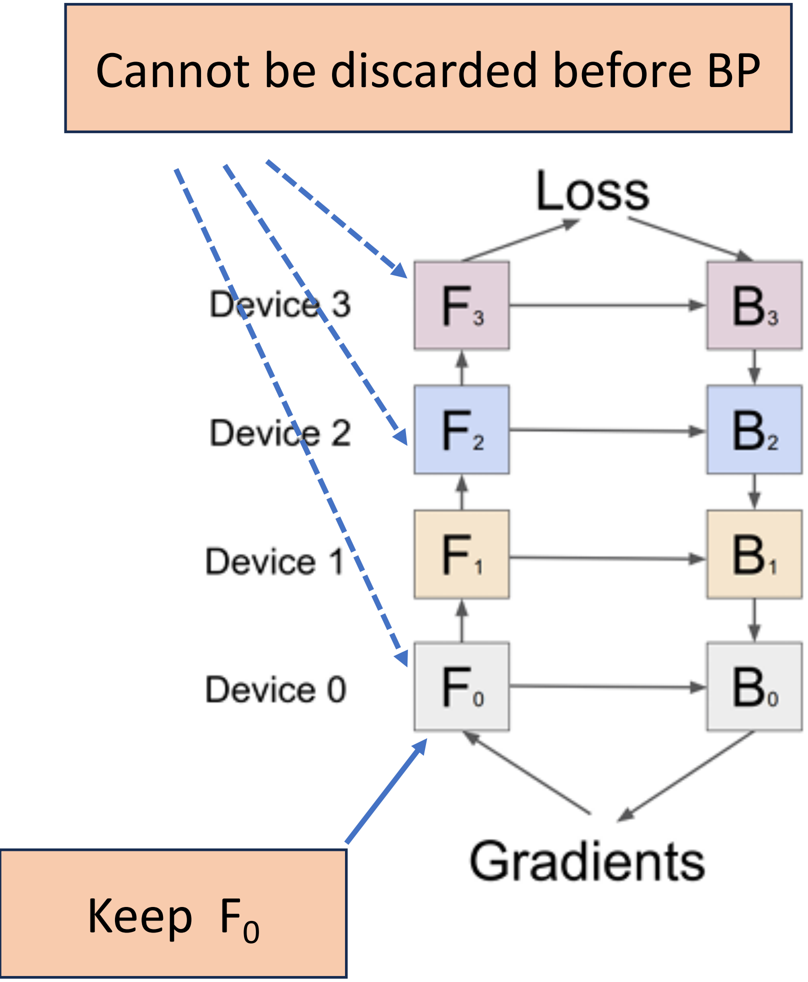


b: batch size
b_m : microbatch size
K: # pipeline stages
M: # microbatches, b = b_m *M

Bubble ratio: $O(\frac{K-1}{K+M-1})$

《GPipe: Efficient Training of Giant Neural Networks using Pipeline Parallelism》

Pipeline Training



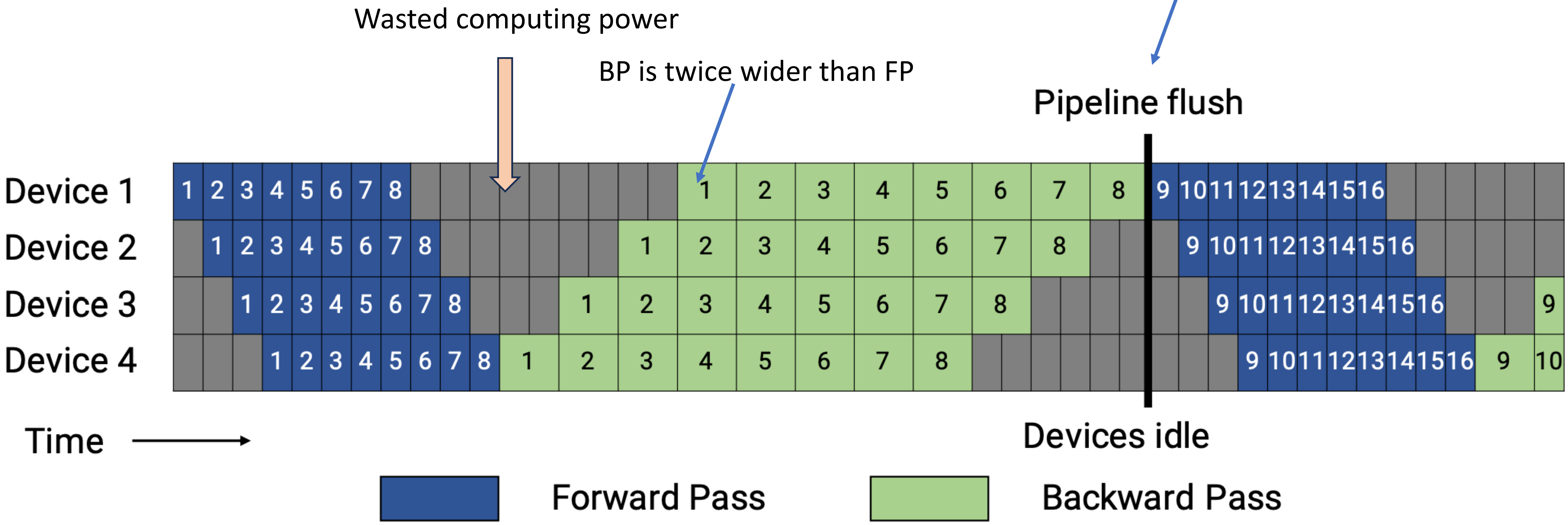
- Computation time
  - Forward time is half of backward time (why?)
- Communication load
  - Intermediate activation between two partitions
- Memory occupation
  - First stage usually needs more memory: $K * M$
  - Re-materialization (re-computation) Further reducing memory consumption

F-then-B: Forward then Backward

《GPipe: Efficient Training of Giant Neural Networks using Pipeline Parallelism》

Pipeline Training

Flush is an important operation to avoid model discrepancy, but also creates more bubbles



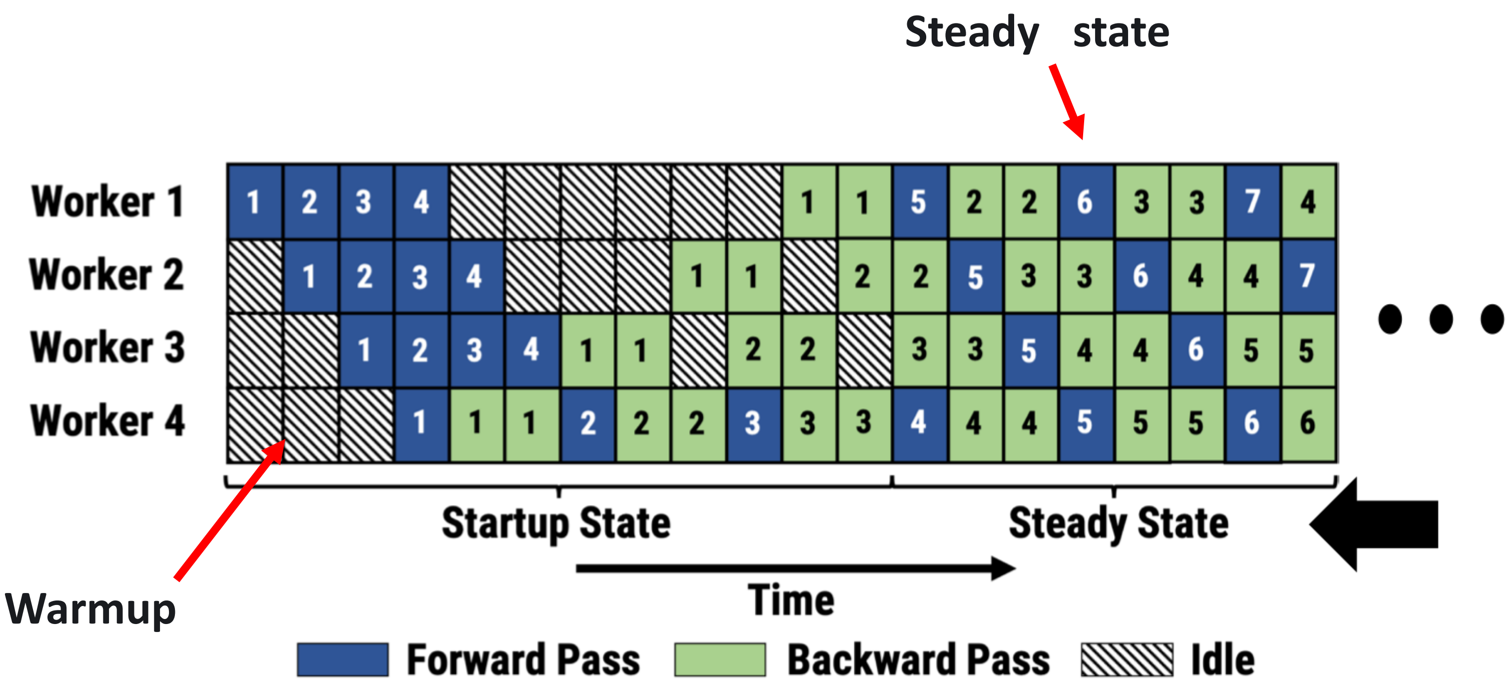
Frequent pipeline flushes lead to increased idle time

Pipeline Training

Orchestrating execution orders of forward and backward passes for microbatch so as to reduce bubbles and peak memory footprints?

- Some KEY concepts
  - Synchronous pipeline versus asynchronous pipeline
  - Interleaved versus non-interleaved

Pipeline Training

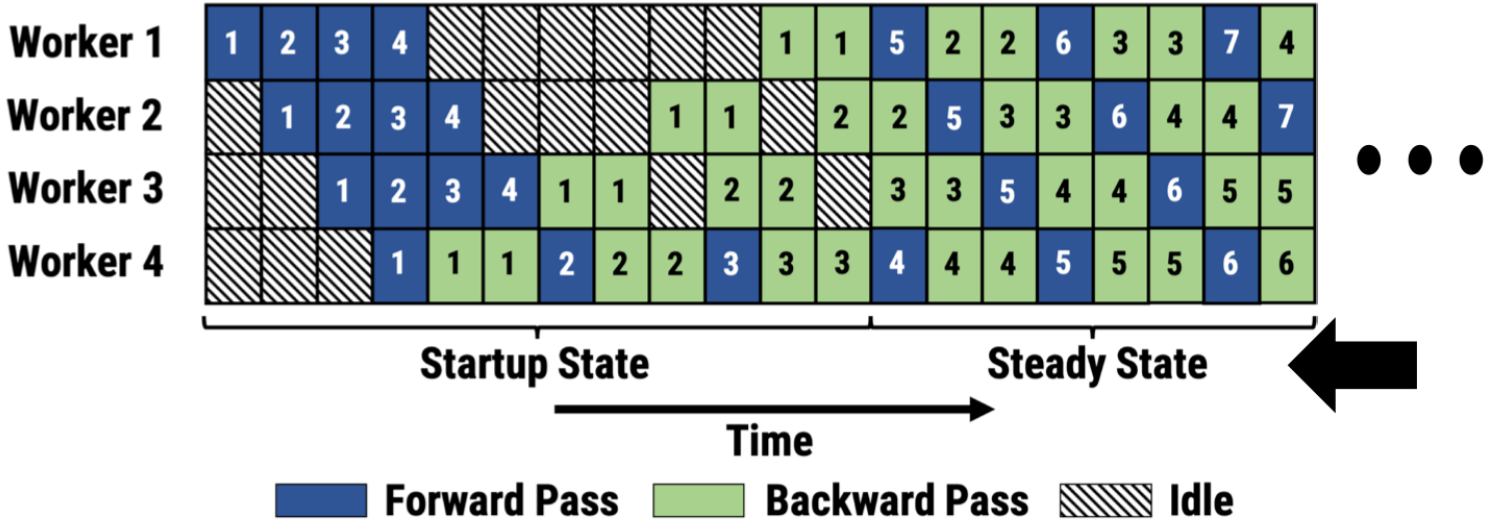


1F1B: One Forward then One Backward

- Minimizing lingering time of activations
  - Immediately executing BP after FP for the first microbatch at the output stage
- At steady state
  - Strictly alternative Forward and Backward operations
- Asynchronous weights

《PipeDream: generalized pipeline parallelism for DNN training》

Pipeline Training



1F1B: One Forward then One Backward

- Challenges
  - How to split a DNN model into different stages?

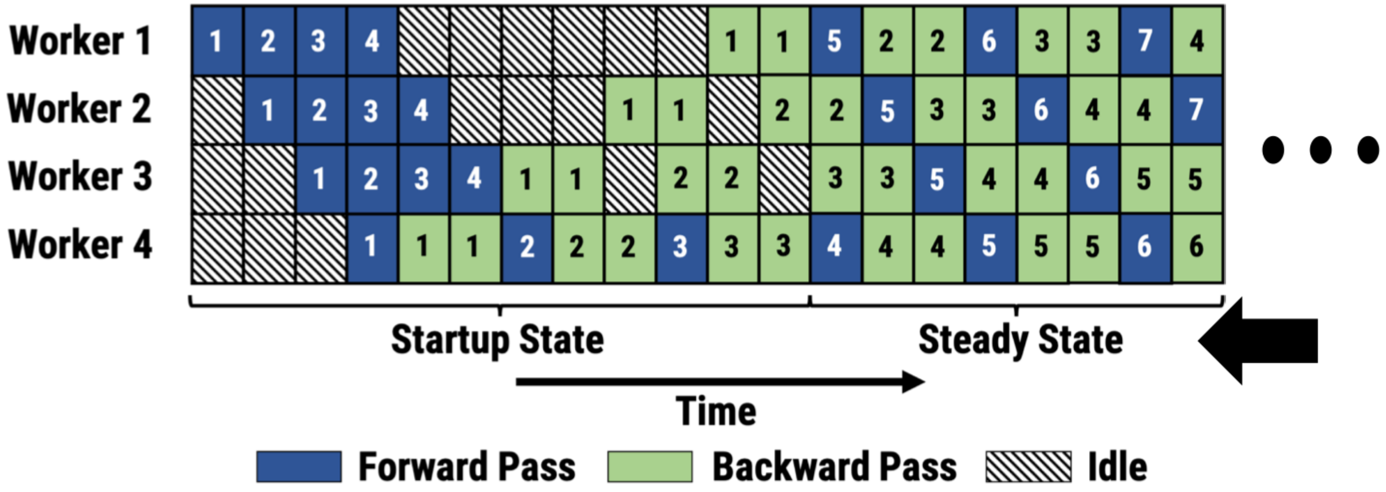
LOAD BALANCING *across stages*

  - How to update parameters?

After backward pass, the gradient can be used ***immediately to update the parameter***

《PipeDream: generalized pipeline parallelism for DNN training》

Pipeline Training



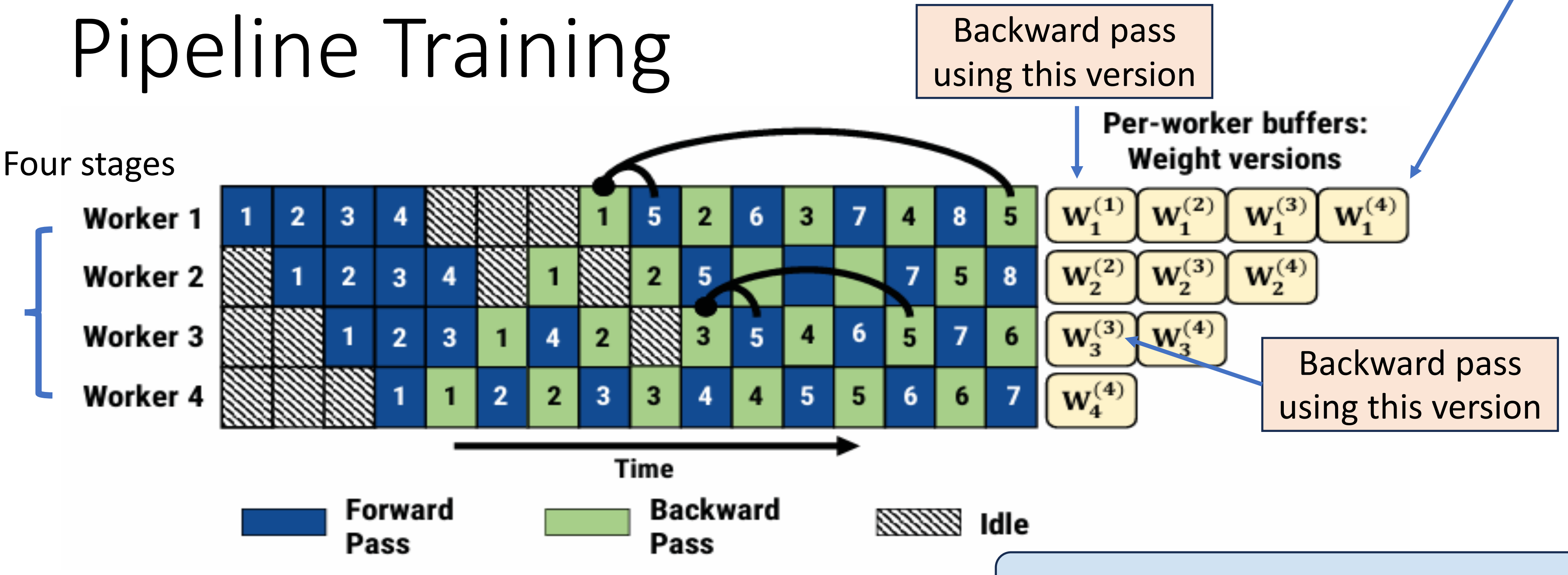
- Observations in stage 1
  - The forward pass of microbatch 5 is performed after the update of **microbatch 1**
  - The backward pass of microbatch 5 is performed after the updates of **microbatches 2, 3 and 4**

Forward and backward passes operate on different versions of parameters!

《PipeDream: generalized pipeline parallelism for DNN training》

Subscript: stage ID, superscript: model version ID

Pipeline Training

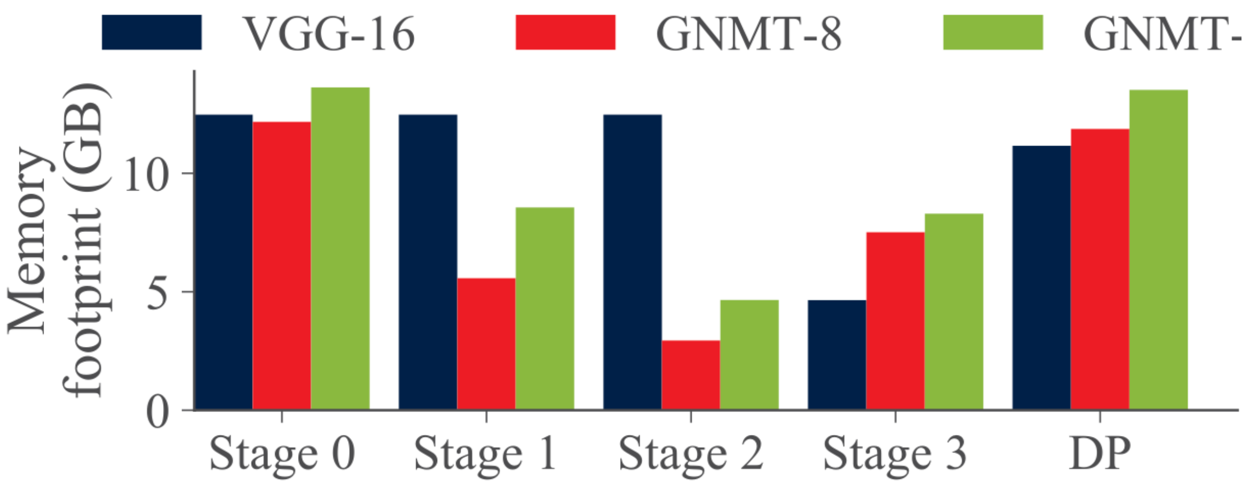


No guarantees across stages!

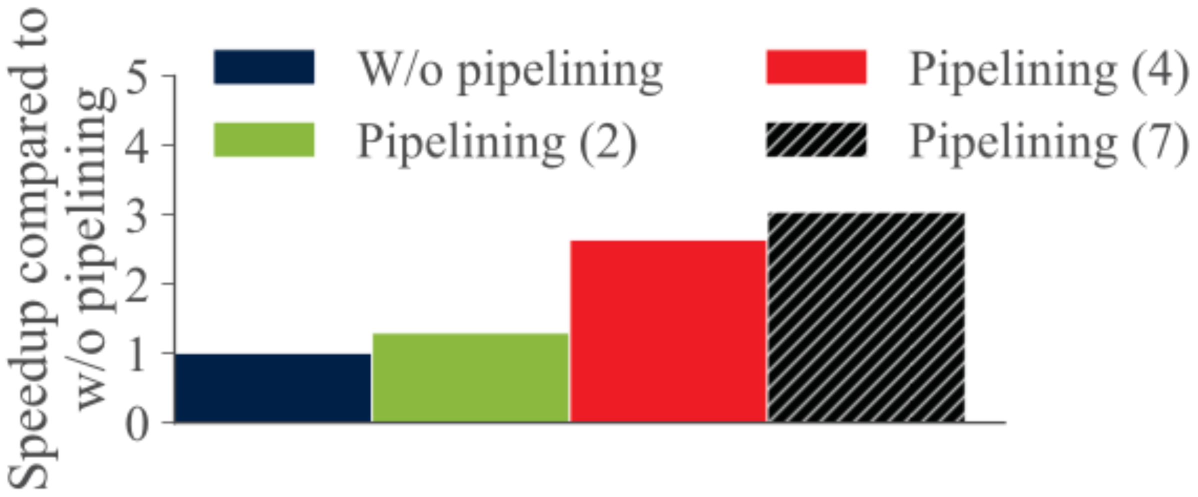
- ***Weight stashing*** of PipeDream
  - keeps multiple versions of model weights—one for each active minibatch.
  - Forward pass: each pipeline stage uses the most recent weight version available. Once the forward pass for a minibatch is completed, the corresponding weight version is stored.
  - Backward pass: the same version is reused to compute gradients and update weights, ensuring that both passes for a given minibatch use identical parameters within a stage.

Pipeline Training

- Experiments



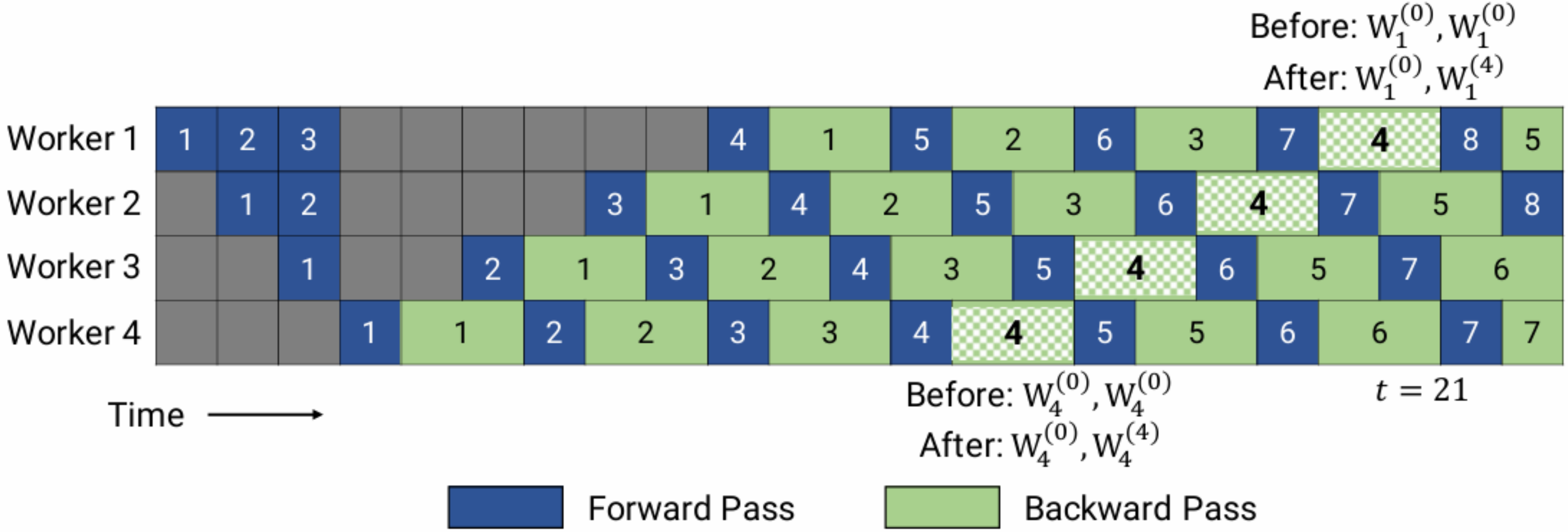
Memory footprint of PipeDream



Speedup ratios on 4 V100 GPUs

PipeDream still saves memory by trading multiple versions of parameters for large activations

Pipeline Training

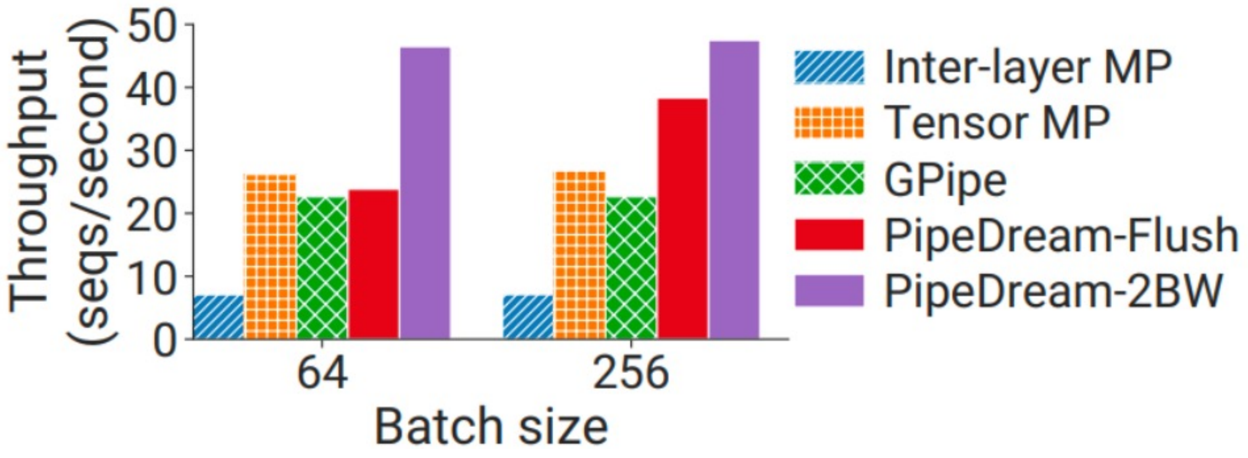


PipeDream-2BW: (1) less frequent flushing than GPipe (2) keeping no more than two model versions.

Think why? How does it related to pipeline depth K and number of microbranches M

《Memory-Efficient Pipeline-Parallel DNN Training》

Pipeline Training



- Flushing causes more bubbles
  - Warmup + Colddown
- Flushing needs less memory

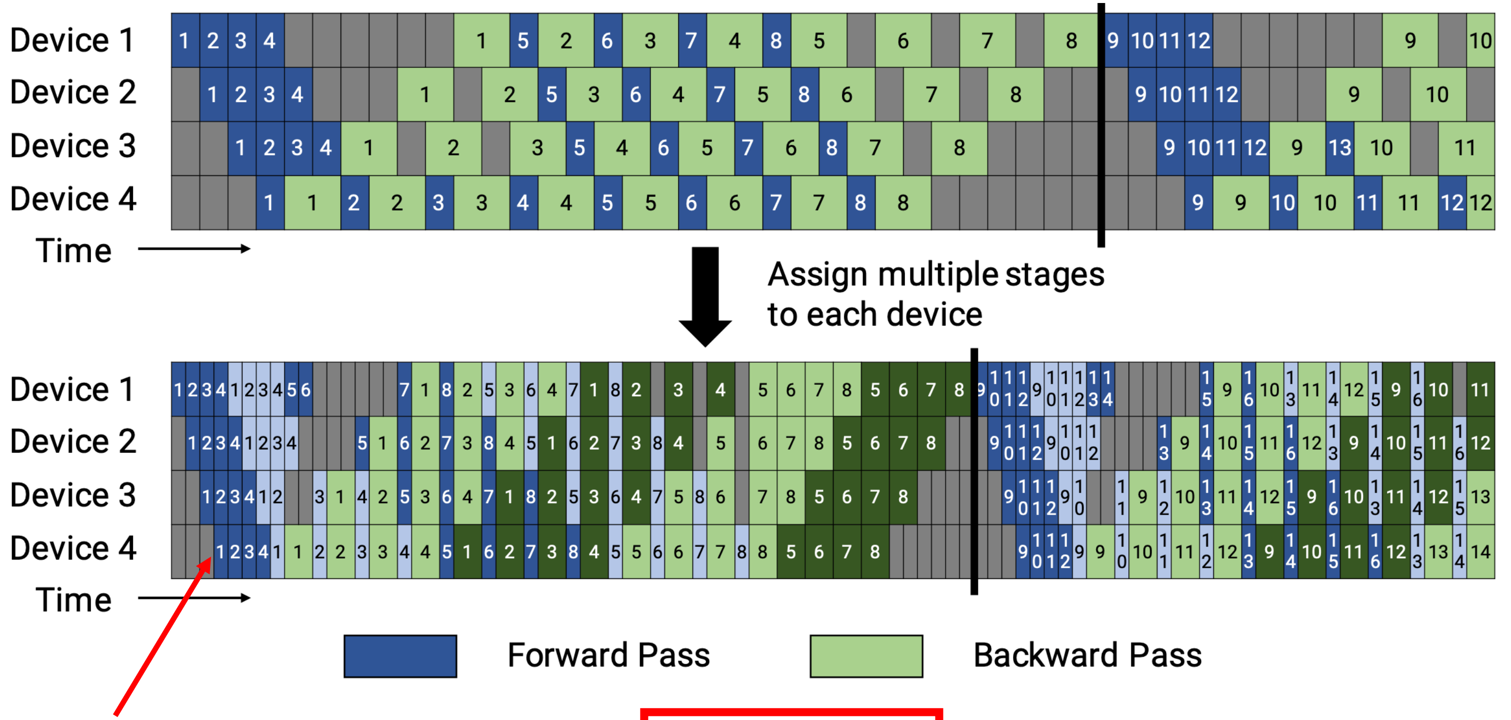
- Training BERT 2.2B with 8-way model parallelism (8 V100 GPU)
- PipeDream is much better than vannila GPipe

《Memory-Efficient Pipeline-Parallel DNN Training》

https://juejin.cn/post/7262274383287484476

Pipeline Training

- IFIB integrated in Megatron-LM later on



Chunk

Interleaved 1F1B

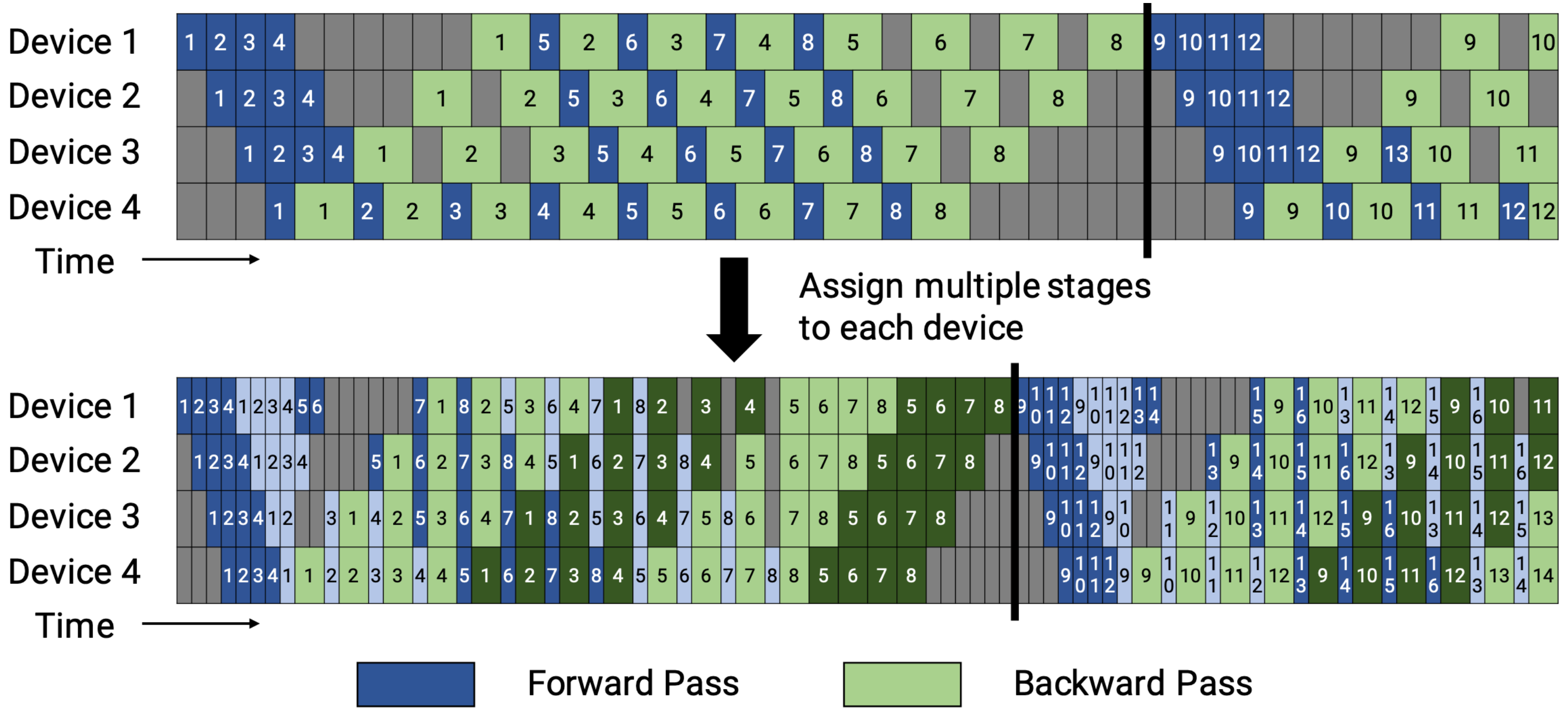
(Stages: 4, Layers: 16, each device assigned with 2 chunks. Layer 0-1, 8-9 on Device 1; Layer 2-3,10-11 on Device 2; Layer 4-5, 12-13 on Device 3 and Layer 6-7, 14-15 on Device4)

- Key ides:
  - Further dividing each microbatch into even smaller chunks
  - Microbatch and small chunks execute alternatively
- Advantages/disadvantages
  - Smaller bubble
  - Higher inter-state traffic load

《Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron-LM》

Pipeline Training

- IFIB integrated in Megatron-LM later on

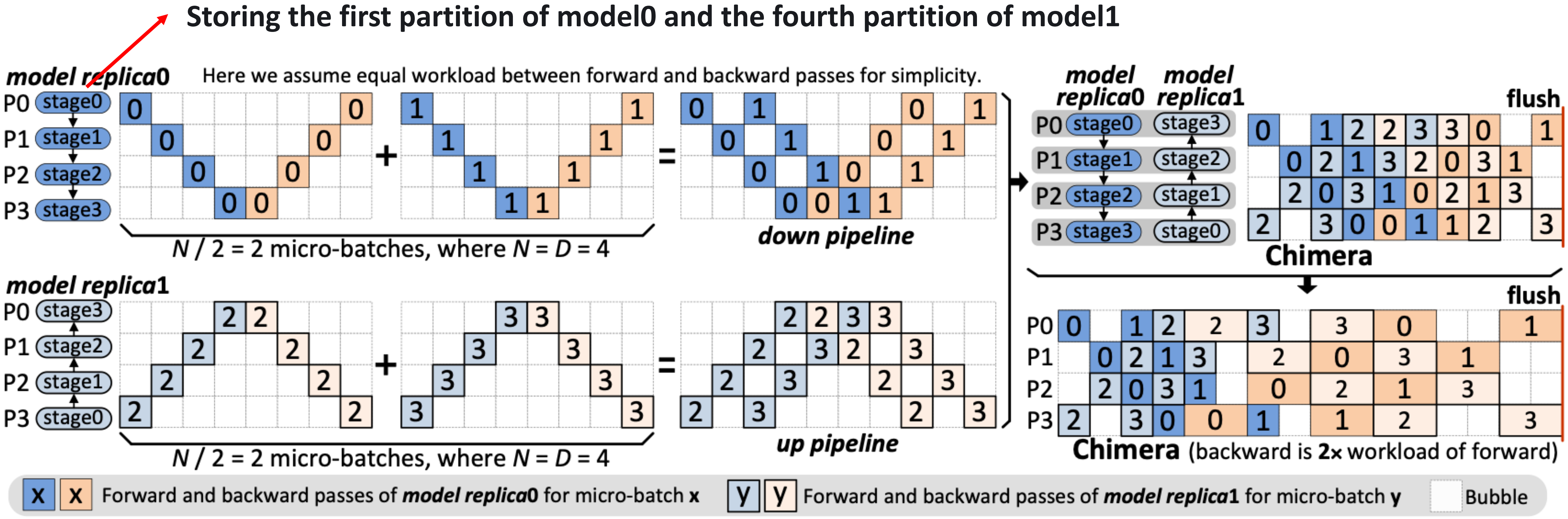


- Model placement:
  - Stages: 4, Layers: 16, each device assigned with 2 chunks
  - Layer 0-1, 8-9 on Device 1
  - Layer 2-3,10-11 on Device 2
  - Layer 4-5, 12-13 on Device 3
  - Layer 6-7, 14-15 on Device 4

Be aware of the communication scheduling!

《Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron-LM》

Pipeline Training



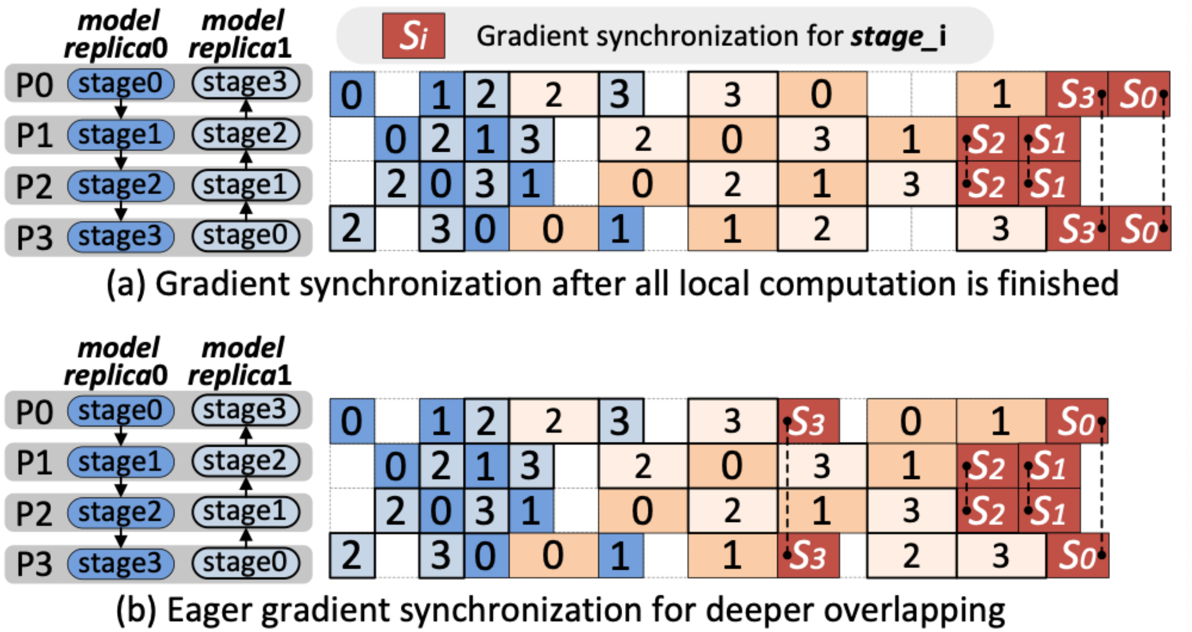
Chimera: double pipeline

- Smaller bubbles yet doubled memory occupation for model replicas

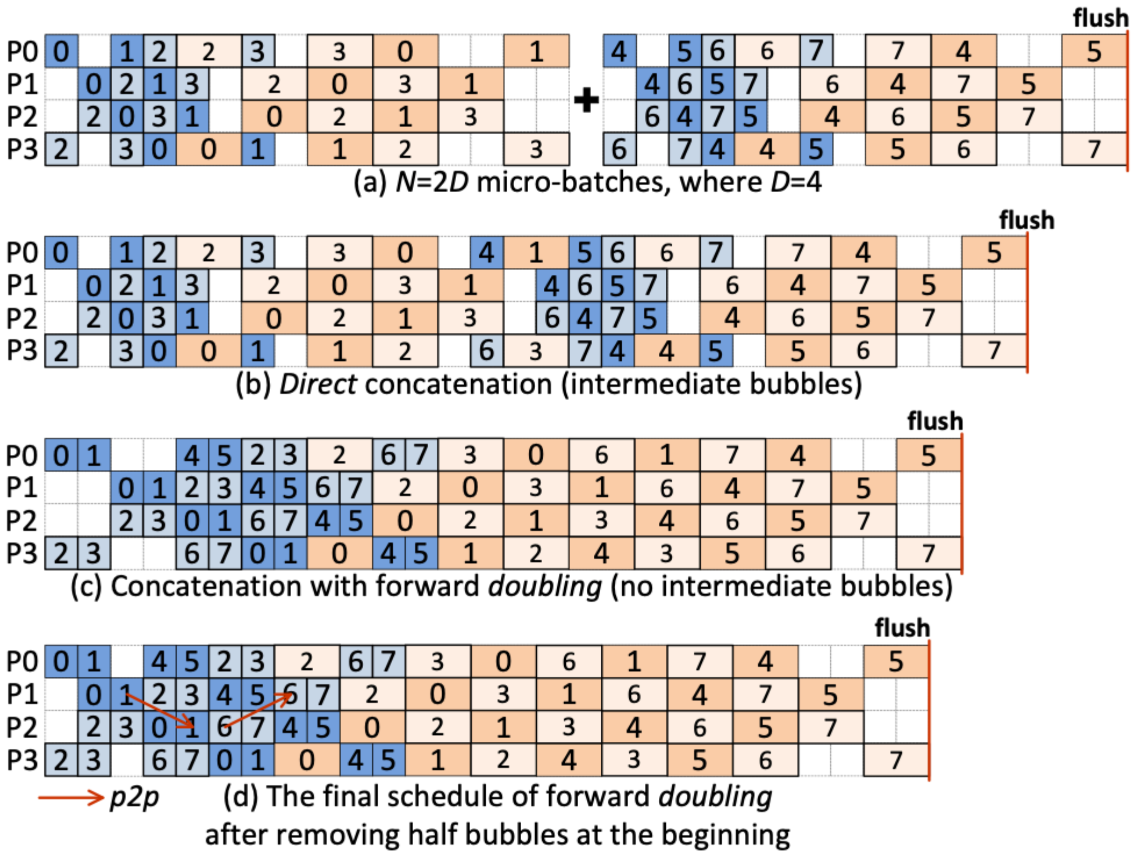
《Chimera: Efficiently Training Large-Scale Neural Networks with Bidirectional Pipelines》

Pipeline Training

- Orchestrating bubble mitigation and data-parallel synchronization



Hide the gradient synchronization overhead by overlapping

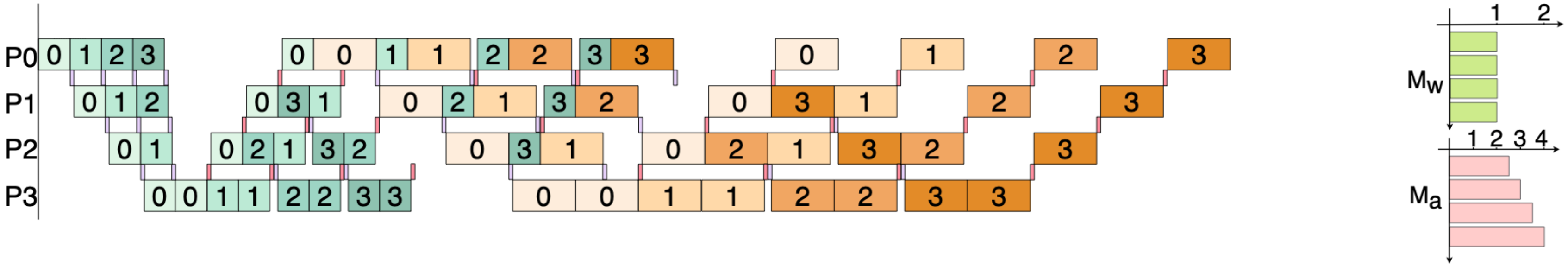


Select # replicas, # stages and Assemble blocks

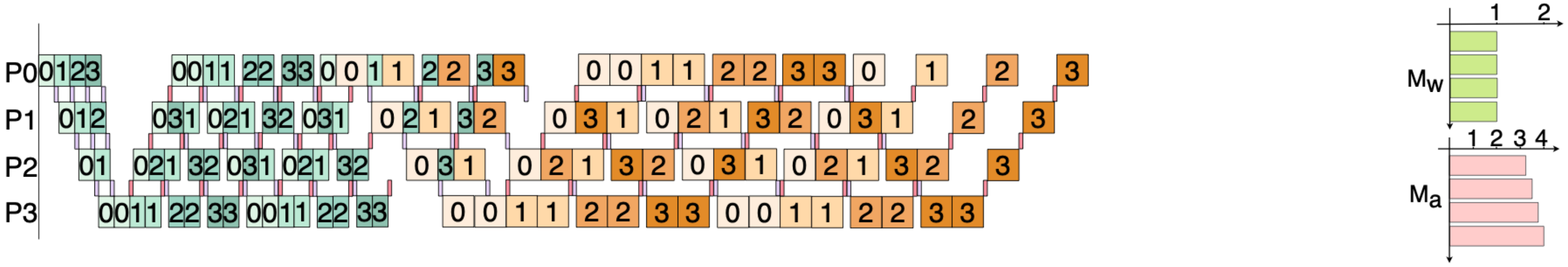
《Chimera: Efficiently Training Large-Scale Neural Networks with Bidirectional Pipelines》

Pipeline Training

- Using multiple WAVEs by dividing a microbatch into multiple smaller microbatches



(d) Hanayo with one wave



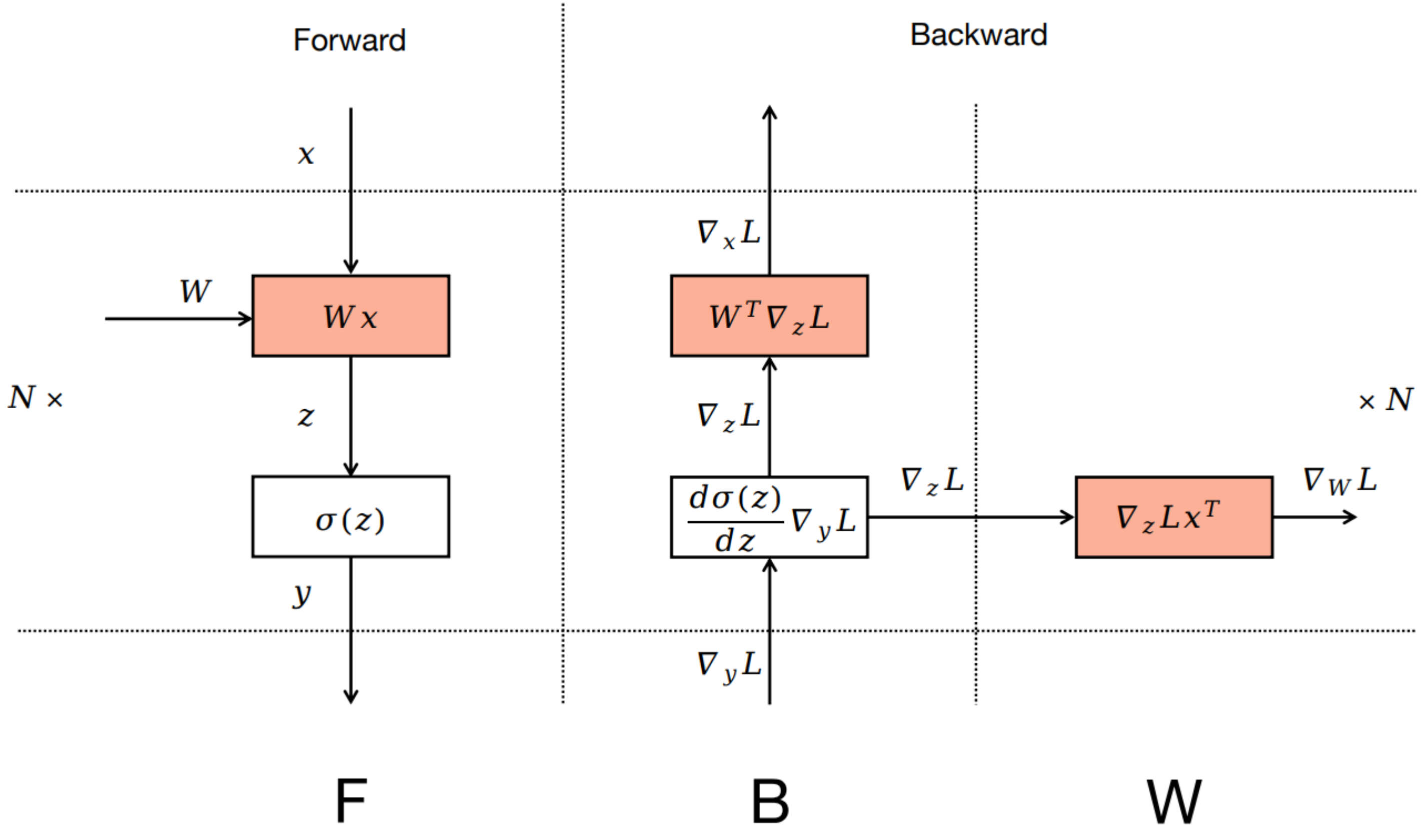
(e) Hanayo with two waves

Forward propagation marked green and Backward propagation marked orange. Back propagation illustrated twice as long as forward propagation. Using purple and pink blocks to represent P2P communication that goes downward and upward. The numbers on the blocks show which of the micro-batches is being processed.

《Hanayo: Harnessing Wave-like Pipeline Parallelism for Enhanced Large Model Training Efficiency》

Pipeline Training

- A microscopic view of BP

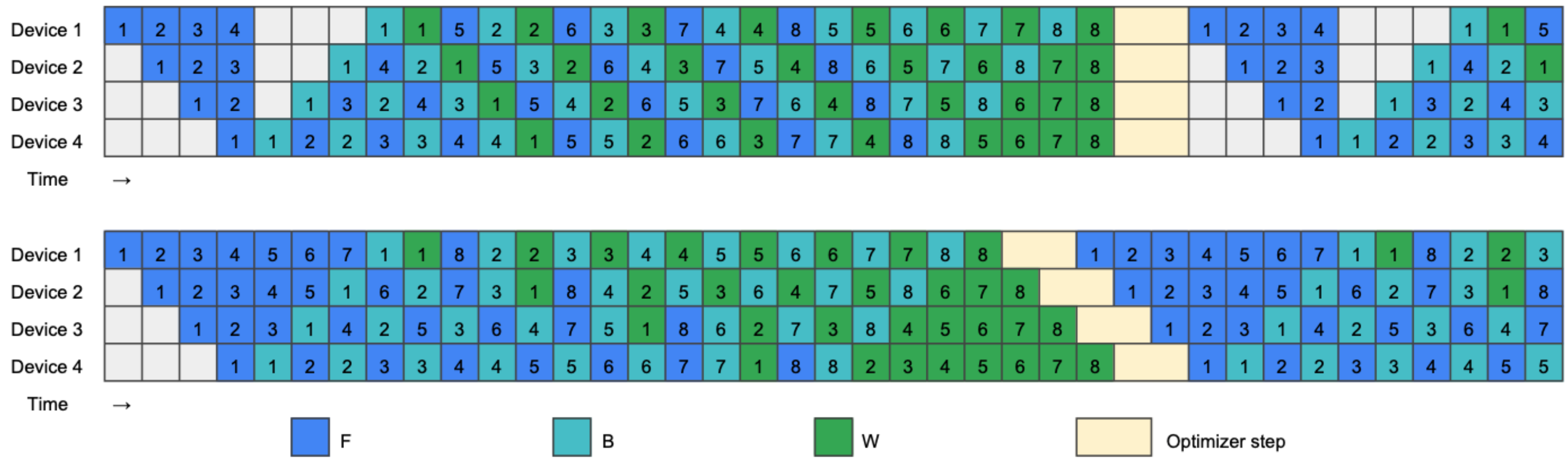


- Derivative over input
  - for backward propagation
- Derivative over weight
  - for updating model directly
- $\nabla_{\boldsymbol{W}} f(\boldsymbol{x}, \boldsymbol{W})^\top \frac{d\ell}{d\boldsymbol{y}}$ computed immediately after $\nabla_{\boldsymbol{x}} f(\boldsymbol{x}, \boldsymbol{W})^\top \frac{d\ell}{d\boldsymbol{y}}$

《Zero Bubble Pipeline Parallelism》

Pipeline Training

- Dividing BP into BP for input (urgent) and BP for weight (not urgent)

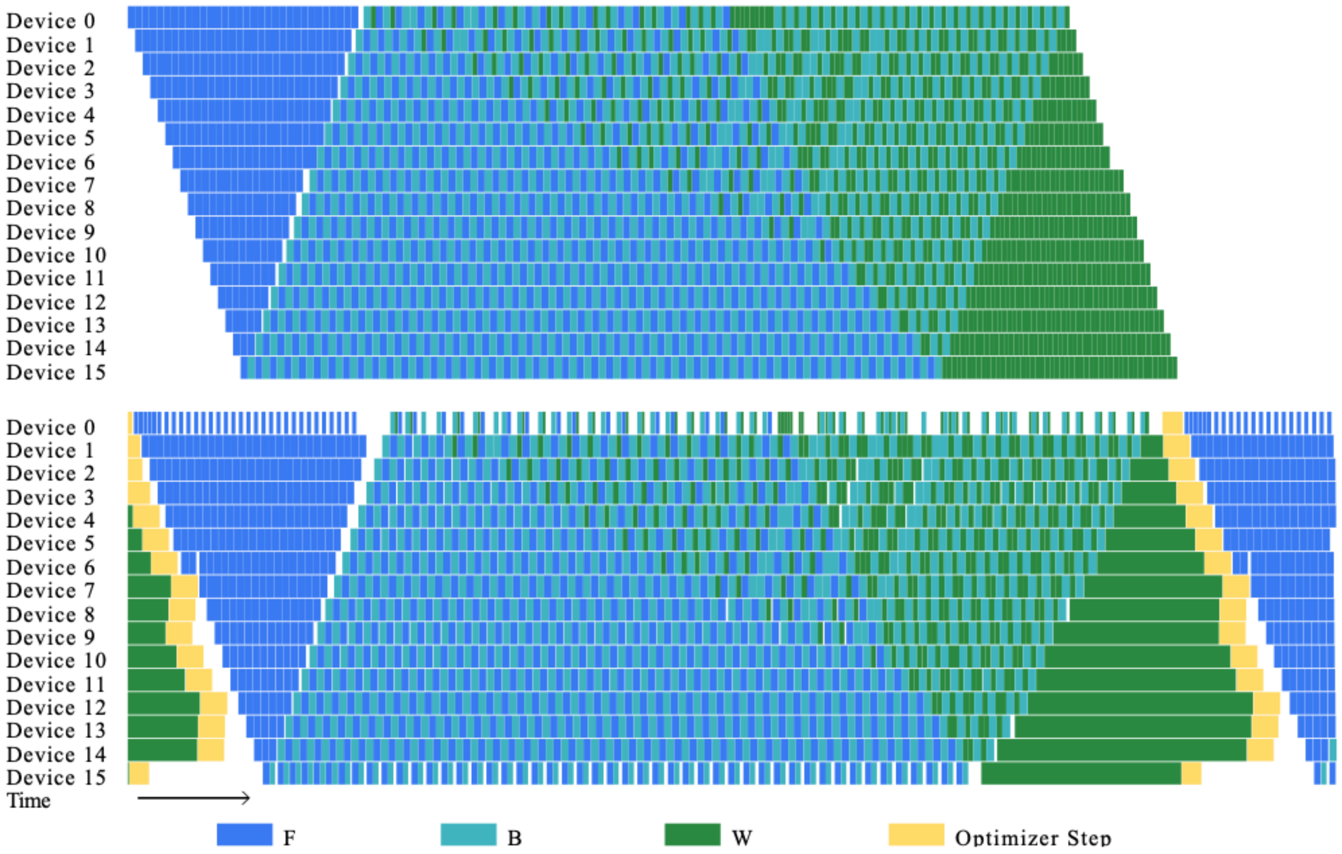


- Upper (ZB-H1): strictly follow 1F1B, and fill W to ensure all workers maintain the same number of in-flight microbatches.
- Lower (ZB-H2): theoretically zero bubble as # of microbatches is sufficiently large; large memory occupation over ZB-H1.

《Zero Bubble Pipeline Parallelism》

Pipeline Training

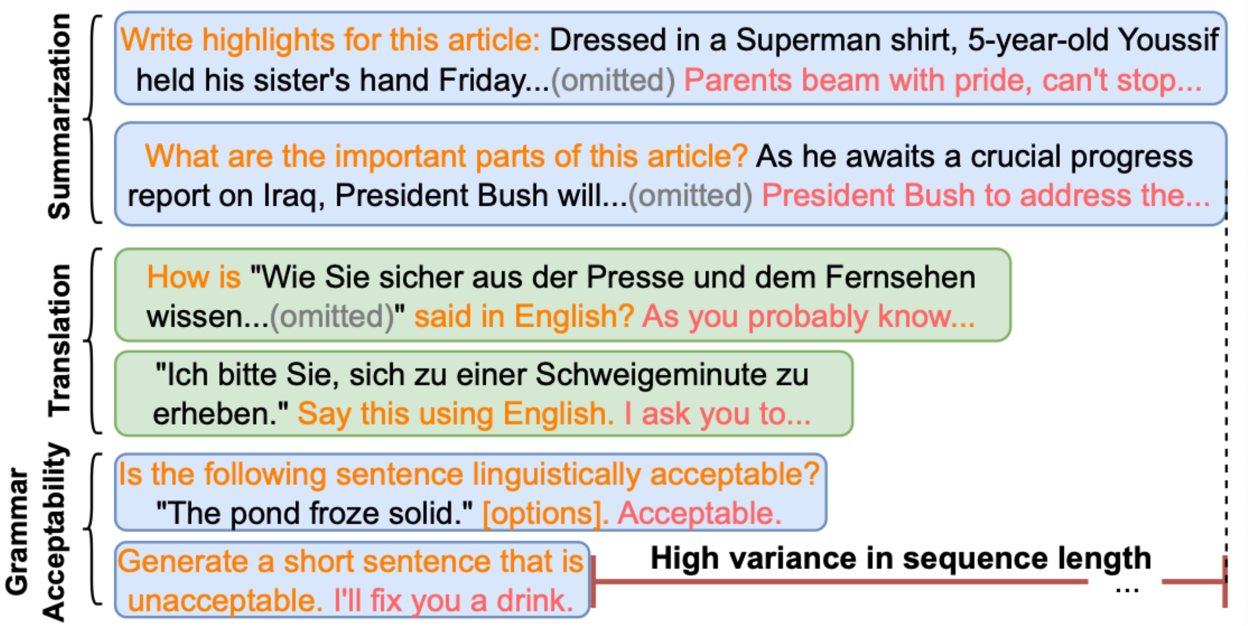
- An illustration of zero bubble with 16 stages: inserting W ops strategically

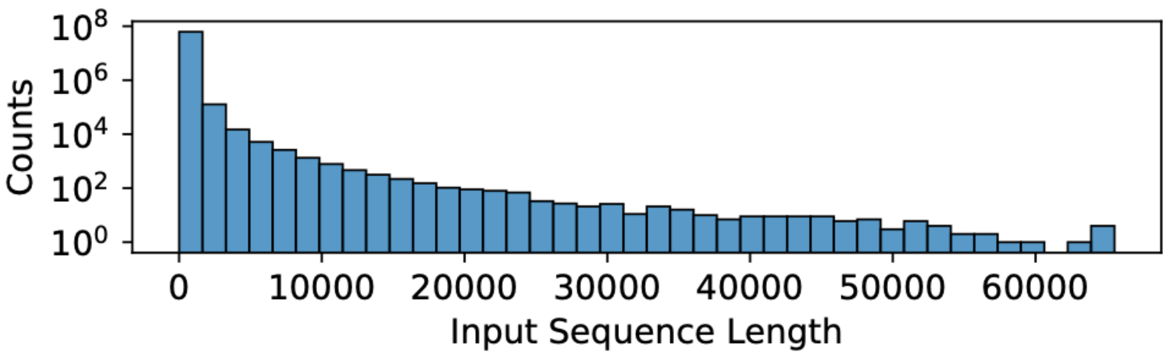


《Zero Bubble Pipeline Parallelism》

Pipeline Training

- Multi-task model pretraining
  - Input sequence lengths are non-uniform





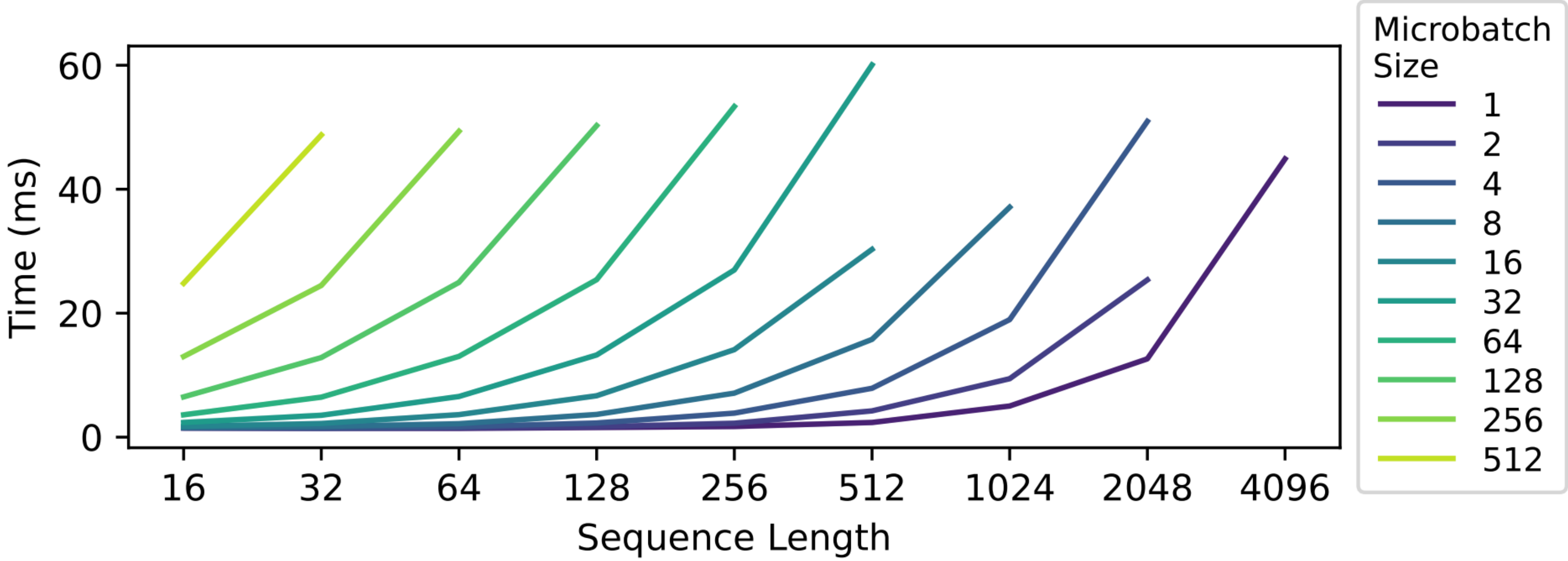
(left) Orange texts are instructions to the model. Inputs to process are colored black. Expected responses are in red.
(upper) The sequence length distribution in dataset (truncated at 65536).

《DynaPipe: Optimizing Multi-task Training through Dynamic Pipelines》

Pipeline Training

- Multi-task model pretraining
  - Execution time exhibits super-linear growth with sequence length

$$FLOPS\ per\ Transformer\ Layer\ =\ 24bsh^2 + 4b\boldsymbol{s^2}h$$



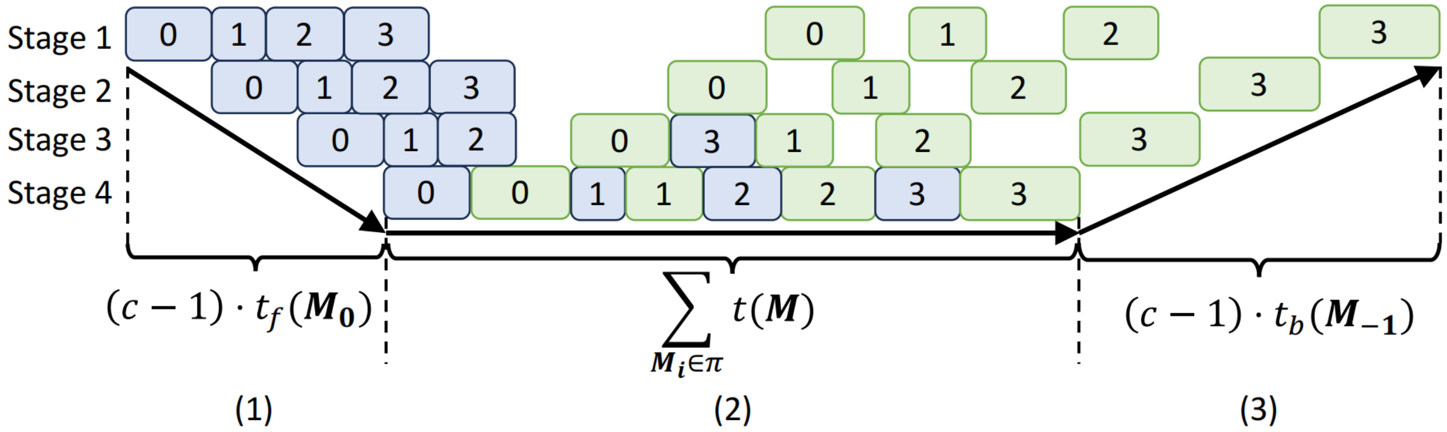
《DynaPipe: Optimizing Multi-task Training through Dynamic Pipelines》

Pipeline Training

- Multi-task model pretraining
  - Seek the best micro-batch assignment $\boldsymbol{\pi}$ to minimize iteration time

$$\min_{\boldsymbol{\pi}} \left\{ (c-1) \cdot \max\{t(\mathbf{M_i}) | \mathbf{M_i} \in \boldsymbol{\pi}\} + \sum_{\mathbf{M_i} \in \boldsymbol{\pi}} t(\mathbf{M_i}) \right\}$$

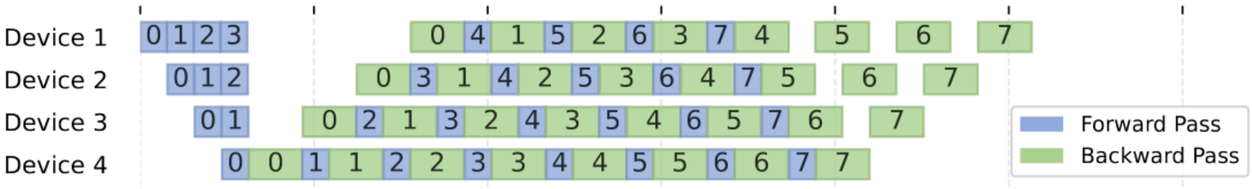
  - Construct a DP algorithm to optimally partition samples



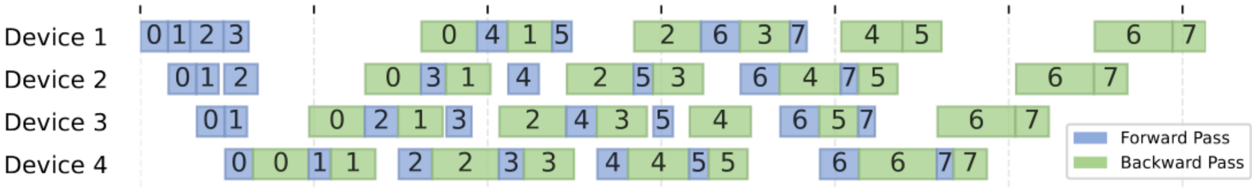
《DynaPipe: Optimizing Multi-task Training through Dynamic Pipelines》

Pipeline Training

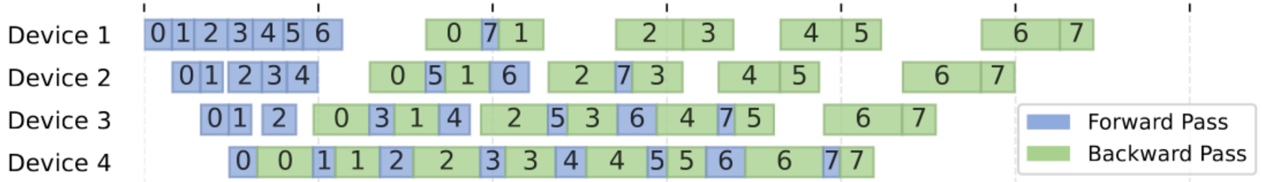
- Multi-task model pretraining
  - Packing and scheduling



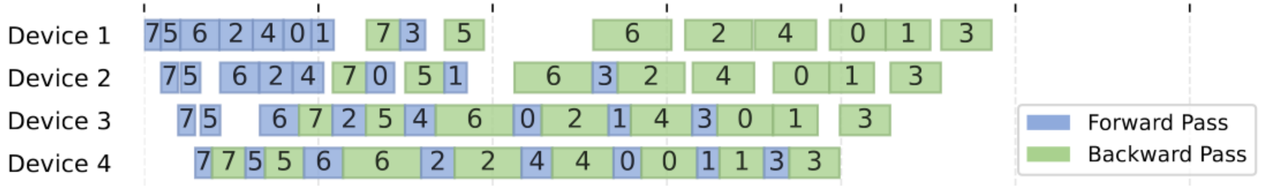
(a) 1F1B (uniform micro-batches)



(b) 1F1B (dynamic micro-batches)



(c) Adaptive Schedule



(d) Adaptive Schedule & Micro-batch Reordering

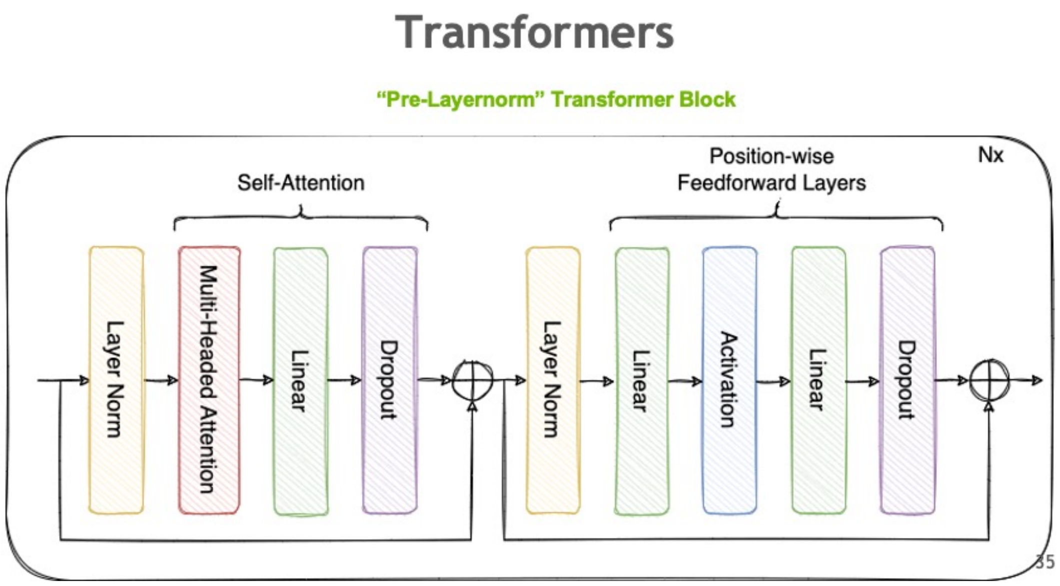
《DynaPipe: Optimizing Multi-task Training through Dynamic Pipelines》

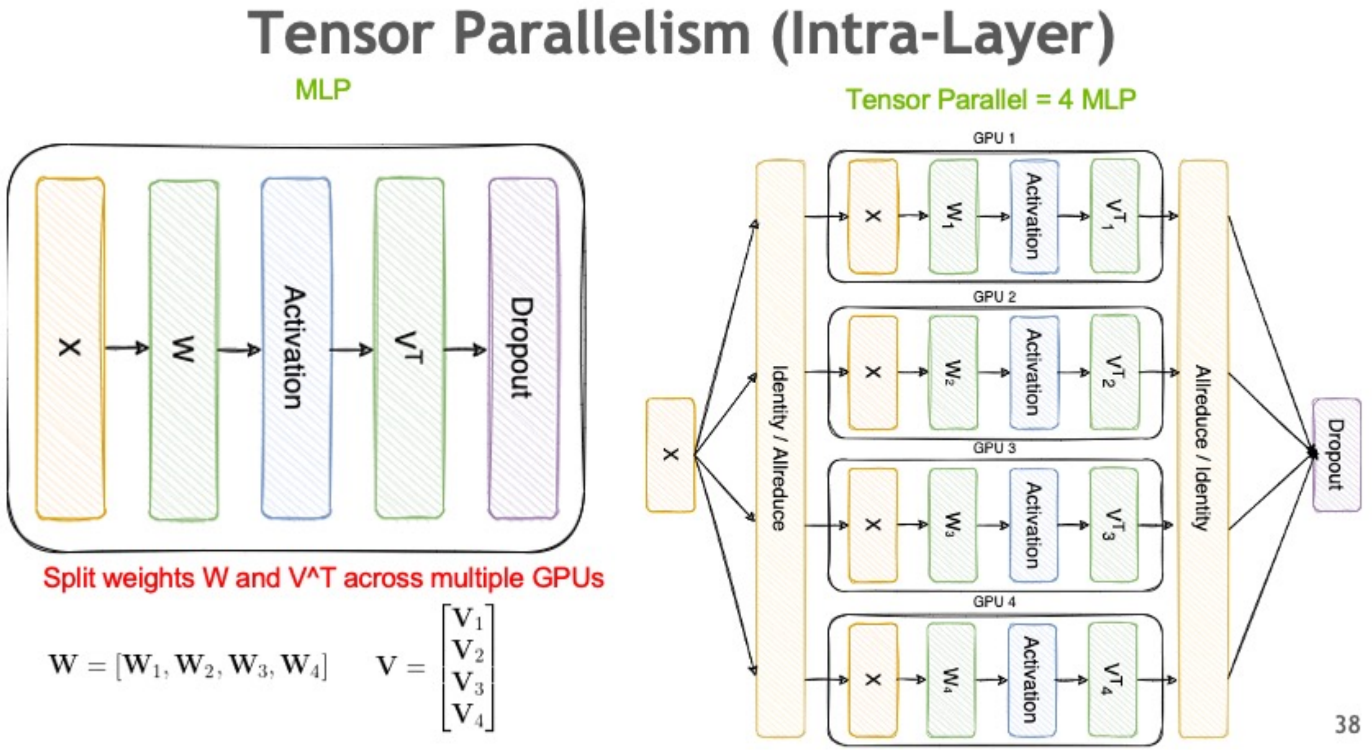
Distributed LLM Training: Outline

- Data Parallelism
  - Parameter-Server
  - All-Reduce
  - Memory Optimization
- Model Parallelism
  - Pipeline Parallelism
  - **Tensor Parallelism**
  - Sequence Parallelism
- Mixture of Experts

Tensor Parallelism

- A visual perception of TP



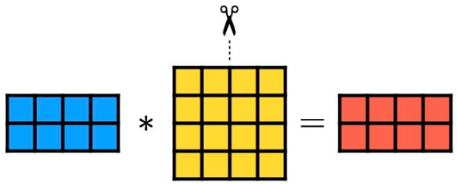


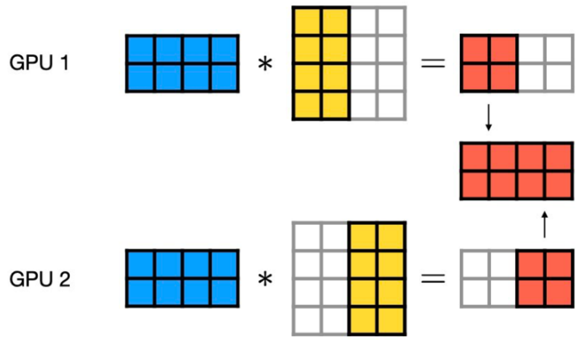
Easily understandable, but how?

《Demystifying Tensor Parallelism》 by *Josh Levy-Kramer*

Tensor Parallelism

- Mathematical principle of tensor parallelism (e.g. GEMM)
  - 1D: segmenting a matrix by **column** or row only
    - *First linear layer* $Y = XA: A = [A_1, A_2] \rightarrow Y = [XA_1, XA_2] = [Y_1, Y_2]$





    - *Second linear layer* $Z = YB:$

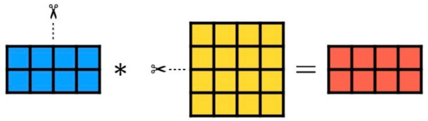
$$Z = [Y_1 \ \ Y_2] \cdot \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

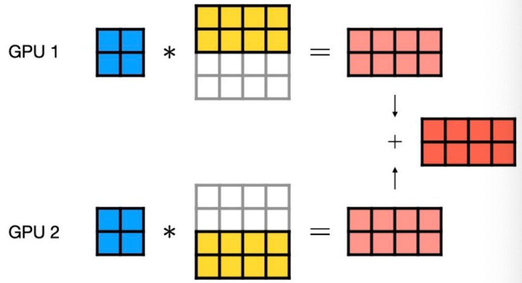
      - *Compute* $Y_i \ B_i$ *independently and aggregate* $Z = Y_1 B_1 + Y_2 B_2$
    - Question: memory usage? communication load? end-to-end latency?
      - $1/P$ parameter, Full activation, $2(P-1)/P$ communication load, 2(P-1) latency

Tensor Parallelism

- Mathematical principle of tensor parallelism (e.g. GEMM)
  - 1D: segmenting a matrix by column or **row** only
    - *First linear layer $Y = XA$:*

$$Y = \begin{bmatrix} X_1 & X_2 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$$





- *Second linear layer Z = YB:*

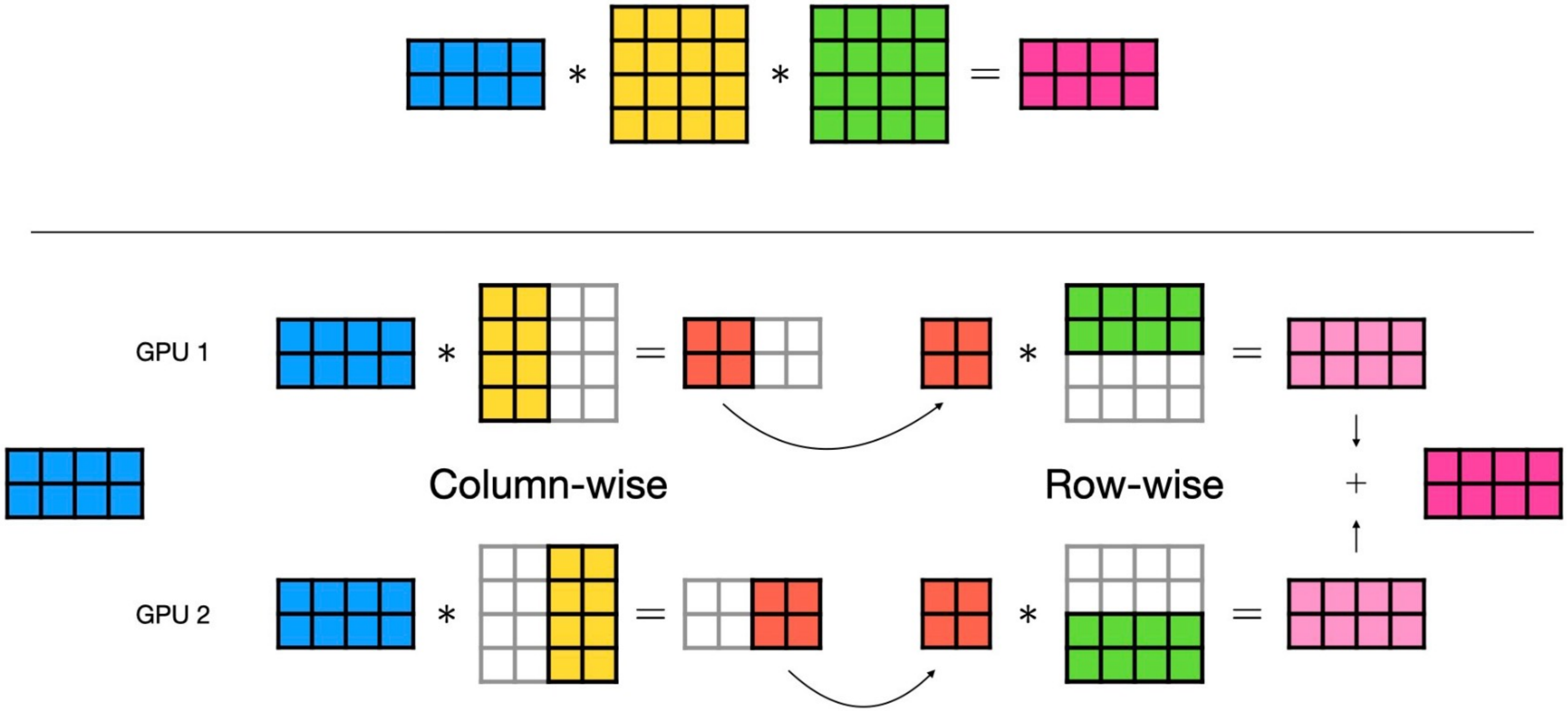
$$Z = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} \cdot \begin{bmatrix} B_1 & B_2 \end{bmatrix}$$

  - *Compute $Y_i\ B_i$ independently and aggregate* $Z = Y_1 B_1 + Y_2 B_2$

- Question: memory usage? communication load? end-to-end latency?
  - 1/P computation, 1/P parameter, Full activation, 2(P-1)/P communication load, 2(P-1) latency

Tensor Parallelism

- How to split matrices
  - The column-wise and row-wise styles can be combined for multiple linear layers

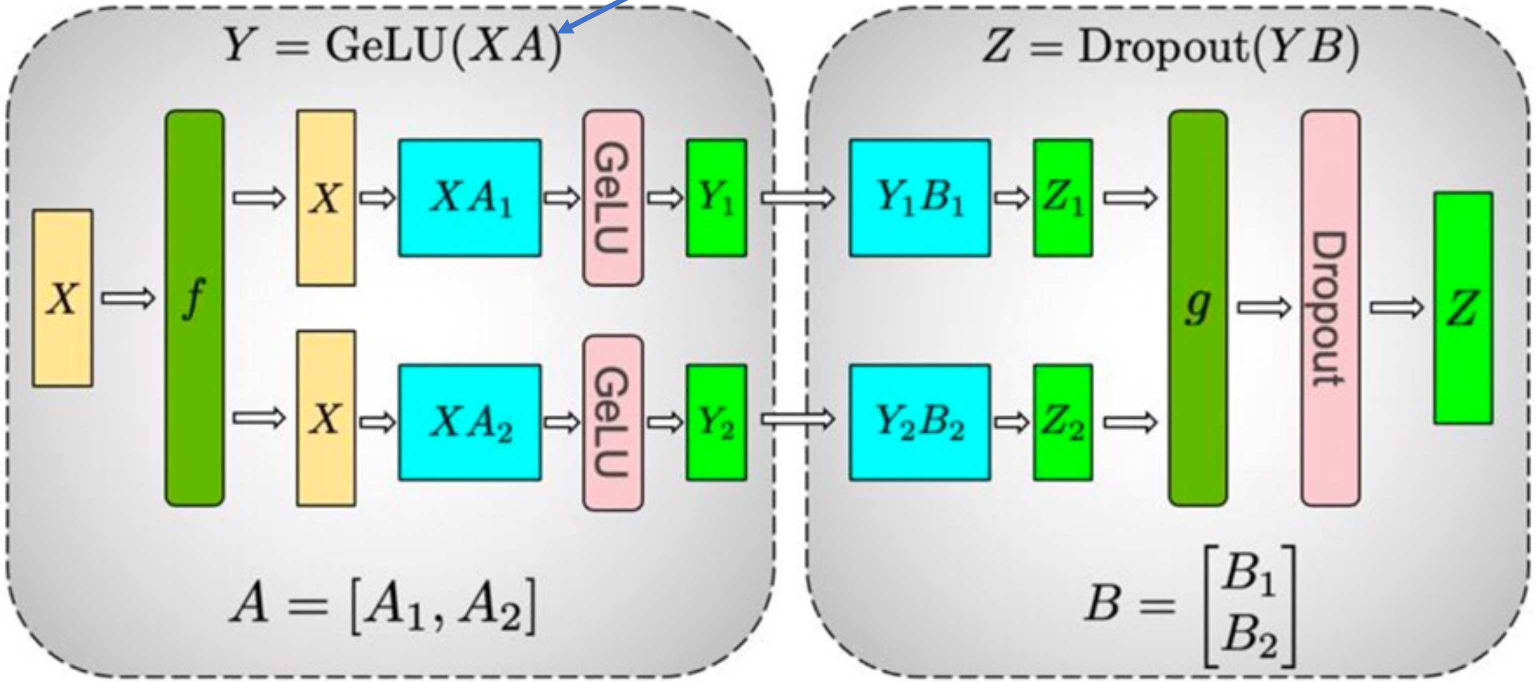


https://lightning.ai/docs/fabric/2.4.0/advanced/model_parallel/tp.html

Tensor Parallelism

- Transformer MLP

Nonlinear column-wise operator

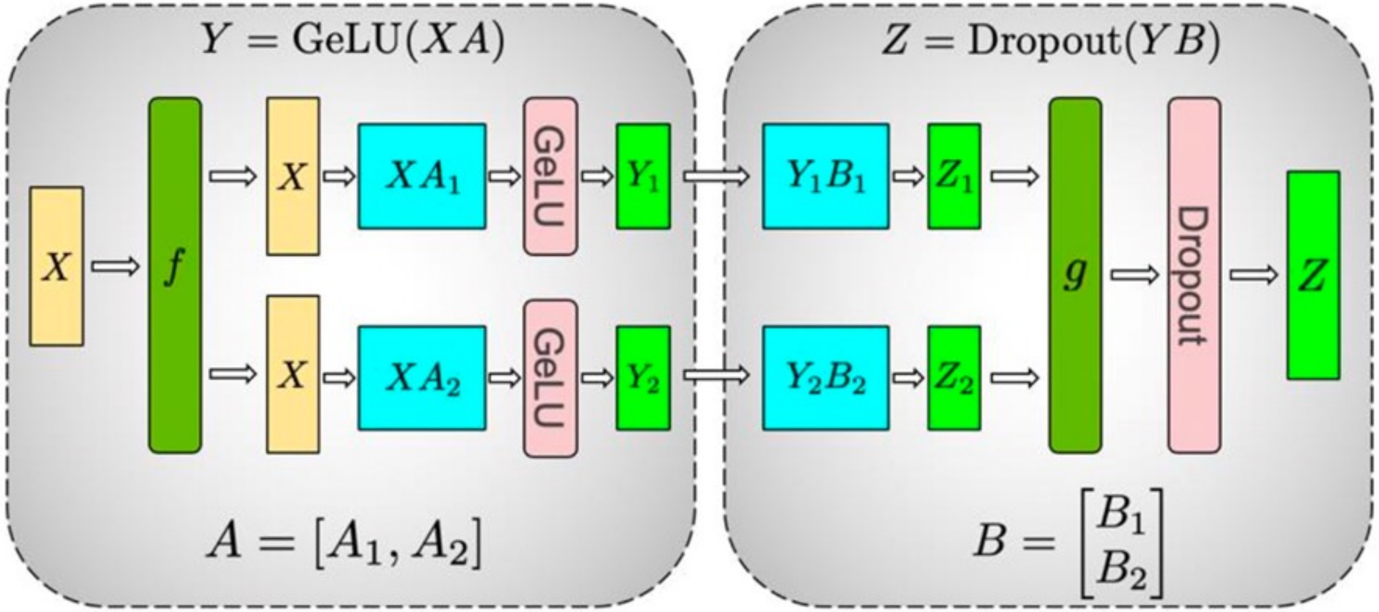


Column-wise sharding **Row-wise sharding**

f and ***g*** are **conjugate**, ***f*** is identity operator in the forward pass and all-reducein the backward pass while ***g*** is all-reduce in forward and identity in backward

Tensor Parallelism

- Transformer MLP

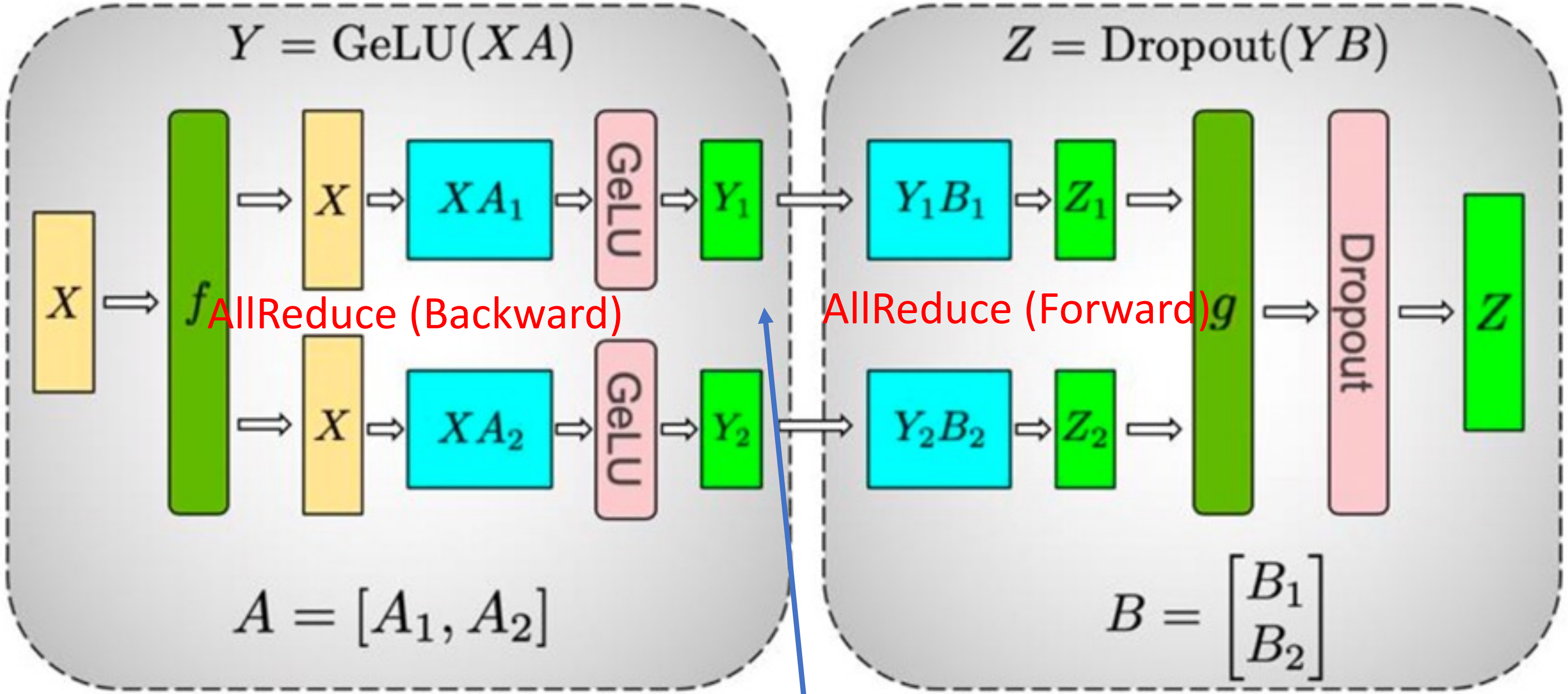


Column-wise sharding **Row-wise sharding**

- Partition A by columns and partition B by rows
- Forward pass
  - "Copy" X to both GPUs via ***f***
  - Execute math operations
  - "Add" Z_1 and Z_2 for a complete output Z via ***g***
- Backward pass
  - "copy" $\partial L/\partial Z$ to both GPUs via ***g***
  - Execute math operations
  - "Add" partial derivatives of X on two paths via ***f***

Tensor Parallelism

- Transformer MLP: *communication analysis*



Φ: *Model Size*

4Φ: 2 *AllReduce*

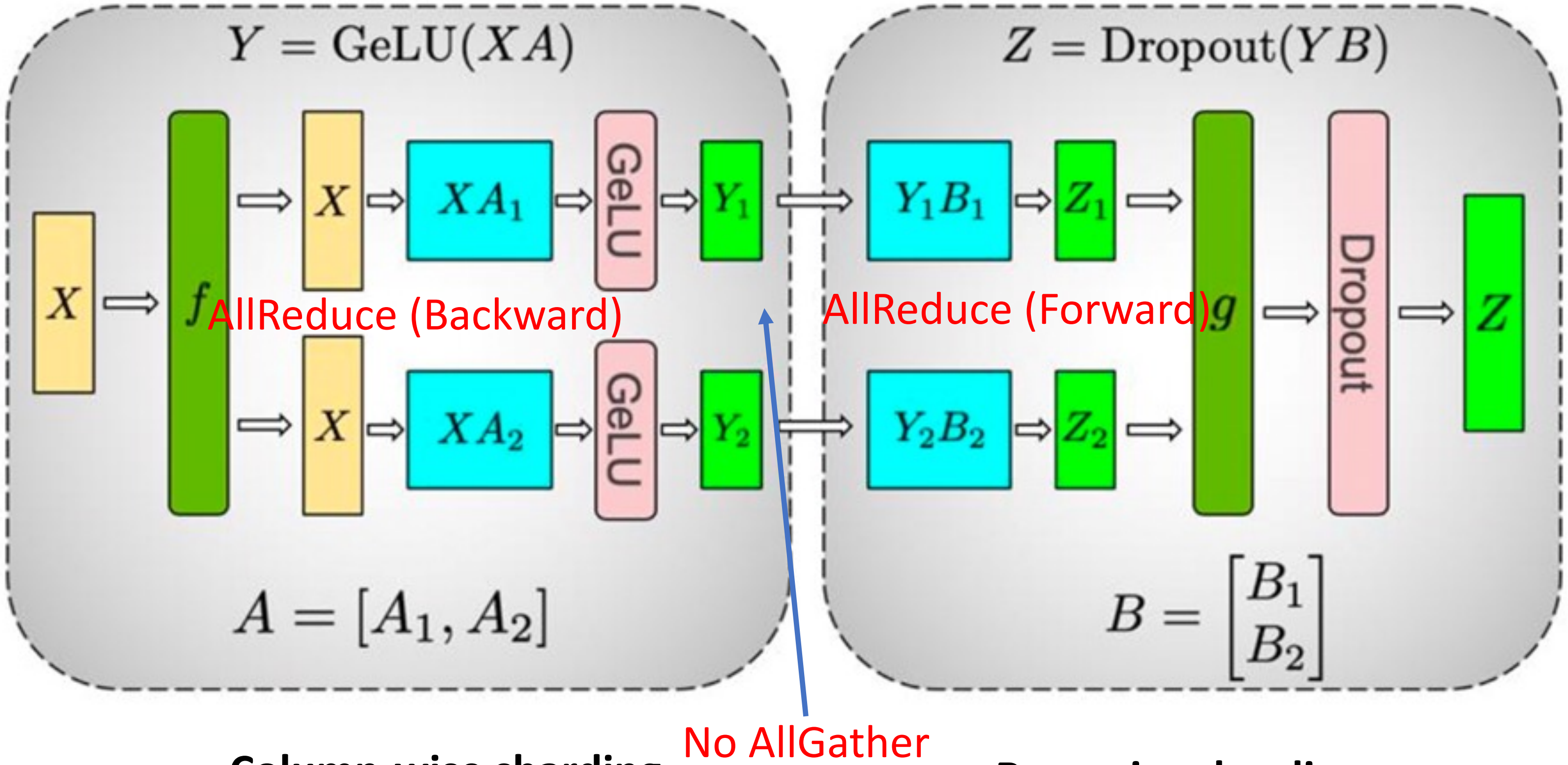
Column-wise sharding

Row-wise sharding

f and ***g*** are **conjugate**, ***f*** is identity operator in the forward pass and all-reducein the backward pass while ***g*** is all-reduce in forward and identity in backward

Tensor Parallelism

- Transformer MLP: *communication analysis*

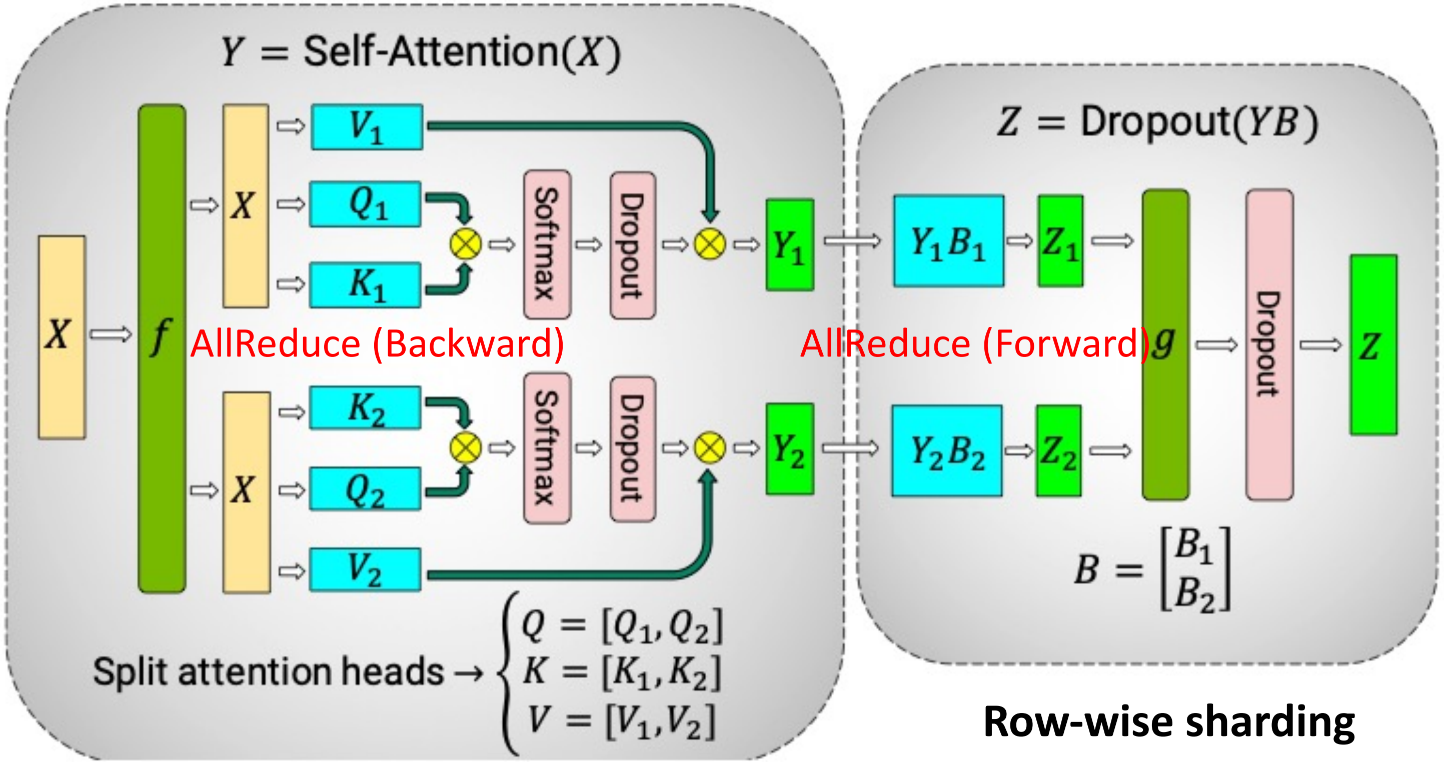


Column-wise sharding **Row-wise sharding**

f and ***g*** are **conjugate**, ***f*** is identity operator in the forward pass and all-reducein the backward pass while ***g*** is all-reduce in forward and identity in backward

Tensor Parallelism

- Transformer Self-Attention



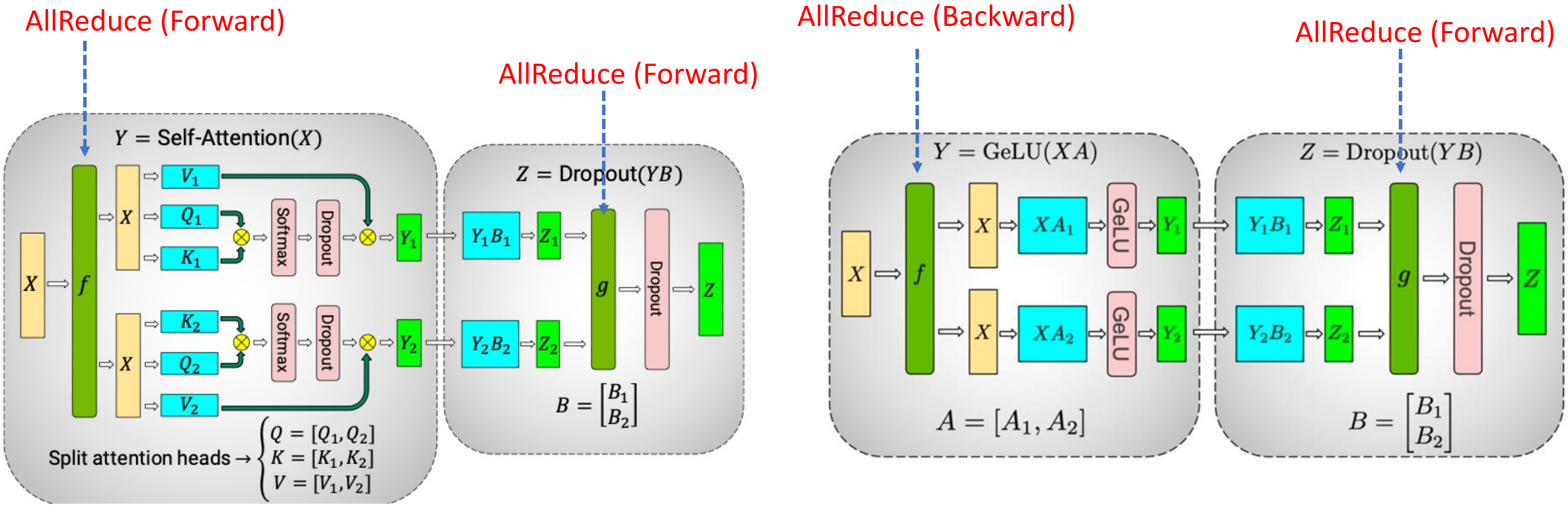
Column-wise sharding

Row-wise sharding

- Partition three parameter matrices W_Q, W_K and W_V by columns and partition the linear layer B by rows
- Forward pass
  - "Copy" X to both GPUs via ***f***
  - Execute math operations such as computing Q, K, V
  - "Add" Z_1 and Z_2 for a complete output Z via ***g***
- Backward pass
  - "copy" $\partial L/\partial Z$ to both GPUs via ***g***
  - Execute math operations
  - "Add" partial derivatives of X on two paths via ***f***

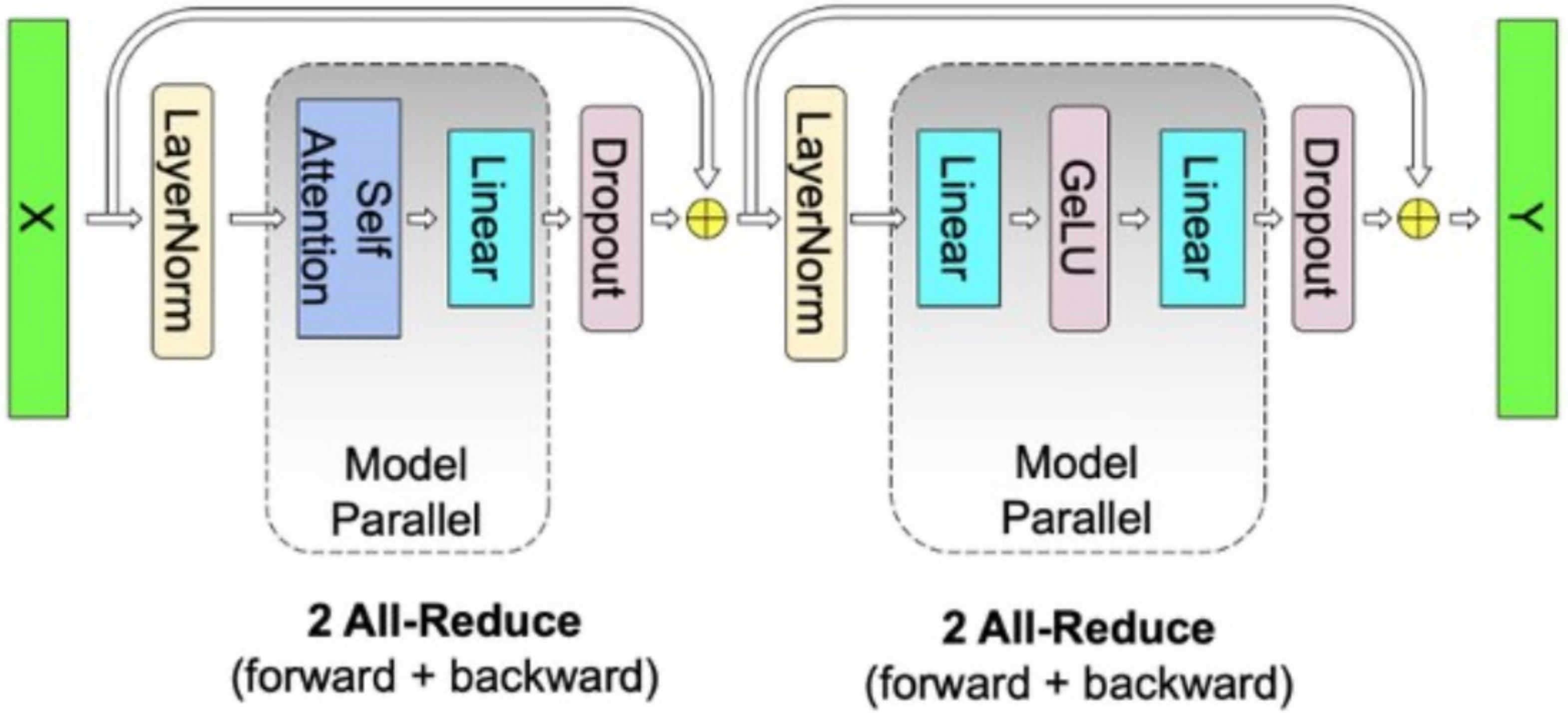
Tensor Parallelism

- TP communication analysis



Tensor Parallelism

- Transformer layer: communication analysis



Total communication complexity: 8 Φ. (Φ is the volume of the corresponding parameters)

《Accelerating Large Language Model Training with Hybrid GPU-based Compression》

Tensor Parallelism

- 2D Tensor Parallelism

$$\begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \cdot \begin{bmatrix} B_{00} & B_{01} \\ B_{10} & B_{11} \end{bmatrix}$$

4 GPUs

Parallel computation

Step 1

$$\begin{bmatrix} A_{00}, B_{00} & A_{00}, B_{01} \\ A_{10}, B_{00} & A_{10}, B_{01} \end{bmatrix} \Rightarrow \begin{bmatrix} A_{00} \cdot B_{00} & A_{00} \cdot B_{01} \\ A_{10} \cdot B_{00} & A_{10} \cdot B_{01} \end{bmatrix}$$

Broadcast #1 column of A to the right, and broadcast #1 row of B down, and multiply two submatrices

Tensor Parallelism

- 2D Tensor Parallelism

$$\begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \cdot \begin{bmatrix} B_{00} & B_{01} \\ B_{10} & B_{11} \end{bmatrix}$$

4 GPUs

Step 2

$$\Rightarrow \begin{bmatrix} A_{01}, B_{10} & A_{01}, B_{11} \\ A_{11}, B_{10} & A_{11}, B_{11} \end{bmatrix} \Rightarrow \begin{bmatrix} A_{01} \cdot B_{10} & A_{01} \cdot B_{11} \\ A_{11} \cdot B_{10} & A_{11} \cdot B_{11} \end{bmatrix}$$

Broadcast #2 column to the left, and broadcast #2 row up, and multiply two matrices

Tensor Parallelism

- 2D Tensor Parallelism

Sequential computation

$$\begin{bmatrix} A_{00} \cdot B_{00} & A_{00} \cdot B_{01} \\ A_{10} \cdot B_{00} & A_{10} \cdot B_{01} \end{bmatrix} + \begin{bmatrix} A_{01} \cdot B_{10} & A_{01} \cdot B_{11} \\ A_{11} \cdot B_{10} & A_{11} \cdot B_{11} \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} A_{00} \cdot B_{00} + A_{01} \cdot B_{10} & A_{00} \cdot B_{01} + A_{01} \cdot B_{11} \\ A_{10} \cdot B_{00} + A_{11} \cdot B_{10} & A_{10} \cdot B_{01} + A_{11} \cdot B_{11} \end{bmatrix}$$

Tensor Parallelism

- 2D Tensor Parallelism: general procedure on $Y = AB$

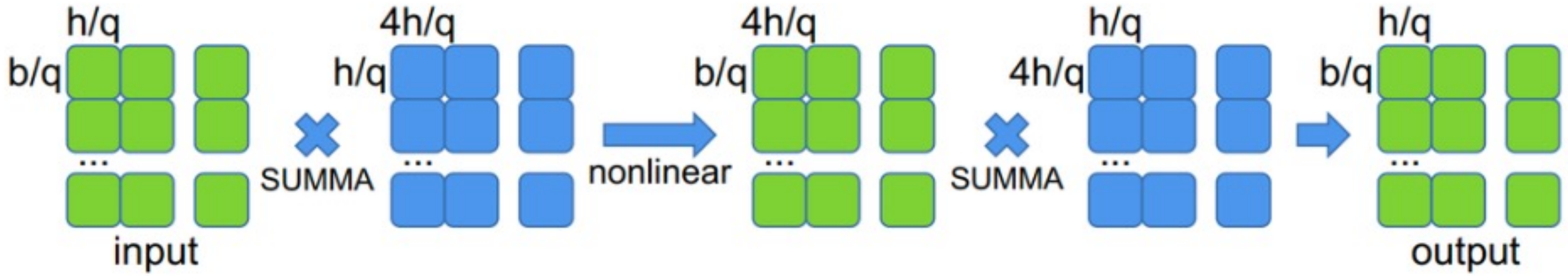
Trading communication for memory:

Less computation, less memory usage, more communication

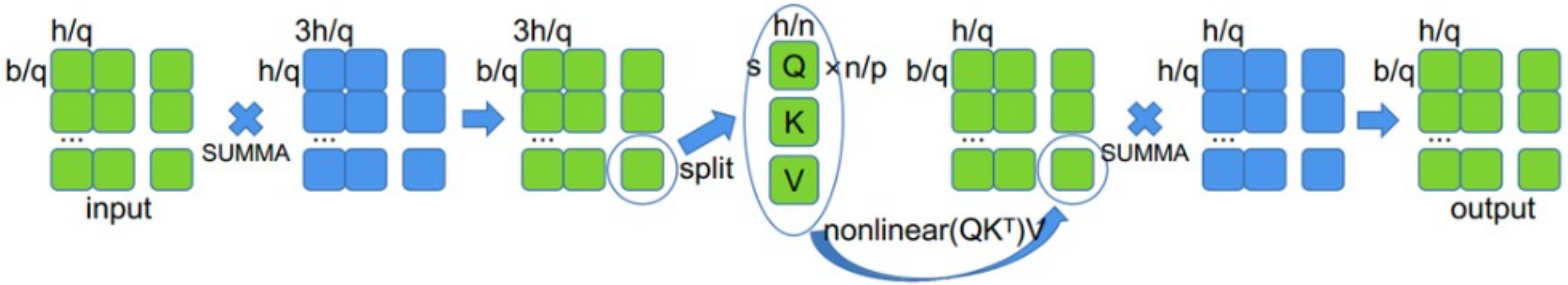
- Computation, memory and communication complexity
  - Computation: $O\left(\frac{1}{P}\right)$
  - Memory: $O\left(\frac{1}{P}\right)$
  - Communication: $O\left(\frac{6(q-1)}{q}\right)$ (a constant factor of model size, but not so sure about this value)

Tensor Parallelism

- 2D Tensor Parallelism



(a) MLP



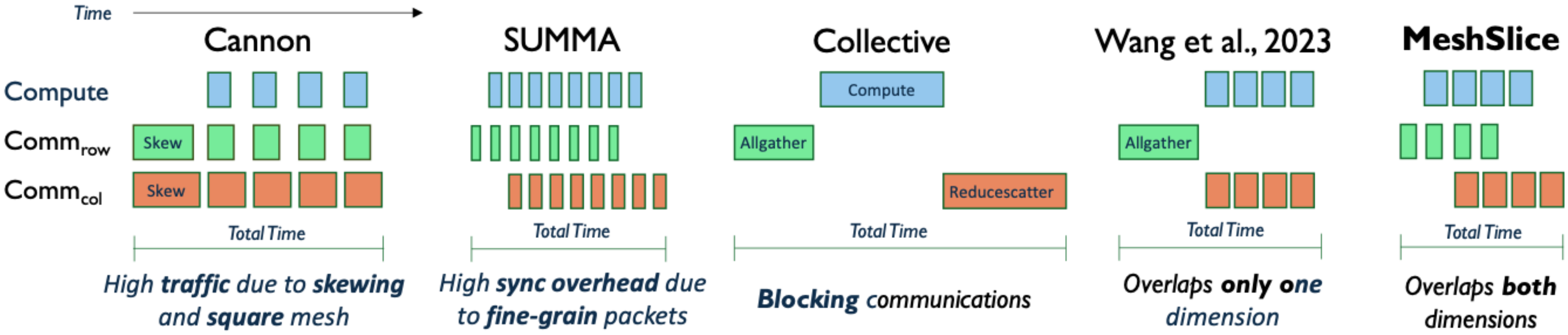
(b) self-attention

Green: activations, Blue: Transformer modules

《SUMMA: Scalable universal matrix multiplication algorithm》

Tensor Parallelism

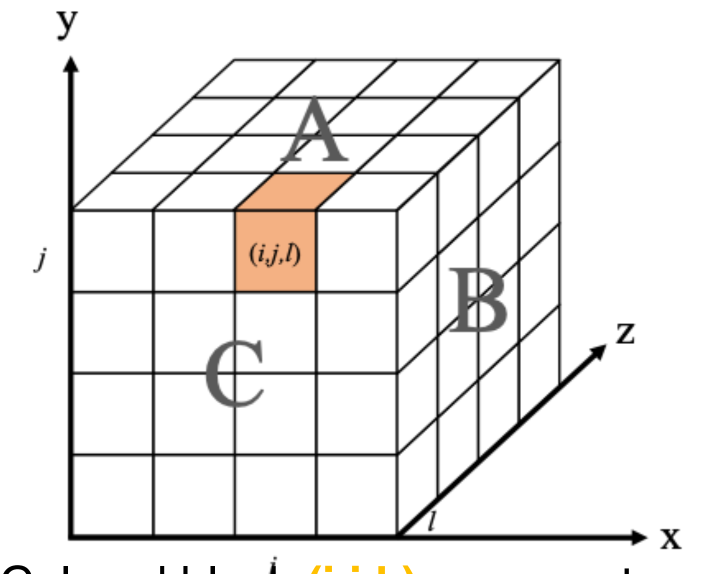
- 2D Tensor Parallelism: still many ongoing works



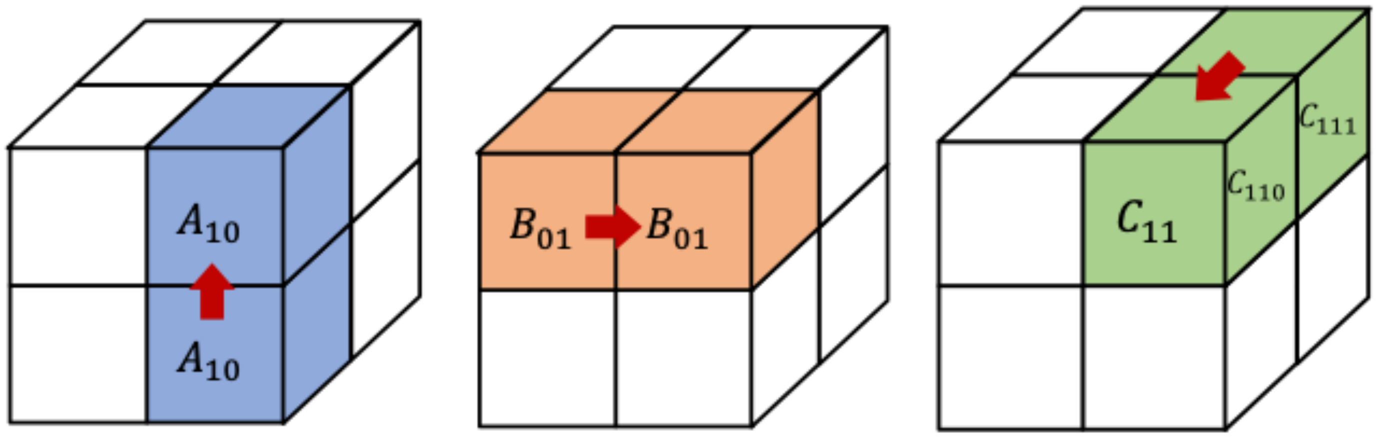
《MeshSlice: Efficient 2D Tensor Parallelism for Distributed DNN Training》

Tensor Parallelism

- 3D Tensor Parallelism: further segmentation for less memory usage



Colored block **(i,j,k)** represents an example of a single processor.



(a) Broadcast A_{il} (b) Broadcast B_{lj} (c) Reduce C_{ij}

An example of the 3-D parallel matrix multiplication

《Maximizing Parallelism in Distributed Training for Huge Neural Networks》

Distributed LLM Training: Outline

- Data Parallelism
  - Parameter-Server
  - All-Reduce
  - Memory Optimization
- Model Parallelism
  - Pipeline Parallelism
  - Tensor Parallelism
  - **Sequence Parallelism**
- Mixture of Experts

Sequence Parallelism

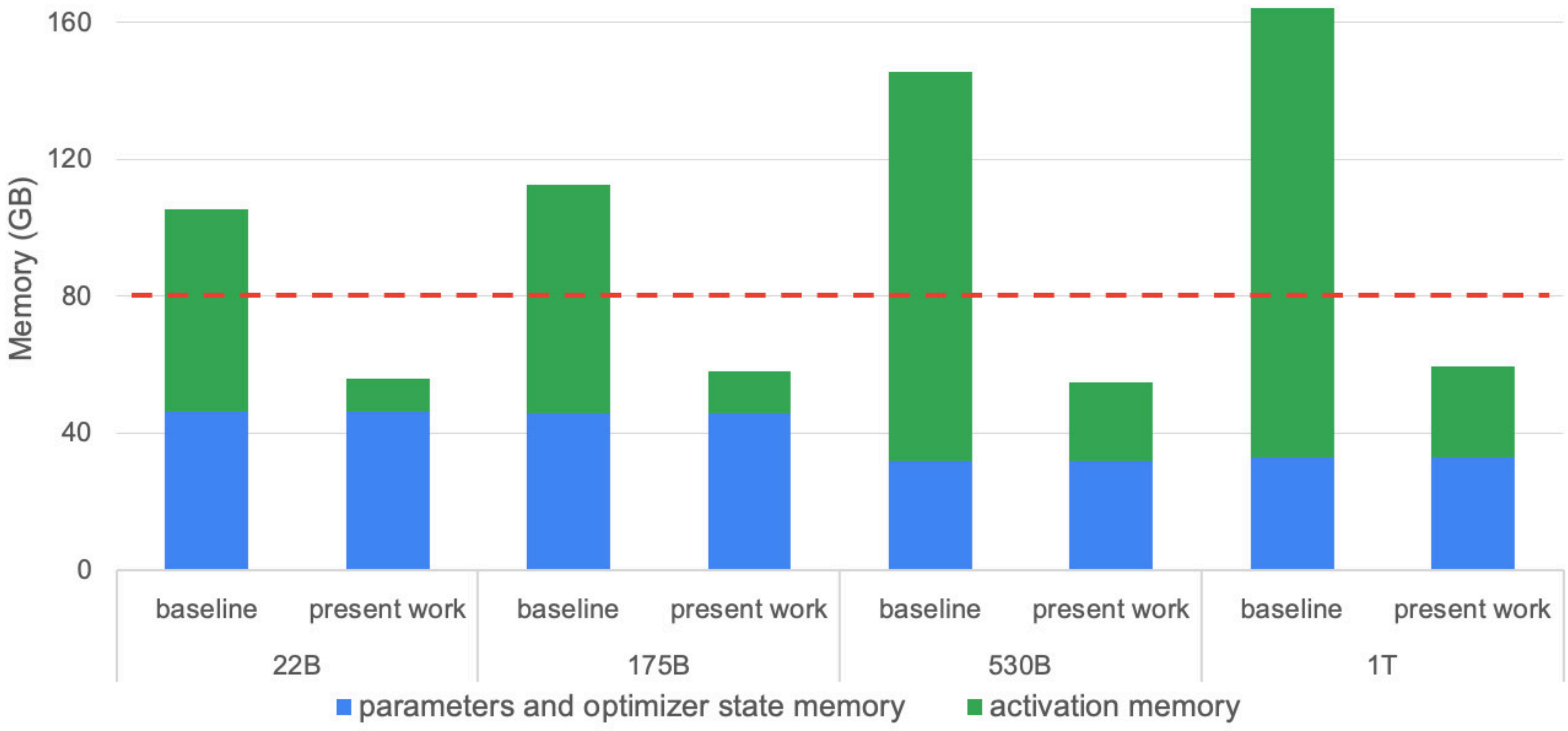
- Sequence Parallelism
  - Tensor parallelism reduces the memory footprint of **model data**, how about **non-model data**?
  - Activation memory is $O(n^2)$ in *self-attention* with *n* being the sequence length
    - Activations include ***Q***, ***K***, ***V*** matrices
    - Not possible of partitioning ***Q***, ***K***, ***V*** in *self-attention*

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

  - *Some layers have not been parallelized: layernorm and dropout*

Sequence Parallelism

- Sequence Parallelism



Parameters, optimizer state, and activations memory

《Reducing Activation Recomputation in Large Transformer Models》

Sequence Parallelism (Recap)

- Activation is non-negligible
  - Forward propagation
  - Backward propagation
  - Size of activation
    - Standard transformer: b - batch size, s - sequence length, h - hidden layer dimension

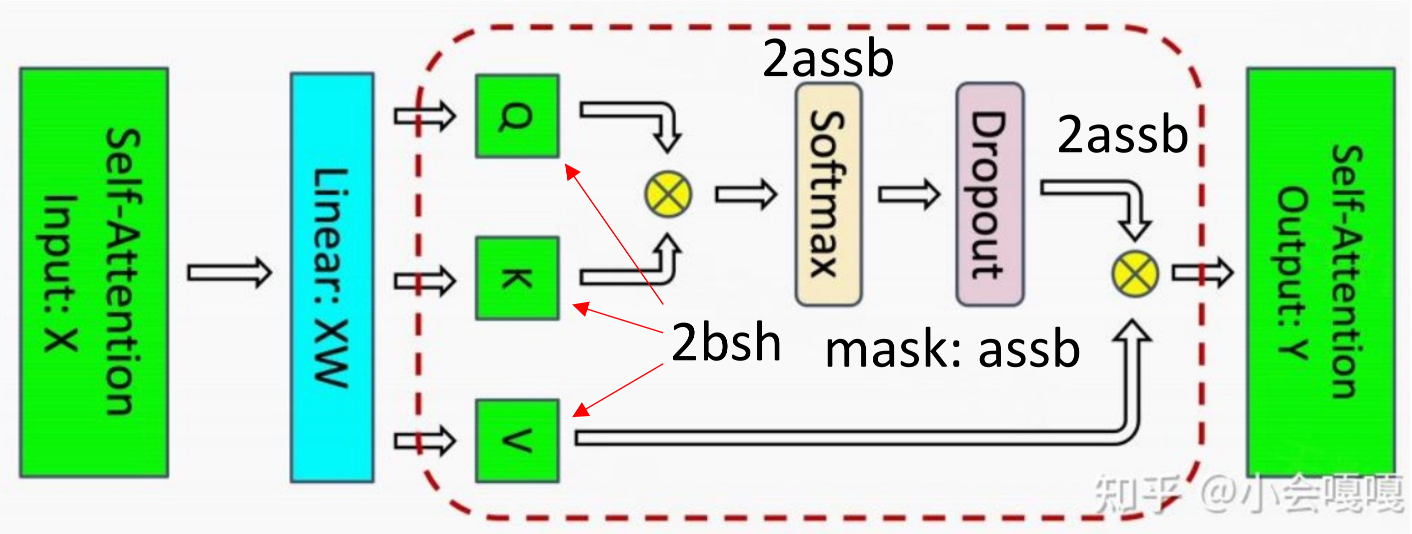
Output of a previous layer, intermediate variable

$$y = Wx + b$$

$$\frac{dL}{dW} = \frac{dL}{dy} x$$

Cannot be discarded after FP

bsh * 2 Bytes



In-total: 5abss + 8sbh

Sequence Parallelism (Recap)

- Activation is non-negligible
  - Forward propagation
  - Backward propagation
  - Size of activation
    - Standard transformer: b - batch size, s - sequence length, h - hidden layer dimension

Output of a previous layer, intermediate variable

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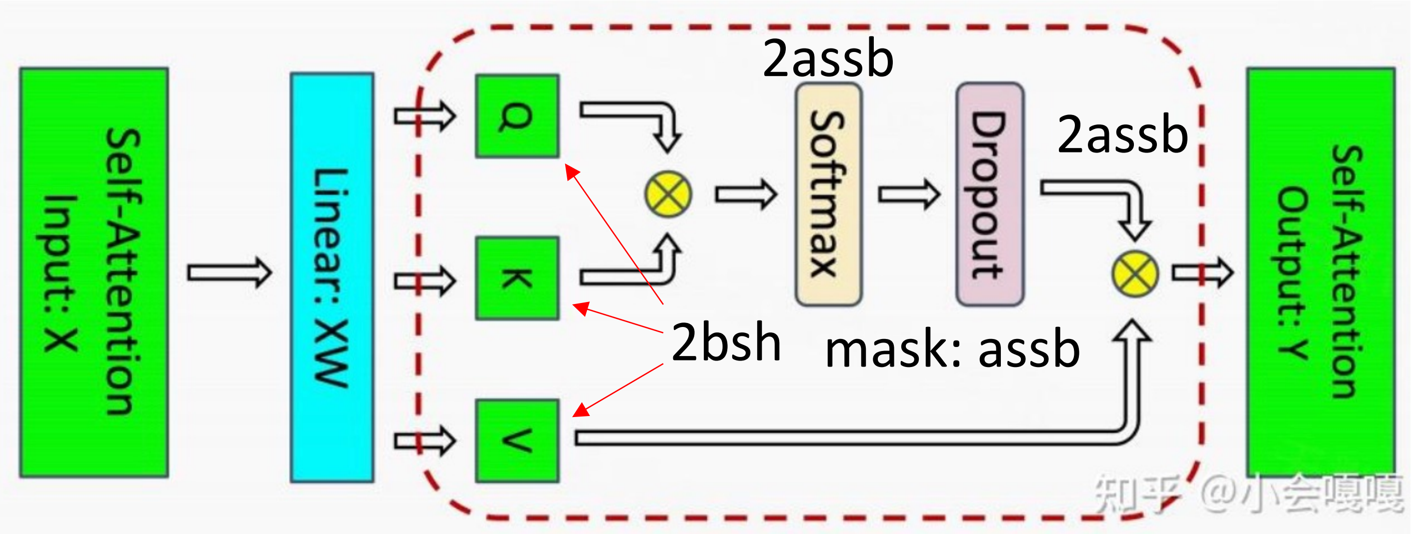
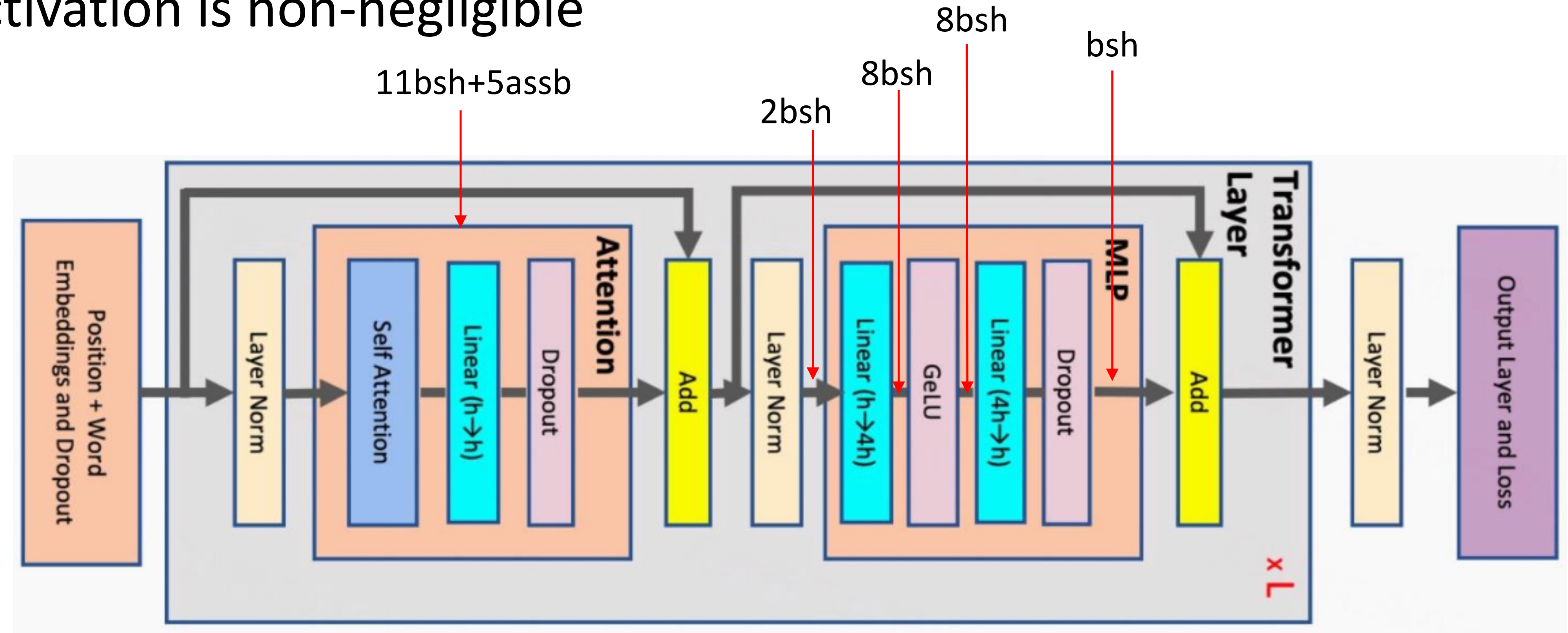


3sbh: Linear layer + dropout layer

In-total: 5abss + 11sbh

Sequence Parallelism (Recap)

- Activation is non-negligible



In-total: 5abss + 34sbh

Sequence Parallelism

- What to keep for a Transformer block in HBM?

  - MLP
    - Input to MLP ✓
    - Input of Dropout ✓
    - Input of GeLU
    - Inputs of linear matrices ✓

  - Attention
    - Input to Attention ✓
    - Q, K and V ✓
    - QK^T
    - Softmax(QK^T) ✓
    - Dropout Mask ✓

Sequence Parallelism

- An exercise: manual gradient computation

  - MLP
    - Input to MLP ✓
    - **Input of Dropout** ✓
    - **Input of GeLU**
    - Inputs of linear matrices ✓

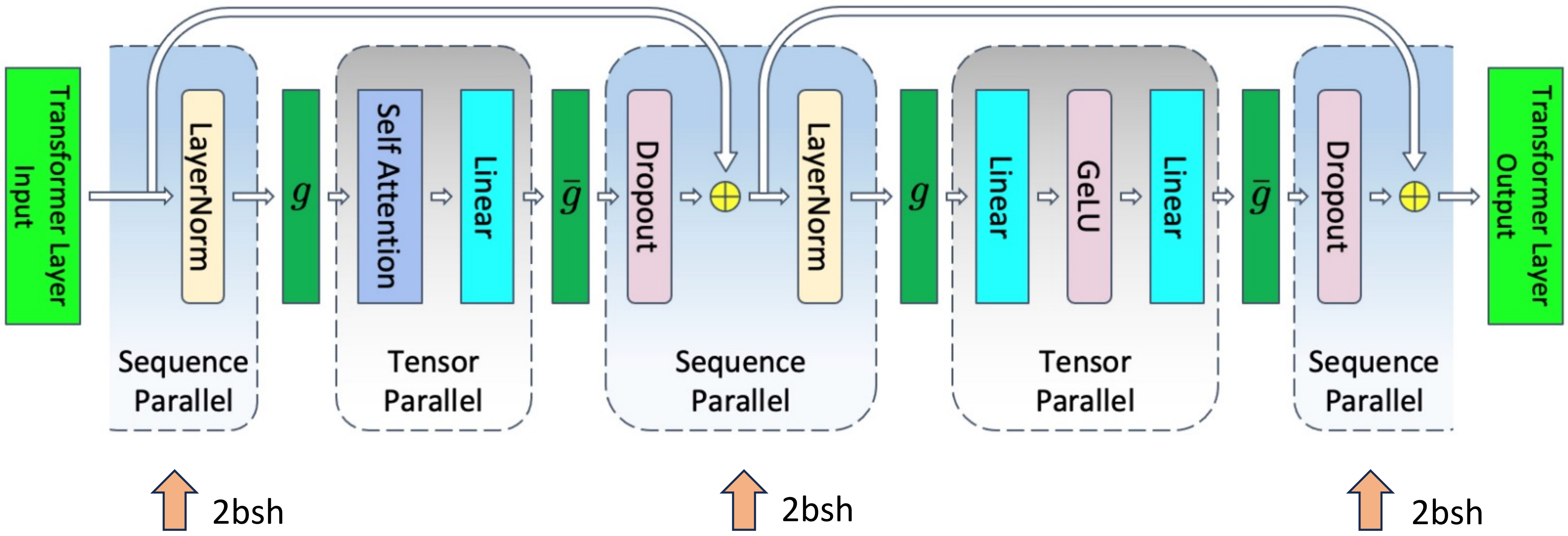
  - Attention
    - Input to Attention ✓
    - Q, K and V ✓
    - **QK^T**
    - **Softmax(QK^T)** ✓
    - Dropout Mask ✓

Sequence Parallelism

- Representative SP frameworks
  - Megatron-LM S
    - 《Reducing Activation Recomputation in Large Transformer Models》
  - DeepSpeed Ulysses
    - 《DeepSpeed Ulysses: System Optimizations for Enabling Training of Extreme Long Sequence Transformer Models》
  - Colossol AI SP
    - 《Colossal-AI: A Unified Deep Learning System For Large-Scale Parallel Training》
  - Megatron Context Parallelism
    - https://docs.nvidia.com/megatron-core/developer-guide/latest/api-guide/context_parallel.html

Sequence Parallelism

- Megatron-LM sequence parallelism



Tensor parallelism only segments inputs and outputs of ***self-attention*** and ***MLP***

Sequence Parallelism

- Megatron-LM sequence parallelism
  - *layernorm* operation
    - The shape of an input matrix: [batch_size, seq_len, hidden_dim]
    - Compute mean, variance and a learnable affine transformation for each ***TOKEN***
    - Can be naturally partitioned by tokens without any cross-device communication
  - *Dropout* operation
    - Each GPU compute dropout independently

Activations (input/output of layernorm) on each GPU: $\frac{2bsh}{t}$

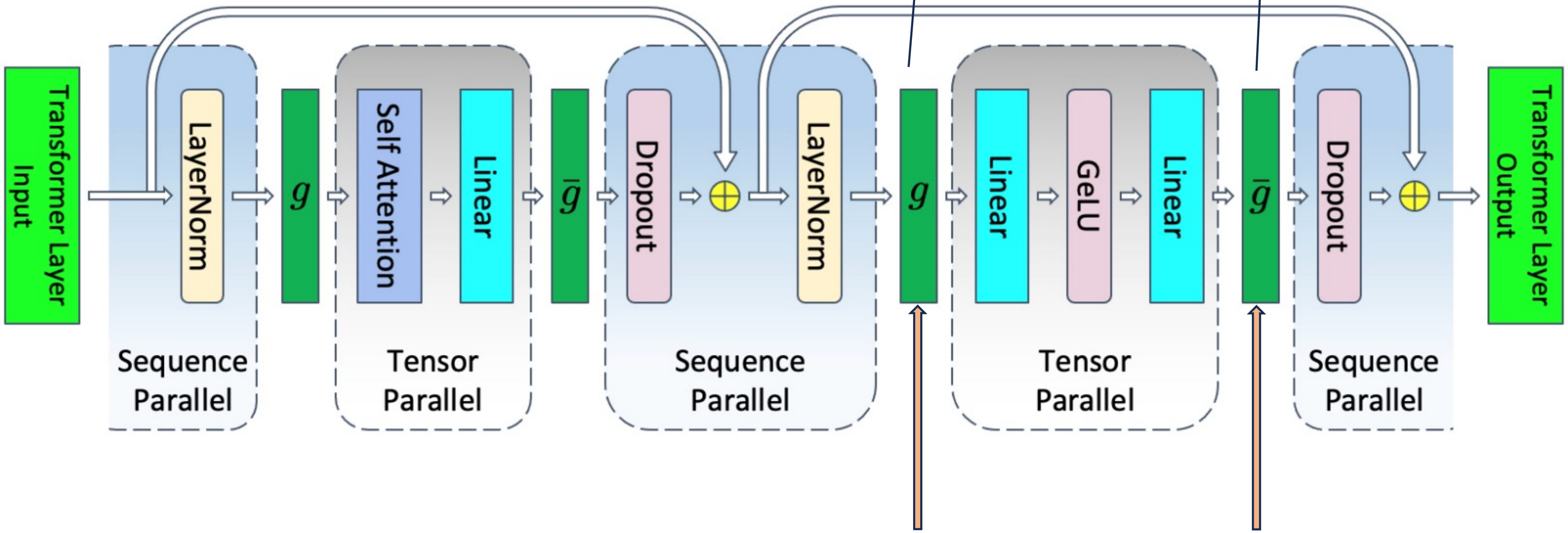
Activations (dropout) on each GPU: $\frac{bsh}{t}$

Sequence Parallelism

Matrix multiplication needs the complete input

The currently buffered output is not what you need to proceed to Dropout

- Megatron-LM sequence parallelism
  - MLP layer (forward pass)



AllGather to obtain the complete input

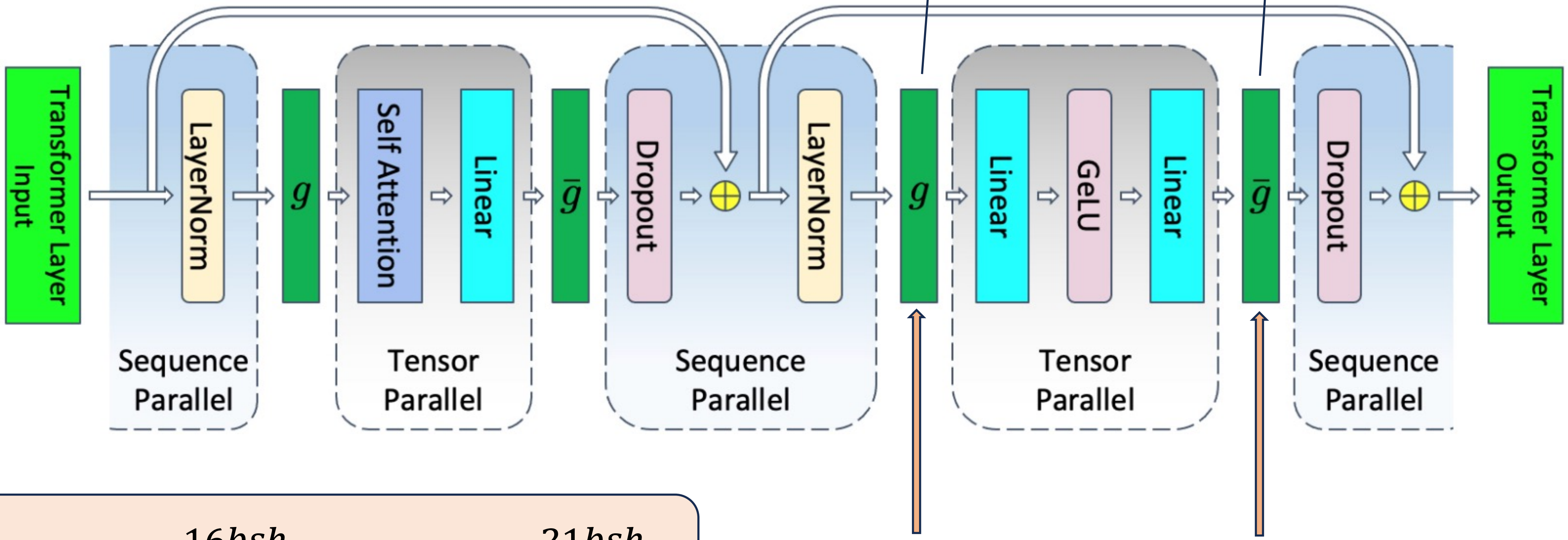
ReduceScatter to obtain the output fraction

Sequence Parallelism

- Megatron-LM sequence parallelism
  - MLP layer (backward pass)

The currently buffered output is not what you need to proceed to Dropout

Matrix multiplication needs the complete input



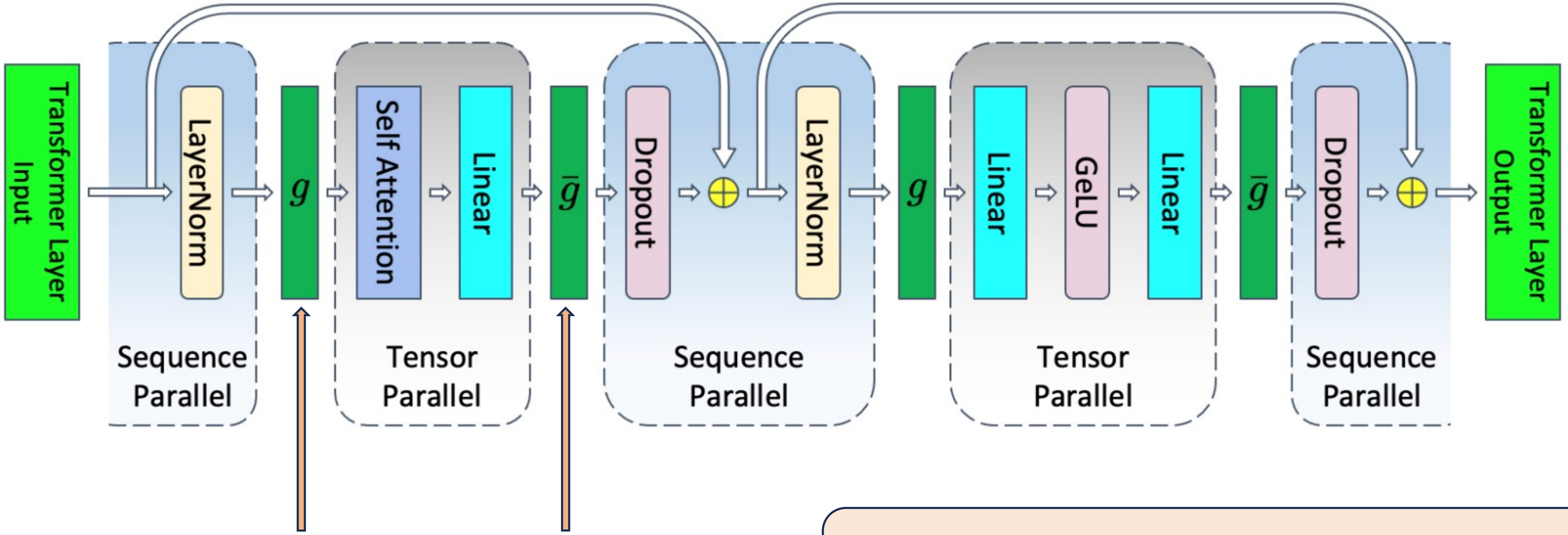
ReduceScatter to obtain the output fraction

AllGather to obtain the complete input

Finally: $5bsh + \frac{16bsh}{t}$ becomes $\frac{21bsh}{t}$

Sequence Parallelism

- Megatron-LM sequence parallelism
  - Attention (forward pass, backward pass omitted)



AllGather to obtain the complete input

ReduceScatter to obtain the output fraction

Finally: $5bsh + \frac{5bass+8bsh}{t}$ becomes $\frac{5bass+13bsh}{t}$

Sequence Parallelism

- Megatron-LM sequence parallelism
  - Slicing input/output of layernorm, and input of dropout
  - Changing AllReduce into ReduceScatter and AllGather
    - Almost **ZERO** extra communication overhead
  - Changing their memory consumption from $10bsh$ to $\frac{10bsh}{t}$
  - An example: batch_size = 64, hidden_size = 4096, seq_len = 32768, sp = 8
    - Memory consumption per-block changes from 86 GB to 10.7 GB

Sequence Parallelism

- Megatron-LM sequence parallelism
  - Further optimization: selective recomputation

Low-hanging Fruits: Recomputing large size but less computation intensive activations, e.g. softmatrix, dot production, dropout, while keeping computation heavy intensive activations, e.g. layernorm, Transformer input/output.

Sequence Parallelism

- Megatron-LM sequence parallelism
  - Setup

Model Size	Attention Heads	Hidden Size	Layers	Tensor Parallel Size	Pipeline Parallel Size	Number of GPUs	Global Batch Size	Micro Batch Size
22B	64	6144	48	8	1	8	4	4
175B (GPT-3)	96	12288	96	8	8	64	64	1
530B (MT-NLG)	128	20480	105	8	35	280	280	1
1T	160	25600	128	8	64	512	512	1

Sequence Parallelism

- Megatron-LM sequence parallelism
  - Setup

Model Size	Attention Heads	Hidden Size	Layers	Tensor Parallel Size	Pipeline Parallel Size	Number of GPUs	Global Batch Size	Micro Batch Size
22B	64	6144	48	8	1	8	4	4
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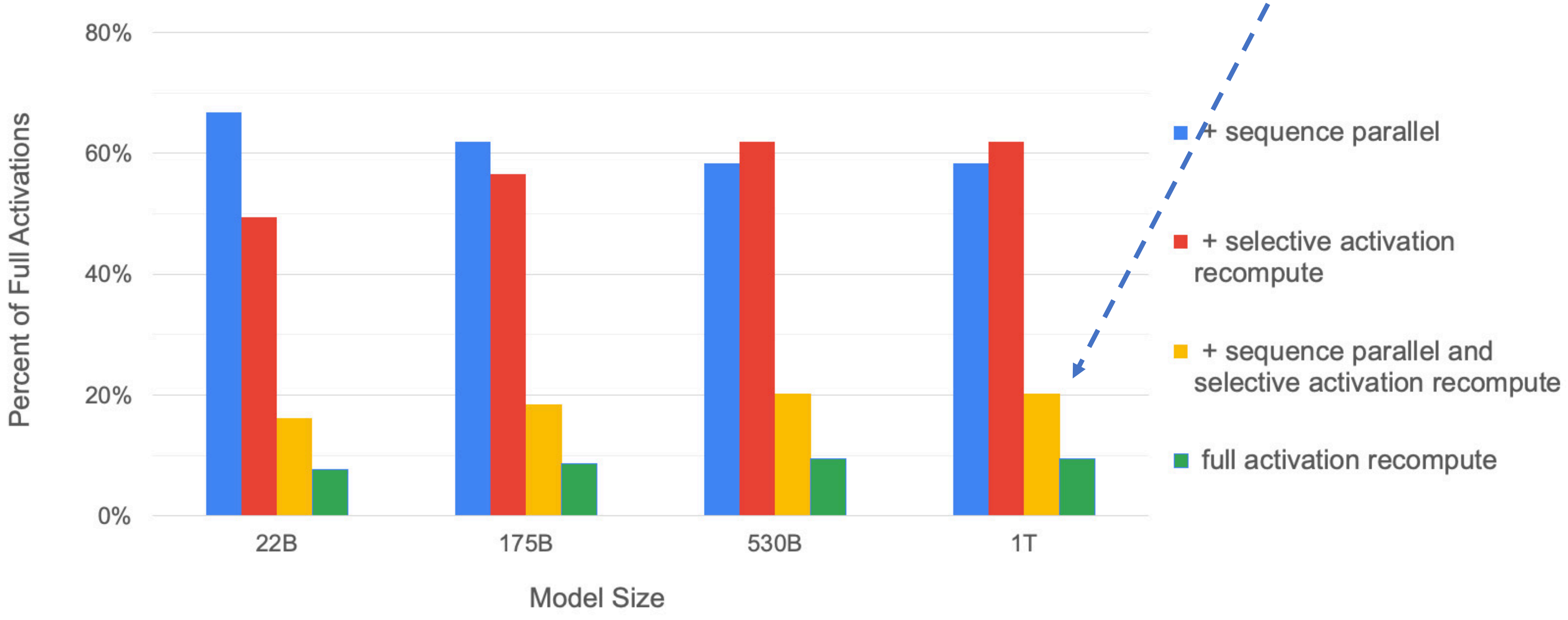
530B (MT-NLG)	128	20480	105	8	35	280	280	1
1T	160	25600	128	8	64	512	512	1

Sequence Parallelism

- Megatron-LM sequence parallelism
  - Experiments

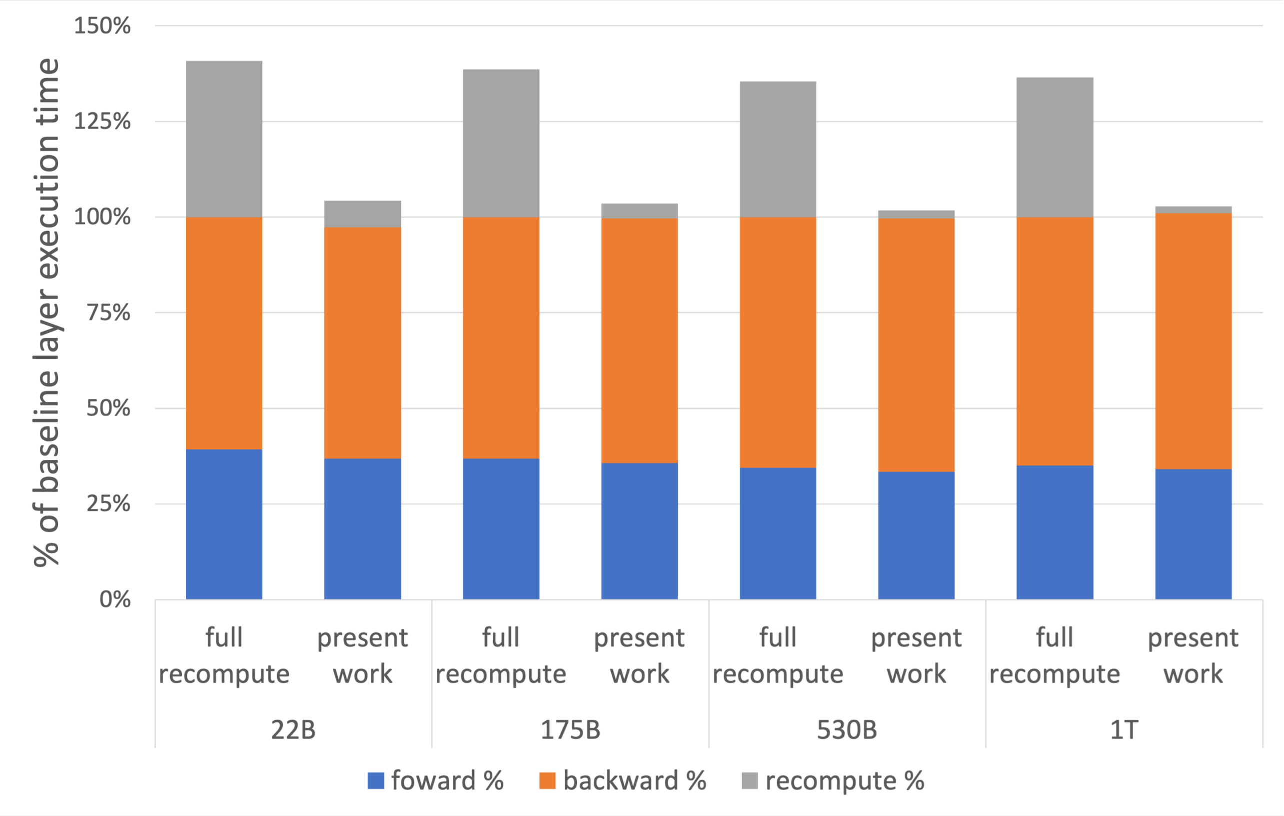
Five times smaller!



Percentage of required memory compared to the tensor-level parallel baseline

Sequence Parallelism

- Megatron-LM sequence parallelism
  - Experiments



Per layer breakdown of forward, backward, and recompute times

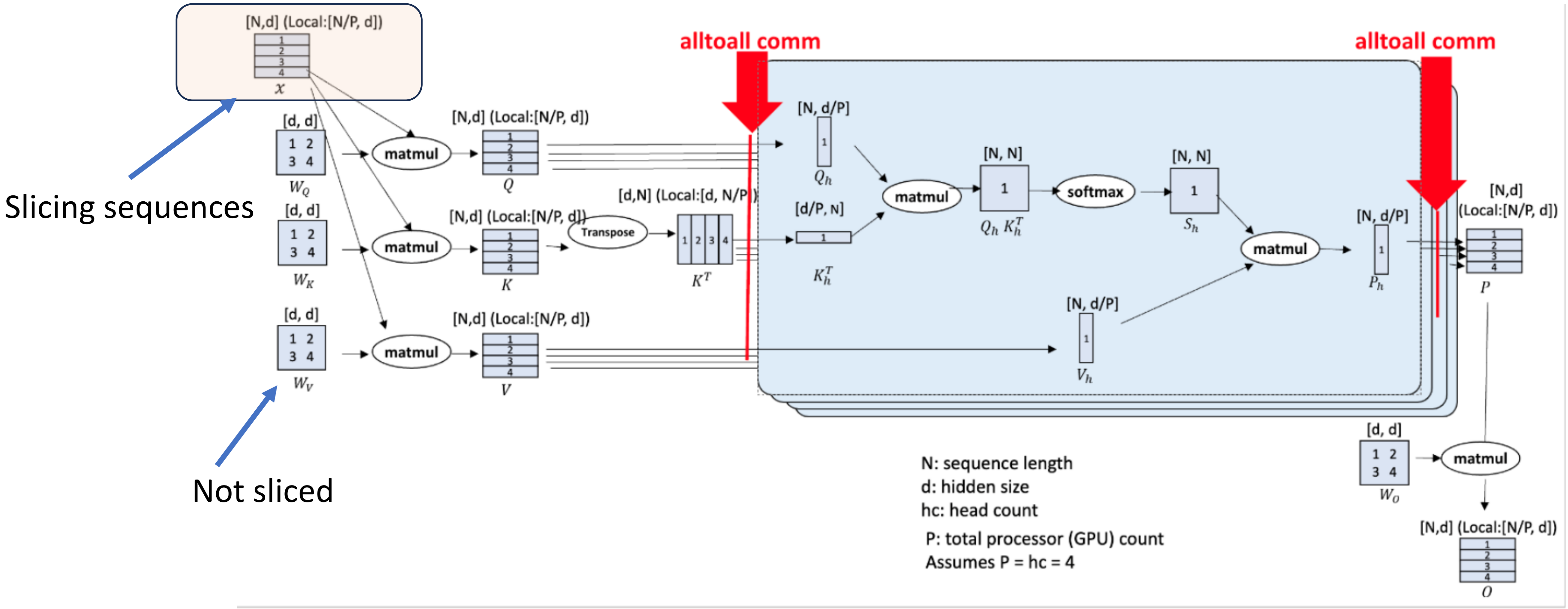
Sequence Parallelism

- DeepSpeed Sequence Parallelism
  - Megatron SP in fact slice activations at the token (sequence) dimension

How about slicing sequences in the very beginning?

Sequence Parallelism

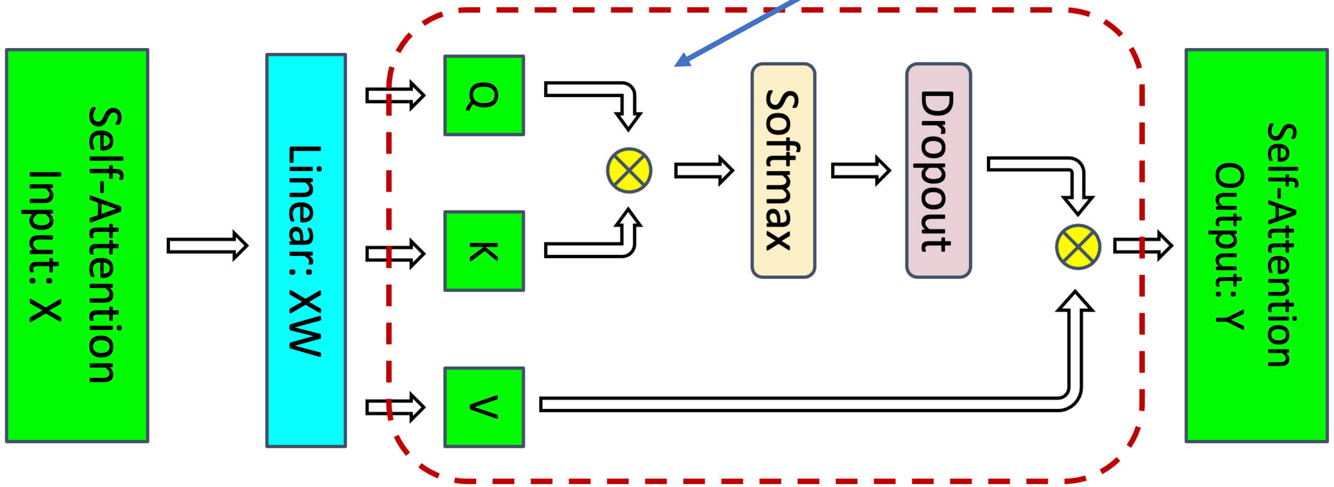
- DeepSpeed Ulysses Sequence Parallelism



Sequence Parallelism

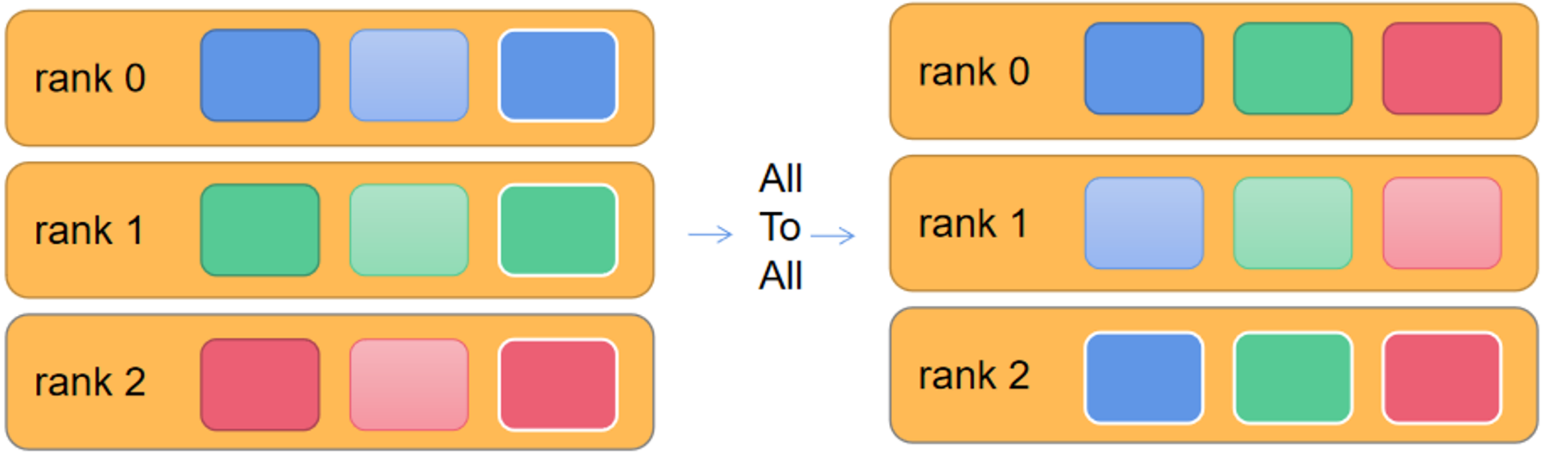
- DeepSpeed Ulysses Sequence Parallelism
  - Slicing input *X=(N, d)*, with *N* for sequence length and *d* for hidden dimension
    - Each GPU has an input ***(N/P, d)***, with *d* for # of GPUs
  - Attention weight matrices W_Q, W_K and $W_V \subseteq \mathbb{R}^{\boldsymbol{d\times d}}$ not partitioned
    - Sizes of Q, K, V **chunks** are $(\boldsymbol{N/P}, \boldsymbol{d})$

Cannot proceed directly, WHY?



Sequence Parallelism

- DeepSpeed Ulysses Sequence Parallelism
  - All-to-All communication
    - Before all-to-all, each GPU holds the SEQUENCE CHUNK of **ALL HEADs** (i.e. $(N/P, d)$)
    - After all-to-all, each GPU holds ALL SEQUENCES on a specific HEAD (i.e. $(N, d/P)$)



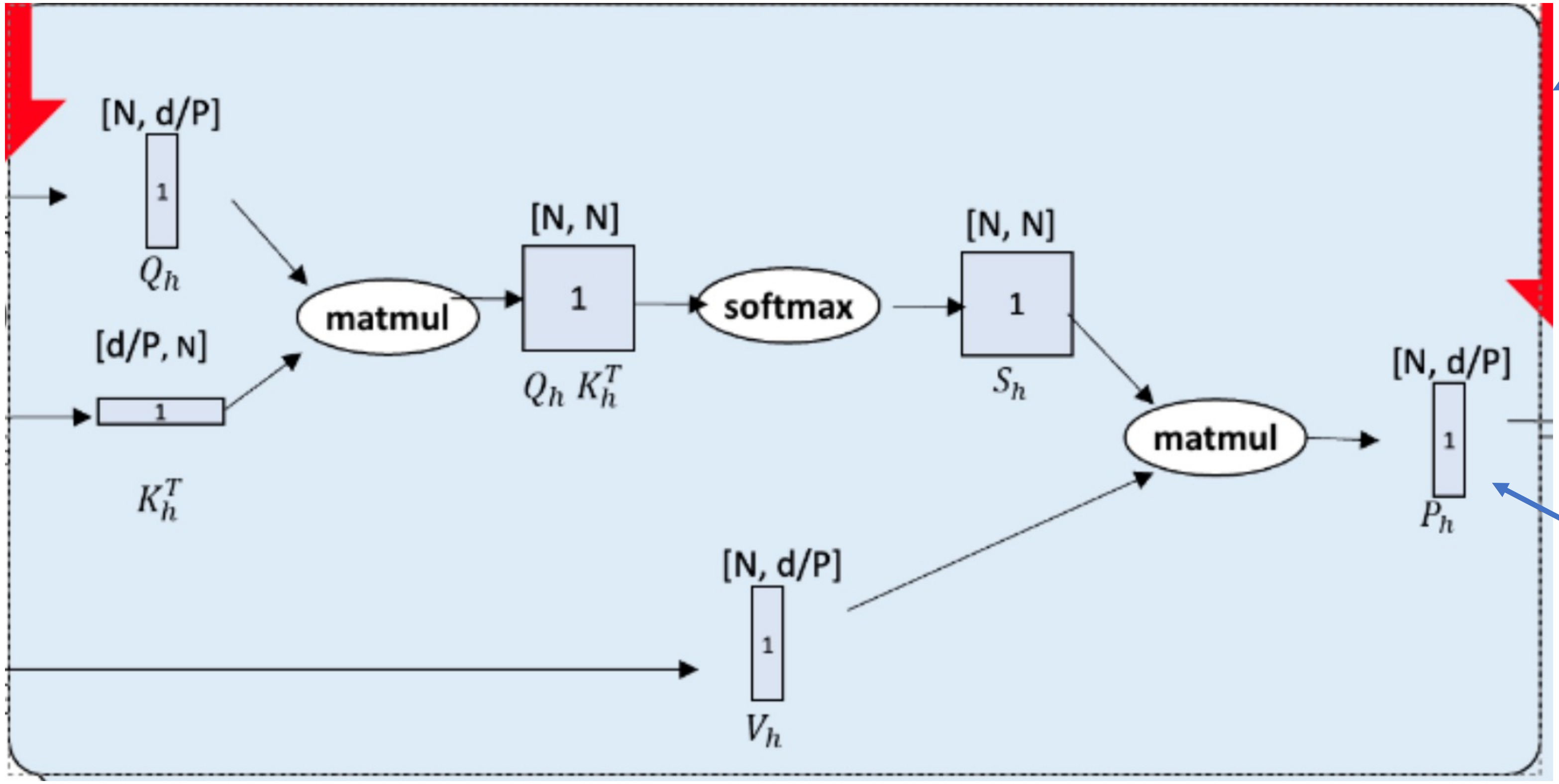
Looks like a transpose operation!

https://mdnice.com/writing/974038f2cc3d46239377b0928e8b9eea

Sequence Parallelism

- DeepSpeed Ulysses Sequence Parallelism
  - All-to-All communication

Multi-head Attention Computation



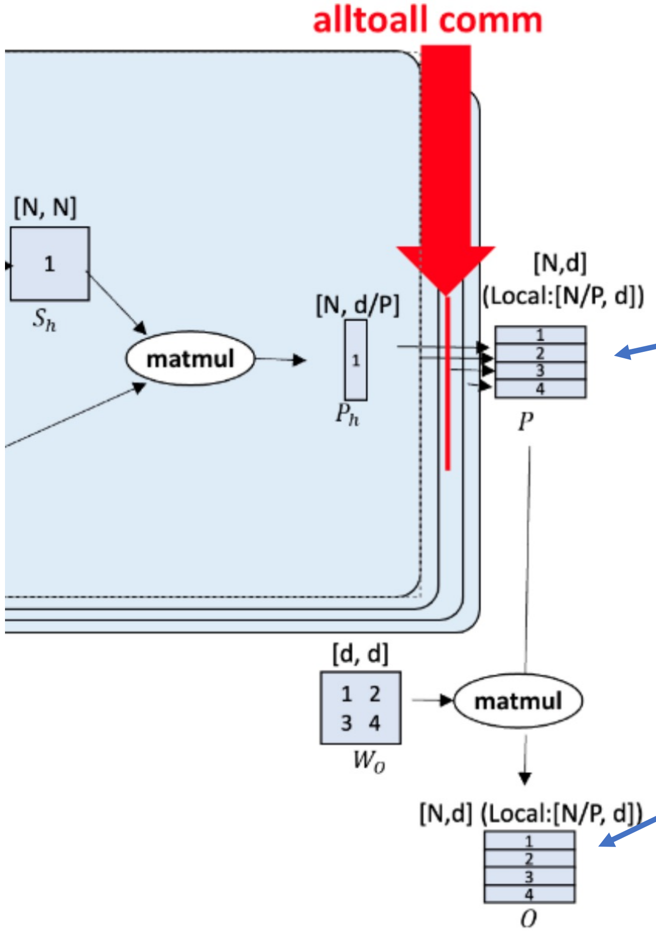
Output dimension: $(N, d/P)$

Sequence Parallelism

- DeepSpeed Ulysses Sequence Parallelism

- MLP can compute at each sequence chunk granularity

- DeepSpeed Ulysses backward pass



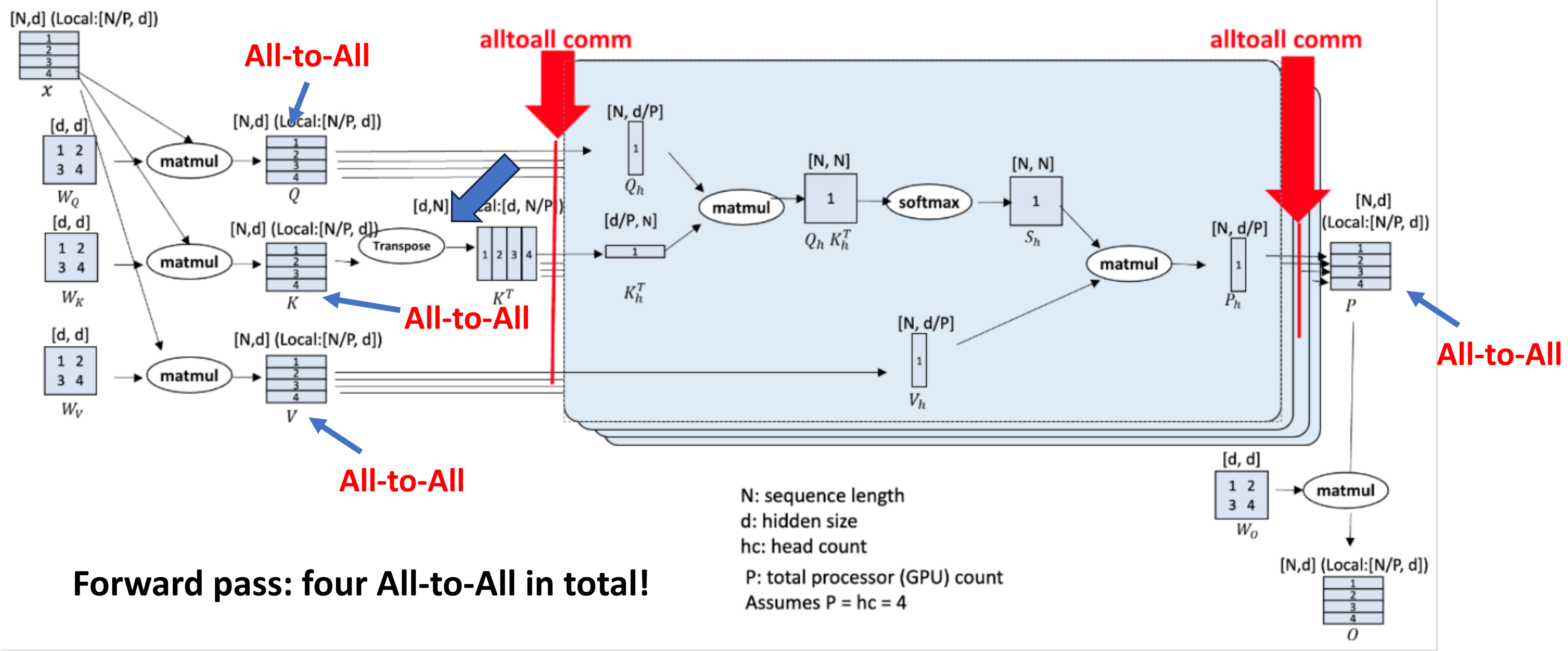
Output dimension: $\left(\frac{N}{P}, d\right)$

after ALL-to-ALL

W_O dimension: $\left(\frac{N}{P}, d\right)$

Sequence Parallelism

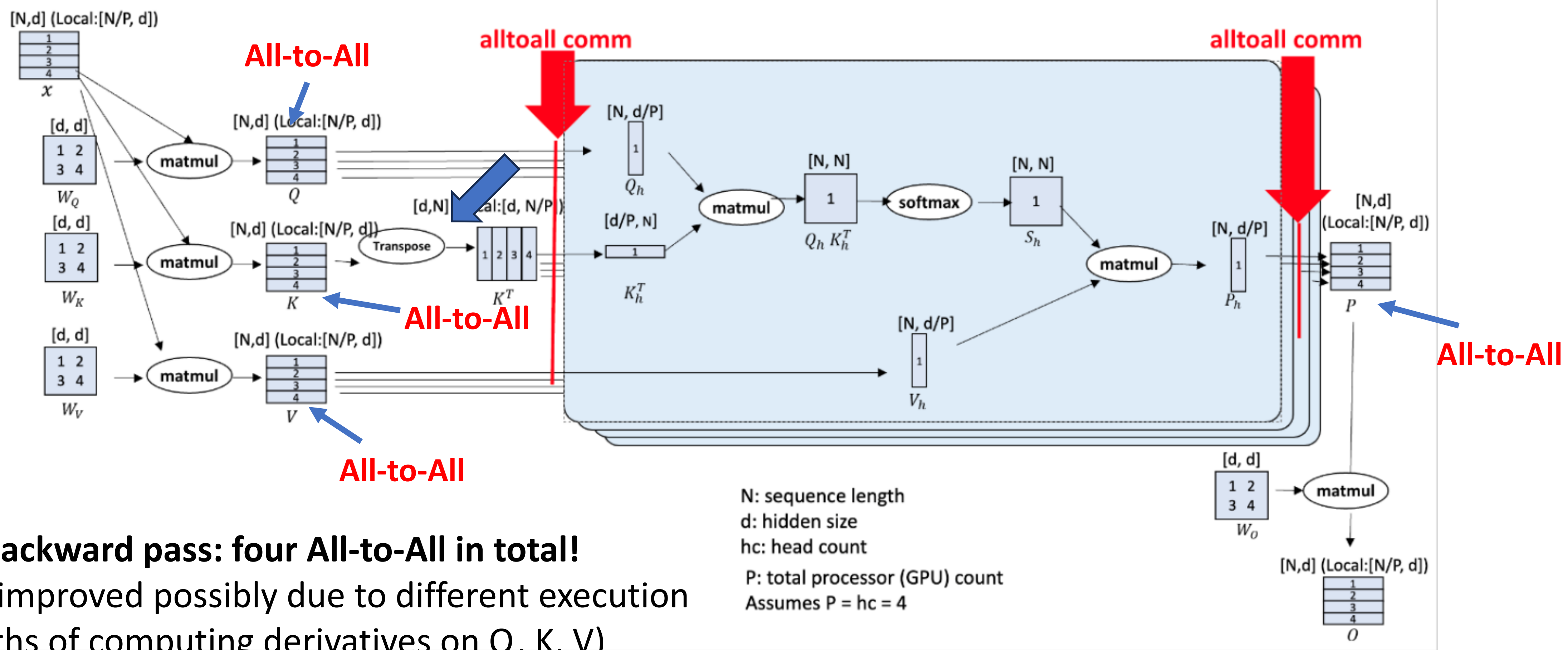
- DeepSpeed Ulysses Sequence Parallelism: Communication Analysis



Forward pass: four All-to-All in total!

Sequence Parallelism

- DeepSpeed Ulysses Sequence Parallelism: Communication Analysis



Backward pass: four All-to-All in total!

(Can be improved possibly due to different execution paths of computing derivatives on Q, K, V)

Sequence Parallelism

- DeepSpeed Ulysses versus NVIDIA Megatron
  - Megatron SP **splits attention and MLP layer parameters** by using tensor parallelism, and splits layernorm and dropout inputs accordingly. The computations are carried out on the heads at different GPUs.
  - Ulysses splits attention and MLP by segmenting the input sequences, and the computations are carried out on the heads at different GPUs. **No segmentation** on the attention layer parameters.

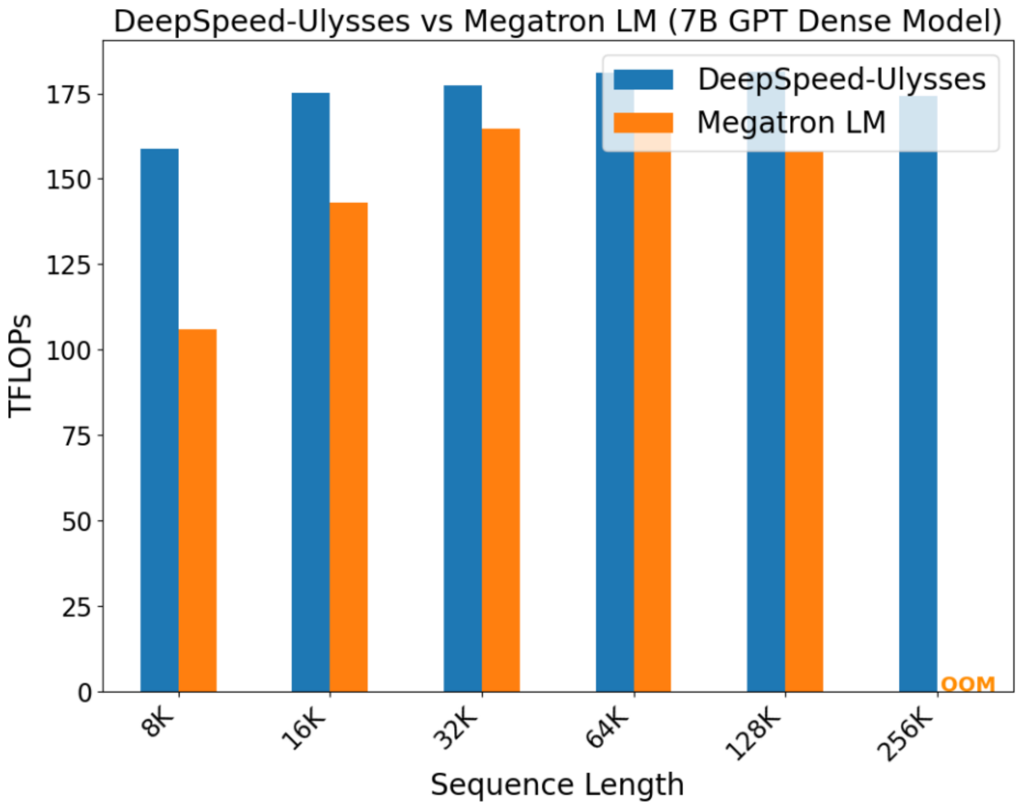
Sequence Parallelism

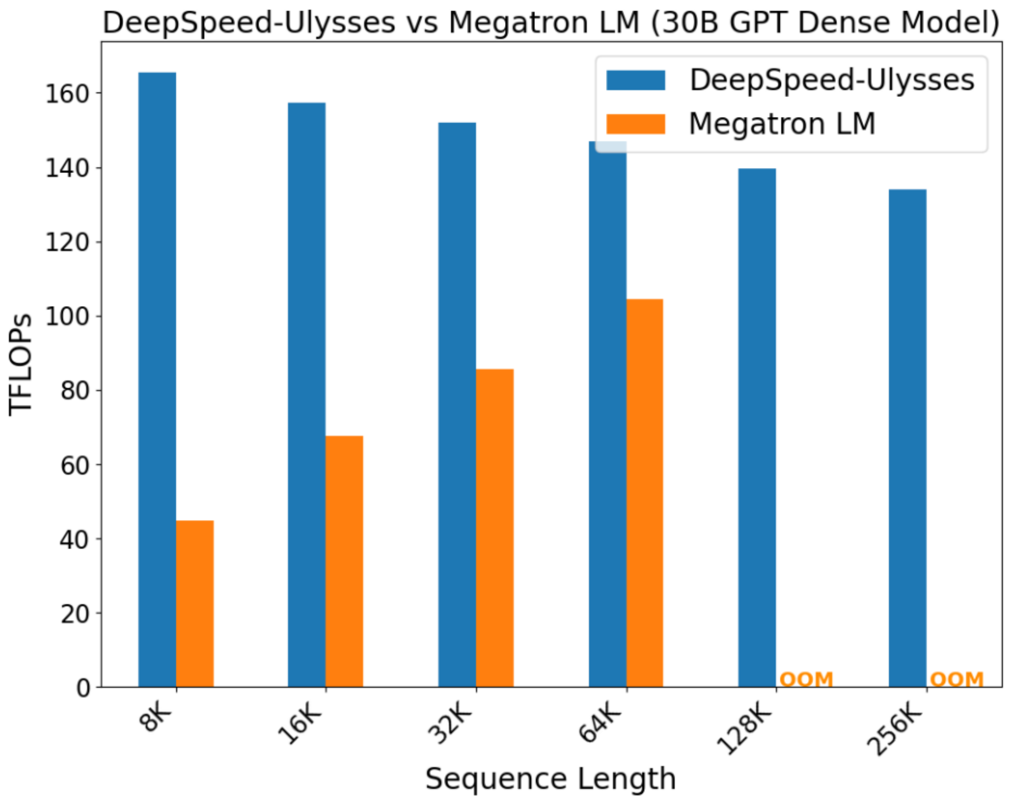
- DeepSpeed Ulysses versus NVIDIA Megatron

	Attention	**MLP**
Megatron-LM (combined with TP)	***4 N d*** Forward: ReduceScatter + AllGather, Backward: ReduceScatter + AllGather	***4 N d*** Forward: ReduceScatter + AllGather, Backward: ReduceScatter + AllGather
DeepSpeed Ulysses	***8 N d/P*** Forward: 4 All-to-All, Backward: 4 All-to-All	***0*** MLP is element-wise, naturally fitting SP

Sequence Parallelism

- DeepSpeed Ulysses versus NVIDIA Megatron





Sequence Parallelism

EXAMPLE

- DeepSpeed Ulysses versus NVIDIA Megatron
  - N = 1M tokens, $d = 4096$, $P = 128$, FP16
  - Megatron: $8 \times N \times d = 6 \times 1e6 \times 4096 \times 2 = 65.536$ GB
  - Ulysses: $8 \times N \times d / P = 65.536$ GB $/ 128 = 0.512$ GB

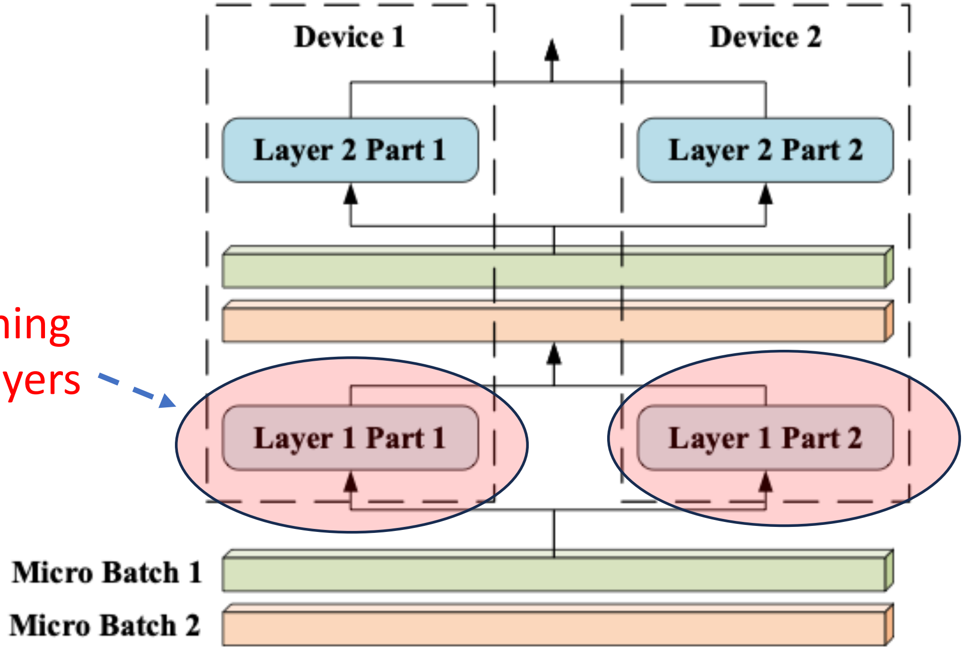
“Ulysses reduces communication volume by eliminating All-Gather in MLP and using All-to-All only in attention, achieving near-linear scaling for sequence lengths up to 1M.”

“Megatron-LM sequence parallelism incurs a communication volume per link of 4Nh which is P times larger than that for DeepSpeed sequence parallelism.”

Sequence Parallelism

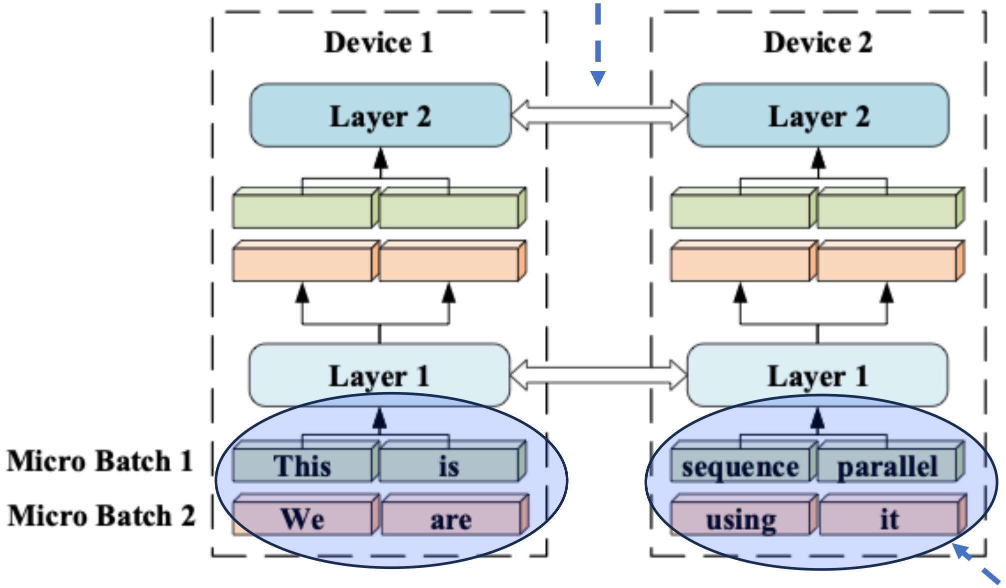
- Colossal AI Sequence Parallelism

Partitioning model layers



Tensor parallelism

Enabling attention computation



Sequence parallelism

Partitioning input sequences

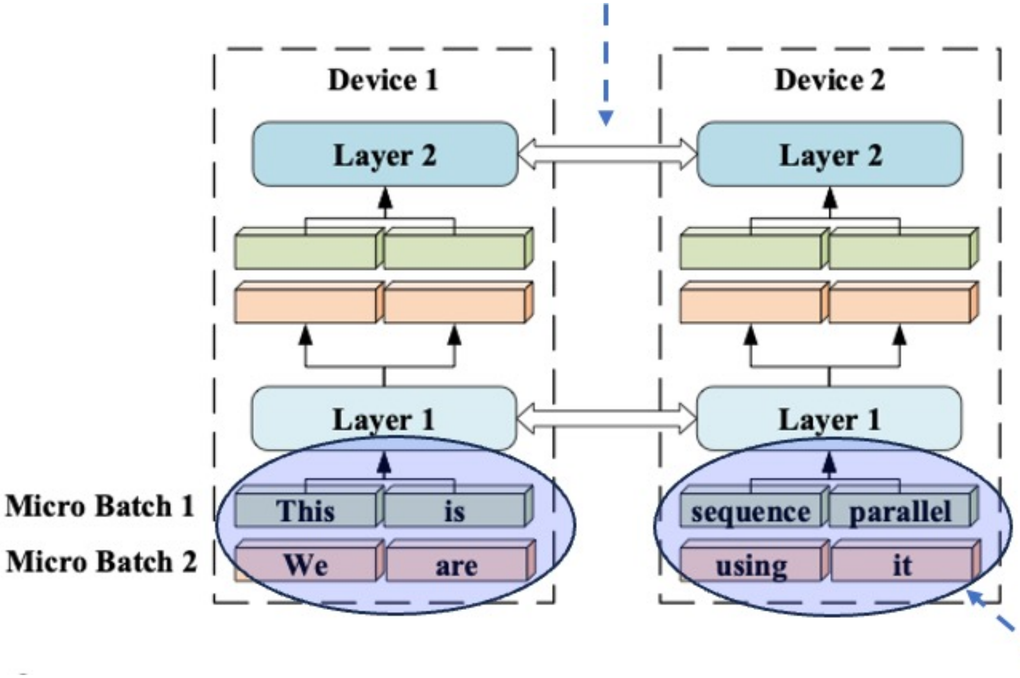
《Sequence Parallelism: Long Sequence Training from System Perspective》

Sequence Parallelism

- Colossal AI Sequence Parallelism
  - Q, K, V partitioned on the sequence
  - Computing attention scores need full Q, K, V on all tokens, in which communication is inevitable

- **Ring Attention** adopted in Colossal-AI

Enabling attention computation



Partitioning input sequences

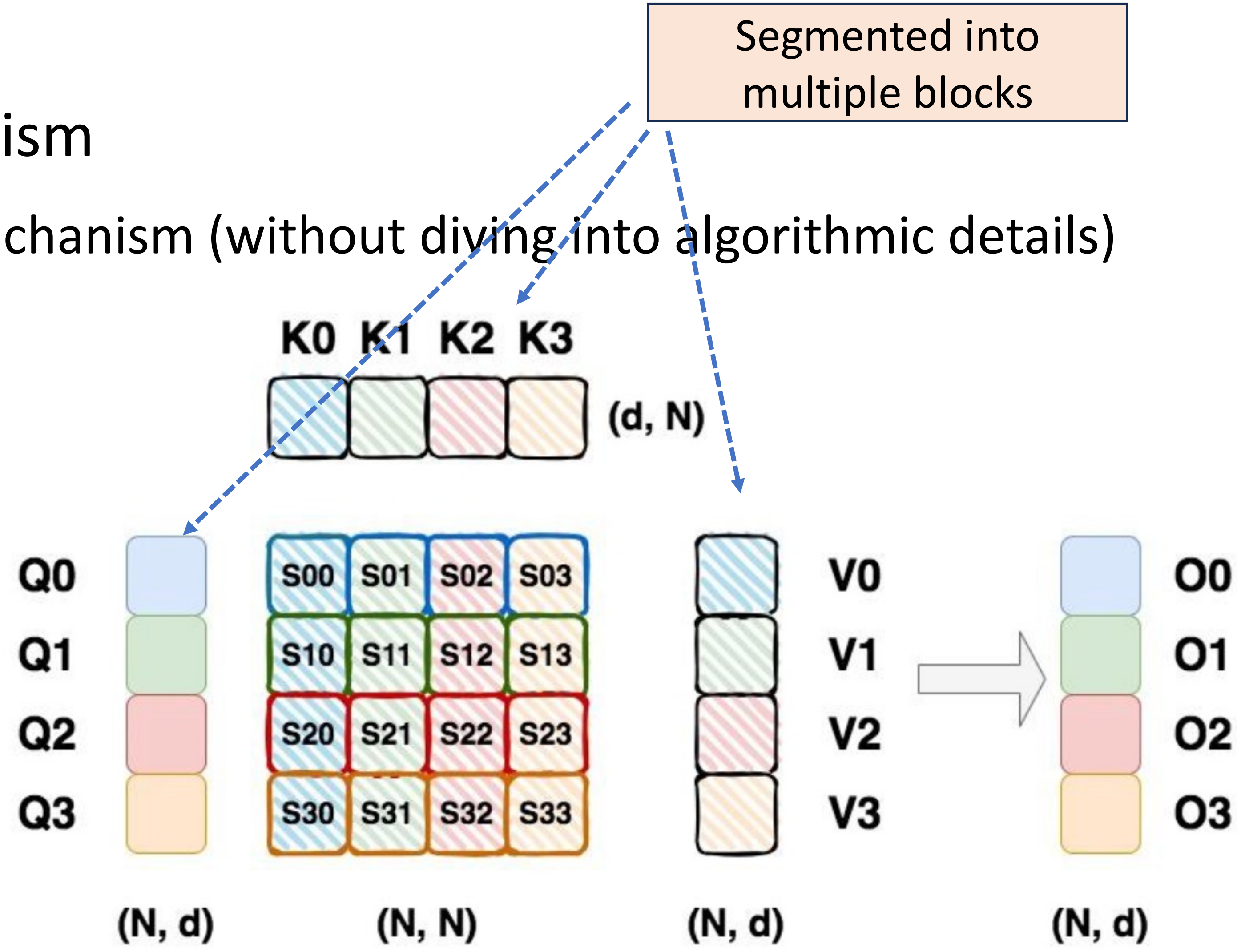
Sequence Parallelism

- Colossal AI Sequence Parallelism
  - Convoluted with Attention mechanism (without diving into algorithmic details)

Step 1: for each Q_i, compute attention score by feeding K_j and V_j, and obtain O_{ij}

Step 2: for each K_j and V_j, repeat Step 1 and update O_{ij}, and eventually obtain O_i

Step 3: change Q_i to Q_{i+1}, repeat Step 1 and Step 2 until Q_C

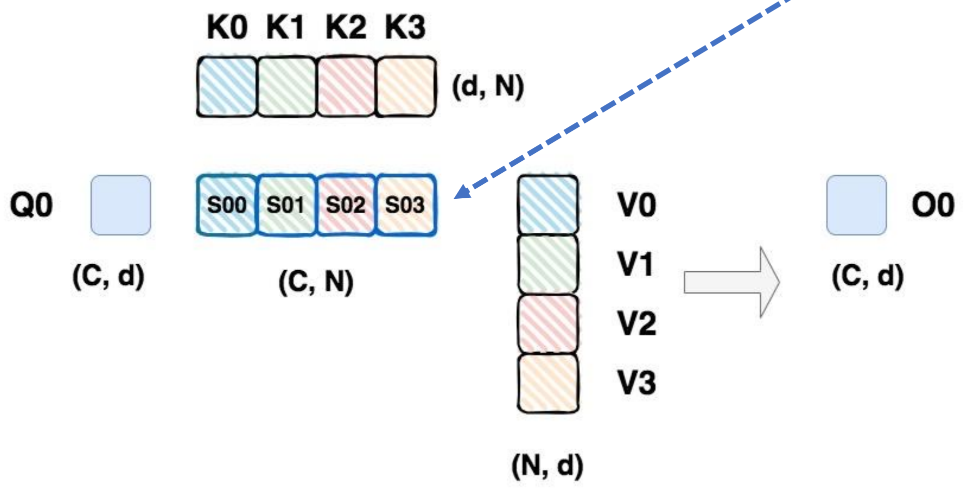


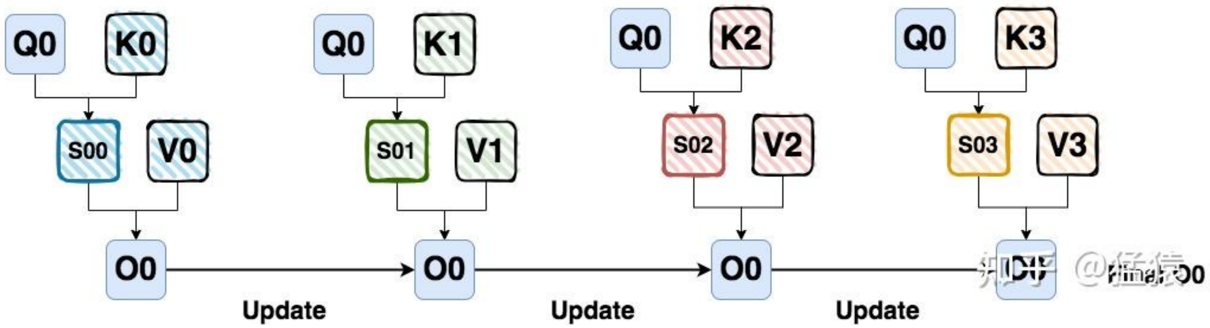
From https://zhuanlan.zhihu.com/p/4963530231

Sequence Parallelism

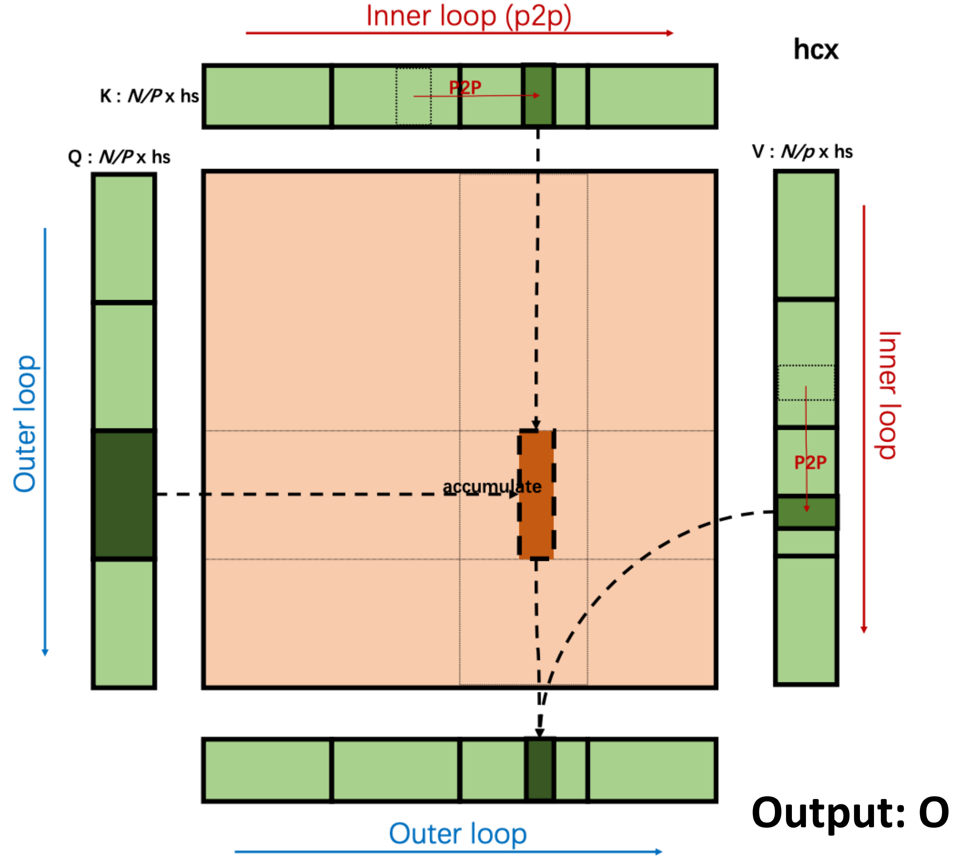
Mathematical deduction shows that S_{00}, S_{01}, S_{02}, and S_{03} can be computed in an arbitrary order!

- Colossal AI Sequence Parallelism





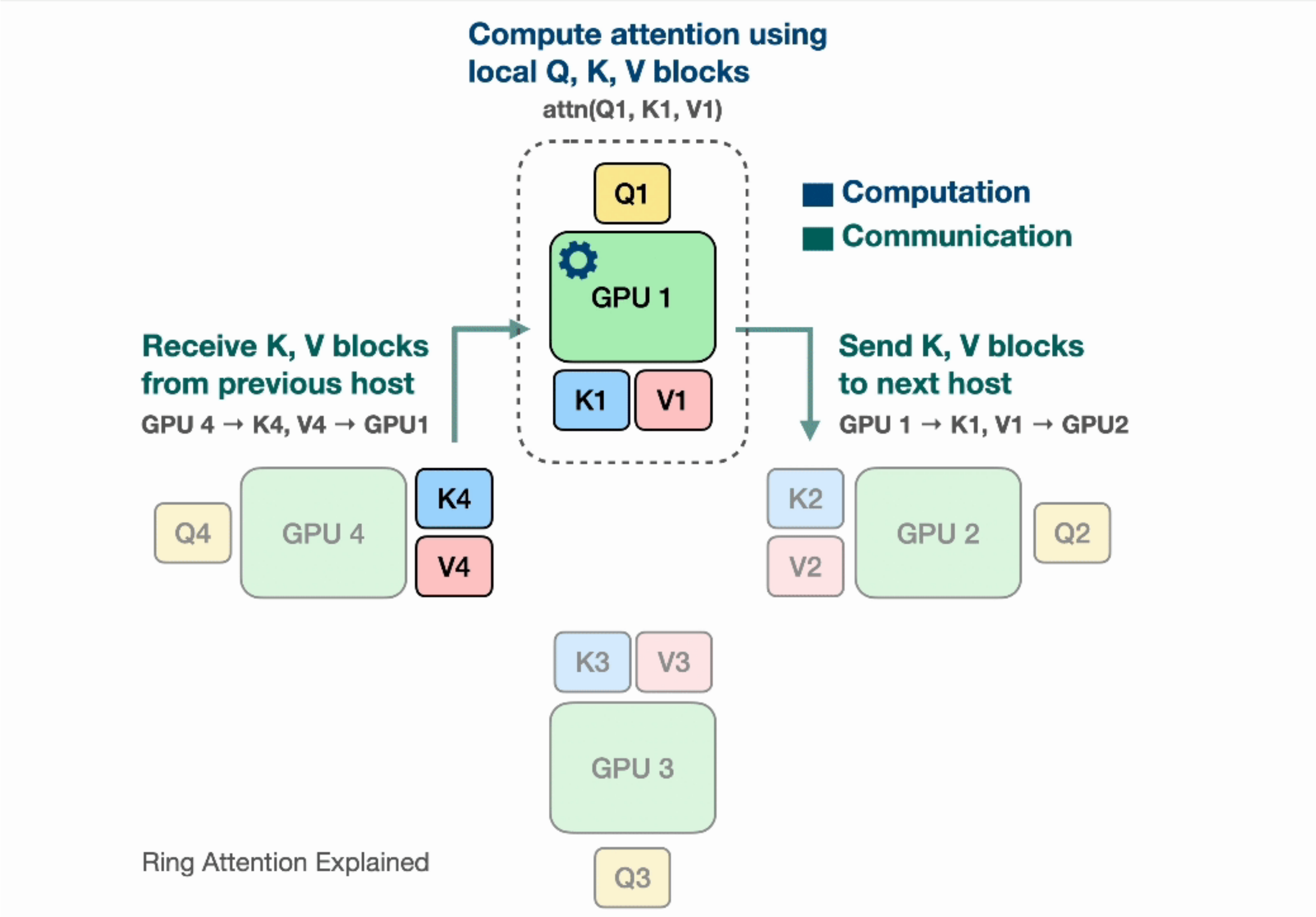
Ring-Attention



From https://zhuanlan.zhihu.com/p/4963530231

Sequence Parallelism

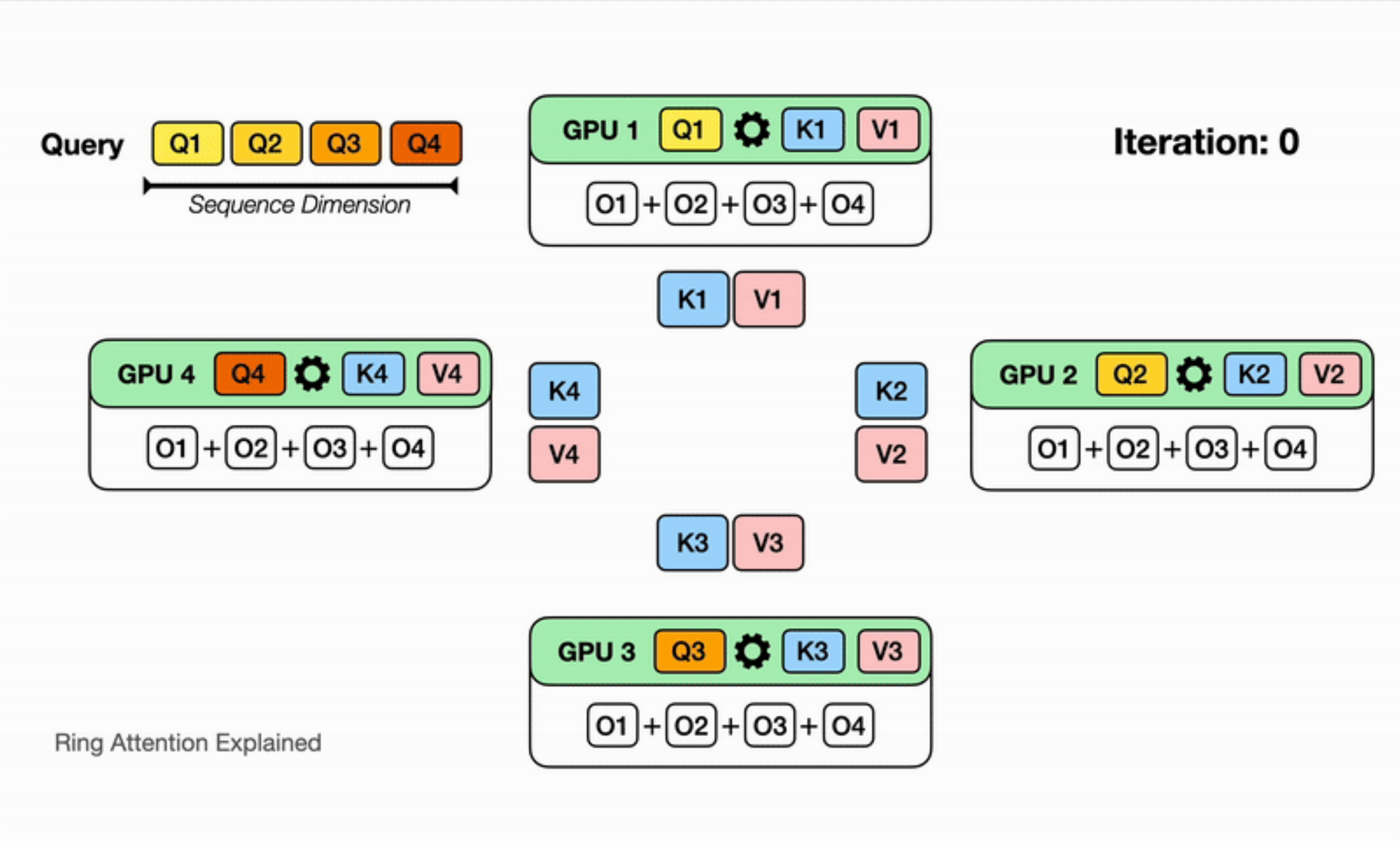
- Colossal AI Sequence Parallelism



Observe that for GPU1, while it is computing output using Q_1 (its local query) and K_1, V_1 (the local K,V blocks that it currently has), it is also receiving K_4, V_4 from GPU4 (previous host int the ring) and sending Q_1, V_1 to GPU2 (next host in the ring). The network communication are illustrated by the blinking arrows. If we select the block size correctly, by the time GPU1 has computed output using Q_1, K_1, V_1, it has received the block K_4, V_4 to compute the output in the next iteration!

From https://coconut-mode.com/posts/ring-attention/

Sequence Parallelism



Illustrate for N devices, it will take N iterations to finish the whole output comptutation. Watch for each iteration on each device, different partial sum of the output is computed with the K, V block it currently has, and it eventually sees all the K, V blocks and has all the partial sum for the output!

From https://coconut-mode.com/posts/ring-attention/

Sequence Parallelism

- Colossal AI Sequence Parallelism
  - Memory Complexity
    - Storing current K and V blocks (in FP16) needs $2dc$ floats or $4dc$ Bytes
    - Receiving new K and V blocks needs $2dc$ floats or $4dc$ Bytes
    - Storing Q_i block needs $2dc$ Bytes
    - Computing the output needs at least $2dc$ Bytes
    - In total at least $12dc$ Bytes, where c is the chunk size
  - Communication Complexity
    - Each step sends $2dc$ floats or $4dc$ Bytes, and the total number of rotations is around $\frac{s}{c}$
    - At each inner loop a GPU sends $4sd$ Bytes, and the outer loop has p rounds
    - Total communication load is around $O(sdp)$, which is much higher than Megatron and Ulysses

Thanks!

Tensor Parallelism

- A step-by-step trial
  - Column-wise sharding

$$\begin{pmatrix} X_0 & X_1 \\ X_2 & X_3 \end{pmatrix}\begin{pmatrix} A_0 & A_1 \\ A_2 & A_3 \end{pmatrix} = \left(\begin{pmatrix} X_0 & X_1 \\ X_2 & X_3 \end{pmatrix}\begin{pmatrix} A_0 \\ A_2 \end{pmatrix} \quad \begin{pmatrix} X_0 & X_1 \\ X_2 & X_3 \end{pmatrix}\begin{pmatrix} A_1 \\ A_3 \end{pmatrix}\right)$$

GPU1 GPU2

$$= \underbrace{\begin{pmatrix} XA_A & XA_B \end{pmatrix}}_{\text{Concatenation}}$$

All_gather

Tensor Parallelism

- A step-by-step trial
  - Row-wise sharding

GPU1

$$\begin{pmatrix} X_0 & X_1 \\ X_2 & X_3 \end{pmatrix}\begin{pmatrix} A_0 & A_1 \\ A_2 & A_3 \end{pmatrix} = \begin{pmatrix} X_0 \\ X_2 \end{pmatrix}\begin{pmatrix} A_0 & A_1 \end{pmatrix} + \begin{pmatrix} X_1 \\ X_3 \end{pmatrix}\begin{pmatrix} A_2 & A_3 \end{pmatrix}$$

GPU2

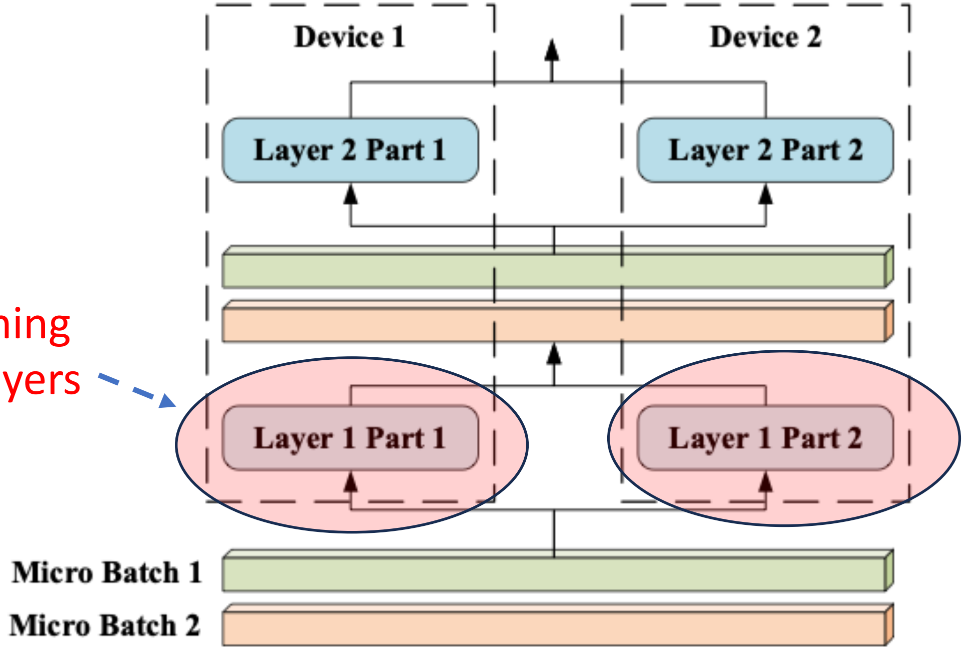
$$= \underbrace{X_A A_A + X_B A_B}_{\text{Summation}}$$

All_reduce

Sequence Parallelism

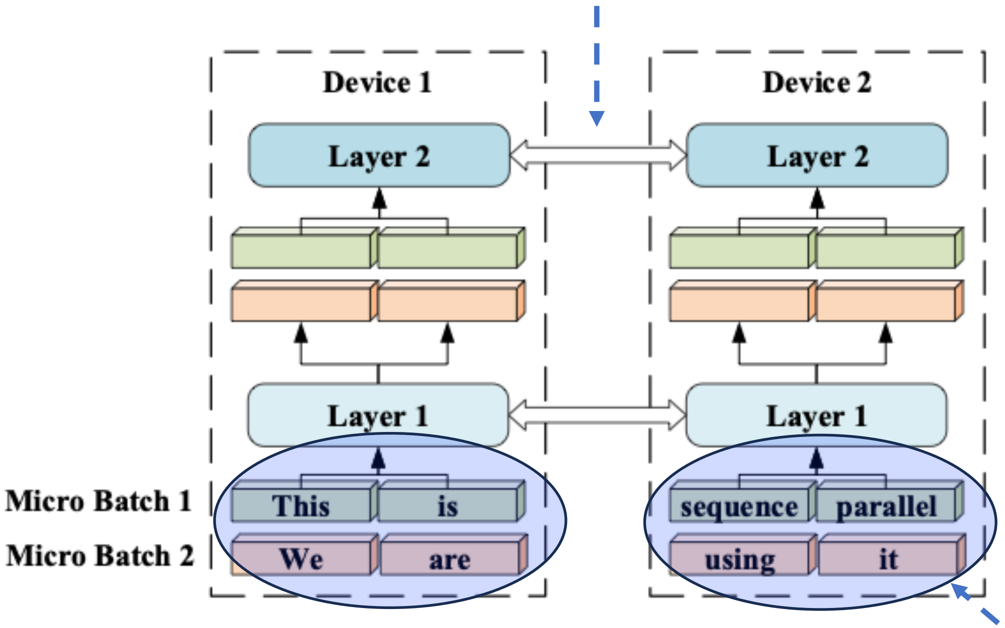
- Sequence Parallelism

Partitioning model layers



Tensor parallelism

Enabling attention computation



Sequence parallelism

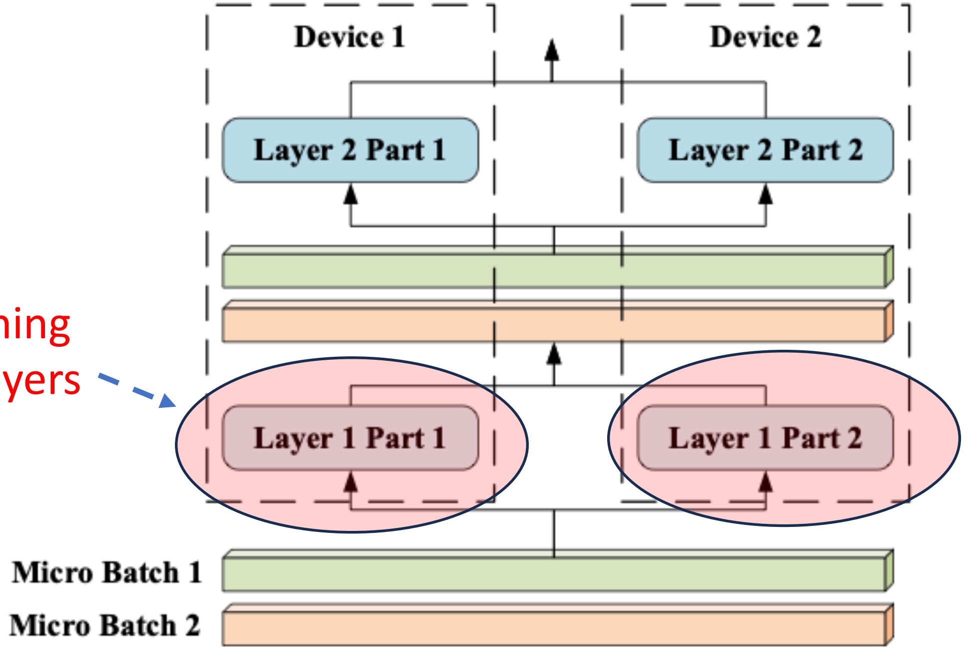
Partitioning input sequences

《Sequence Parallelism: Long Sequence Training from System Perspective》

Sequence Parallelism

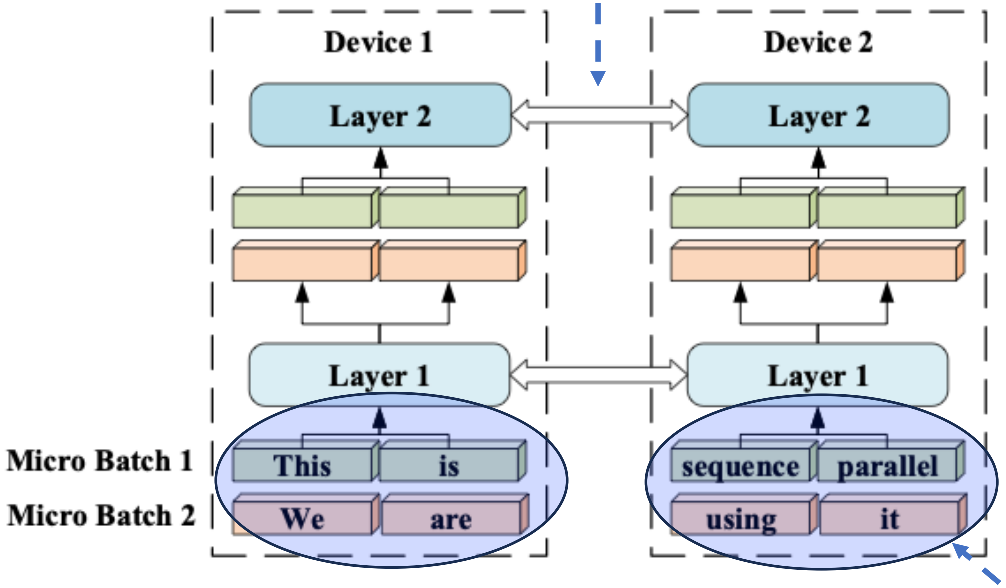
- Sequence Parallelism

Partitioning
model layers



Tensor parallelism

Enabling attention
computation



Sequence parallelism

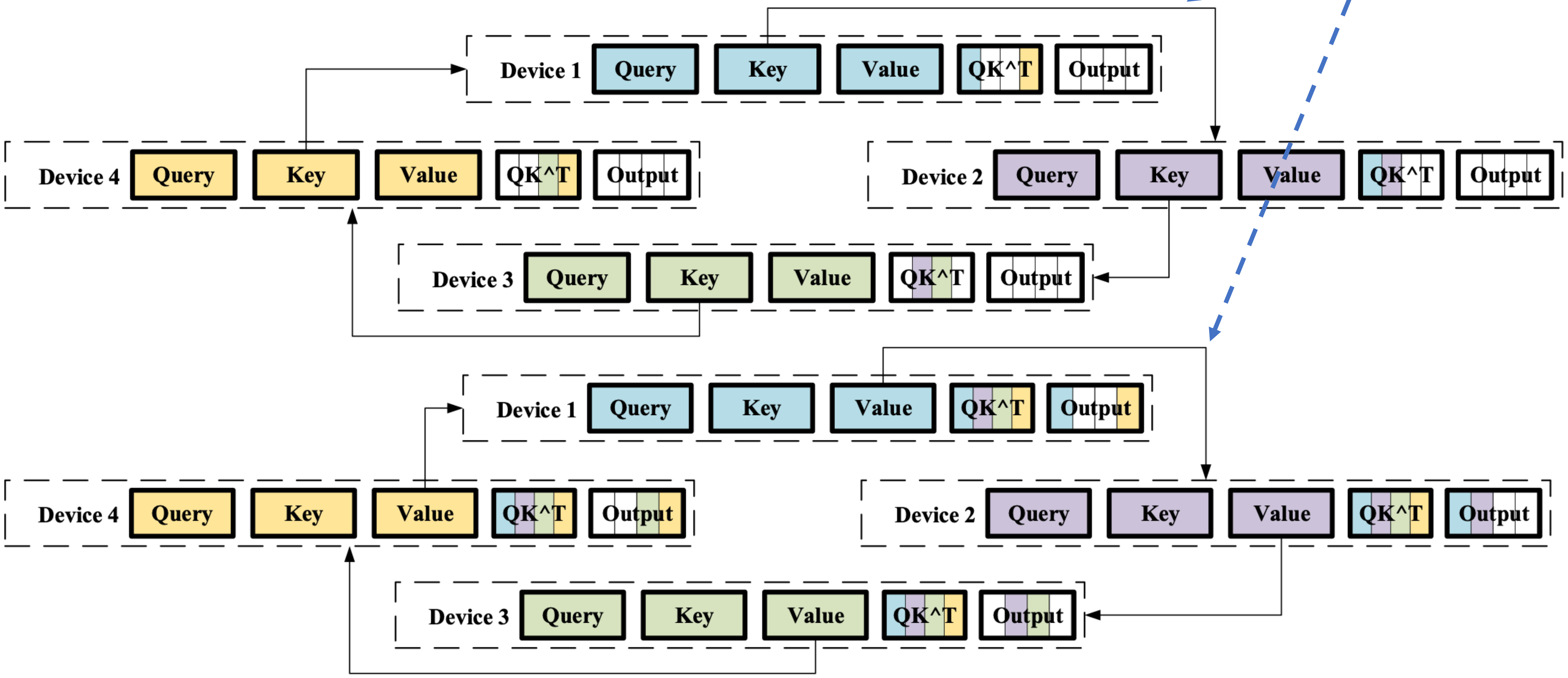
Partitioning
input sequences

《Sequence Parallelism: Long Sequence Training from System Perspective》

Sequence Parallelism

- Sequence Parallelism

Ring-Self-Attention: similar to ring all-reduce



RingQK: Concatenation before *softmax*

RingAV: Query values and sum the values

《Sequence Parallelism: Long Sequence Training from System Perspective》